STATISTICAL INFERENCE

BAYESIAN INFERENCE

Full-Syllabus Detailed Step-by-Step Solutions

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135 application-based problems

7 chapters × 45 topics × 3 problems per topic

How this manual is organised. Every problem is reproduced in its original wording from the supplied *BI Full-Syllabus Critical Thinking Practice Set*. Each solution begins with the relevant teacher-explanation and board-focus theory reproduced verbatim from the supplied *Bayesian Inference Book V3*, followed by a step-by-step derivation and contextual interpretation.

Notation. Gamma distributions use shape–rate parameterisation. The inverse-Gamma kernel is $v^{-a-1}\exp(-b/v)$. Variance and precision are distinguished explicitly. When a problem omits numerical information required for a unique number, the solution gives the exact symbolic result and identifies the missing input rather than inventing it.

Solution edition – 2026–2027

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Chapter 1

The Bayesian framework: probability, models, and updating

1.1 What Bayesian inference is

Problem 1

Original question (verbatim).
A screening programme is introduced in two districts. In District A the disease prevalence is believed to be 1%, while in District B it is believed to be 8%. The same test has sensitivity 0.96 and specificity 0.93 in both districts. A report gives only the overall positive-test rate and then claims that a positive result has the same meaning in both districts. Construct the full Bayesian model for each district, derive the two posterior disease probabilities after a positive result, and explain precisely why the test characteristics alone do not determine these posteriors. State which quantities are prior information, likelihood information, and posterior information.

Required theory (verbatim from the supplied lecture manual).
Let us begin with the central question of this entire course: when we do not know a quantity, how should we describe what we know about it? In a Bayesian analysis, we do not begin by pretending that one estimate tells the whole story. We describe uncertainty with a probability distribution. Before looking at the current data, we have a prior distribution. We then describe how the data would arise for each possible parameter value; that is our sampling model, and once the observed data are fixed it is also the likelihood as a function of the parameter. Finally, Bayes' rule combines those two ingredients and gives us the posterior distribution. The posterior is our updated description of uncertainty after seeing the data.
Please notice the difference between the three objects. The prior is about the unknown parameter before the current data. The likelihood is about the observed data conditional on a proposed parameter value. The posterior is about the parameter conditional on the data we actually observed. We must not swap the direction of conditioning. For example, the probability of a positive test given disease is not the probability of disease given a positive test. That reversal is exactly why Bayes' rule is needed.
When I write the equation on the board, I want you to read it aloud as: posterior density equals prior density multiplied by likelihood, divided by the marginal probability of the observed data. The denominator makes the result integrate or sum to one. When we are deriving a posterior only up to proportionality, we can temporarily omit that denominator because it does not depend on the unknown parameter. But we must remember that the final posterior is a properly normalised probability distribution. **Board equations from the notes.**

$$p(\theta \mid y) = \frac{p(\theta)p(y \mid \theta)}{p(y)}, \qquad p(y) = \int p(\theta)p(y \mid \theta)\,\mathrm{d}\theta.$$

$$\boxed{p(\theta \mid y) \propto p(\theta)\,p(y \mid \theta).}$$

Detailed solution and reasoning.
Step 1: Specify the event and the district-specific prior. Let D denote disease and $+$ a positive test. The prior probabilities are

$$\Pr(D \mid A) = 0.01, \qquad \Pr(D \mid B) = 0.08.$$

The test likelihood is the same in both districts:

$$\Pr(+ \mid D) = 0.96, \qquad \Pr(+ \mid D^c) = 1 - 0.93 = 0.07.$$

Step 2: Apply Bayes' theorem separately. For a district with prevalence π,

$$\Pr(D \mid +) = \frac{0.96\pi}{0.96\pi + 0.07(1-\pi)}.$$

For A,

$$\Pr(D \mid +, A) = \frac{0.0096}{0.0096 + 0.0693} = 0.121673 \text{ (about } 12.17\%).$$

For B,

$$\Pr(D \mid +, B) = \frac{0.0768}{0.0768 + 0.0644} = 0.543909 \text{ (about } 54.39\%).$$

Step 3: Explain the difference. Sensitivity and specificity specify the likelihood, not the prior prevalence. The same positive evidence is combined with different prior odds, so the posterior meanings differ substantially. An overall positive-test rate is a mixture depending on the proportions of residents sampled from A and B; without those mixture weights it does not identify either district's posterior.
Classification of information. The prevalences are prior information; sensitivity and false-positive probability are likelihood information; the two conditional probabilities just calculated are posterior information. The overall positive rate is a prior-predictive/marginal quantity once district composition is specified.

Problem 2

Original question (verbatim).
A machine has an unknown probability p of producing a defective item. Before observing the current batch, an engineer assigns a distribution to p, observes 12 defects in 200 items, and then predicts the number of defects in the next 50 items. A colleague says that the posterior distribution is simply the probability distribution of the next batch. Write the complete sequence of probability statements needed to distinguish the unknown parameter, observed data, likelihood, prior, posterior, and future data. Explain what additional randomness enters the prediction beyond uncertainty about p.

Required theory (verbatim from the supplied lecture manual).
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distribution. Before looking at the current data, we have a prior distribution. We then describe how the data would arise for each possible parameter value; that is our sampling model, and once the observed data are fixed it is also the likelihood as a function of the parameter. Finally, Bayes' rule combines those two ingredients and gives us the posterior distribution. The posterior is our updated description of uncertainty after seeing the data.
Please notice the difference between the three objects. The prior is about the unknown parameter before the current data. The likelihood is about the observed data conditional on a proposed parameter value. The posterior is about the parameter conditional on the data we actually observed. We must not swap the direction of conditioning. For example, the probability of a positive test given disease is not the probability of disease given a positive test. That reversal is exactly why Bayes' rule is needed.
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$$p(\theta \mid y) = \frac{p(\theta)p(y \mid \theta)}{p(y)}, \qquad p(y) = \int p(\theta)p(y \mid \theta)\,\mathrm{d}\theta.$$

$$\boxed{p(\theta \mid y) \propto p(\theta)\,p(y \mid \theta).}$$

Detailed solution and reasoning.
Step 1: Separate the random objects. Before data, p is unknown and has prior density $\pi(p)$ on $[0,1]$. Given p, the current defect count X satisfies

$$X \mid p \sim \text{Binomial}(200, p), \qquad \Pr(X = 12 \mid p) = \binom{200}{12} p^{12}(1-p)^{188}.$$

After observing $X = 12$, the likelihood as a function of p is proportional to $p^{12}(1-p)^{188}$ and the posterior is

$$\pi(p \mid X = 12) = \frac{\pi(p)\Pr(X = 12 \mid p)}{\int_0^1 \pi(u)\Pr(X = 12 \mid u)\,du}.$$

Step 2: Define the future batch. Conditional on p, let K be the number of defects among the next 50 items:

$$K \mid p \sim \text{Binomial}(50, p).$$

The posterior predictive distribution is the mixture

$$\Pr(K = k \mid X = 12) = \int_0^1 \binom{50}{k} p^k (1-p)^{50-k} \pi(p \mid X = 12)\,dp.$$

It is a distribution for future data, not the posterior distribution of the parameter.
Step 3: Identify the extra randomness. Prediction contains (i) uncertainty about p, represented by $\pi(p \mid X = 12)$, and (ii) item-to-item Bernoulli variation even if p were known. If the prior is $\text{Beta}(\alpha, \beta)$, then $p \mid X$ is $\text{Beta}(\alpha + 12, \beta + 188)$ and $K \mid X$ is Beta–Binomial with those updated parameters. Its variance exceeds the plug-in binomial variance because it includes variation in p.

Problem 3

Original question (verbatim).
Two analysts use the same sampling model and observe the same data but begin with different prior distributions. One reports a posterior concentrated near a small parameter value; the other reports a posterior concentrated near a larger value. Give a mathematically coherent example in which this can happen. Identify the circumstances under which the data will eventually dominate the prior, and construct a counterexample or limiting argument showing why a large sample does not automatically rescue a badly misspecified sampling model.

Required theory (verbatim from the supplied lecture manual).
Let us begin with the central question of this entire course: when we do not know a quantity, how should we describe what we know about it? In a Bayesian analysis, we do not begin by pretending that one estimate tells the whole story. We describe uncertainty with a probability distribution. Before looking at the current data, we have a prior distribution. We then describe how the data would arise for each possible parameter value; that is our sampling model, and once the observed data are fixed it is also the likelihood as a function of the parameter. Finally, Bayes' rule combines those two ingredients and gives us the posterior distribution. The posterior is our updated description of uncertainty after seeing the data.
Please notice the difference between the three objects. The prior is about the unknown parameter before the current data. The likelihood is about the observed data conditional on a proposed parameter value. The posterior is about the parameter conditional on the data we actually observed. We must not swap the direction of conditioning. For example, the probability of a positive test given disease is not the probability of disease given a positive test. That reversal is exactly why Bayes' rule is needed.
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$$\boxed{p(\theta \mid y) \propto p(\theta)\, p(y \mid \theta).}$$

Detailed solution and reasoning.
Step 1: Give a coherent conjugate example. Suppose $X \mid p \sim \text{Binomial}(n, p)$ and x successes are observed. Analyst A uses $p \sim \text{Beta}(1, 19)$ while Analyst B uses $p \sim \text{Beta}(19, 1)$. Their posteriors are respectively

$$p \mid x, A \sim \text{Beta}(1 + x, 19 + n - x), \qquad p \mid x, B \sim \text{Beta}(19 + x, 1 + n - x).$$

For a modest sample, the prior centres (near 0.05 and 0.95) can keep the posterior masses far apart even though the likelihood is identical. This is mathematically coherent: Bayes' rule combines the same likelihood with different prior densities.
Step 2: State when data dominate. Under a correctly specified, identifiable regular model, with the true parameter in the support of both priors and positive prior density in its neighbourhood, independent data accumulate likelihood information at order n. A fixed prior contributes order-one log-density, so the posterior typically concentrates near the true parameter as n grows.
Step 3: Show why misspecification is not cured by sample size. If the observations are

actually clustered, but the model assumes iid Bernoulli observations with one common p, the model's likelihood does not represent the dependence. A large sample can make the posterior very concentrated around a pseudo-parameter that best approximates the misspecified family, while its uncertainty statements and predictions remain wrong. More data reduce uncertainty conditional on the assumed model; they do not validate that model.

1.2 The three steps of Bayesian data analysis

Problem 4

Original question (verbatim).
A public-health team models the number of cases in each district using independent Poisson distributions with a common rate per person-year. The observed counts are far more variable across districts than the fitted model predicts. Lay out the three stages of Bayesian data analysis for this investigation: model construction, posterior computation, and model evaluation. Propose at least three posterior predictive checks that would expose the failure, explain what each check diagnoses, and describe a principled model revision without treating a larger sample or more MCMC iterations as a cure for misspecification.

Required theory (verbatim from the supplied lecture manual).
A Bayesian analysis is not just one calculation. It is a three-stage process, and all three stages matter. First, we construct the full probability model. This means stating what is random, what is unknown, how observations are generated, what is assumed independent, and what prior information is being used. A model that is mathematically convenient but scientifically unreasonable is still a poor model.
Second, after the data have been observed, we condition on those data and calculate the posterior distribution. In simple conjugate problems we may obtain a closed-form expression. In a more complicated model we may need numerical integration or simulation. The computational method changes, but the target remains the posterior distribution.
Third, we evaluate the model and the consequences of the posterior. We ask whether replicated data from the model look plausible, whether the conclusions make scientific sense, and whether they change substantially under reasonable alternative prior assumptions. If not, we revise the model and repeat the process. I want to emphasise that a posterior can be calculated perfectly and still be misleading if the first-stage model is inappropriate. Computation does not repair a bad model automatically.

Detailed solution and reasoning.
Stage 1 – construct the model. Write $Y_j \mid \lambda \sim \text{Poisson}(e_j\lambda)$, where e_j is district person-time, and state conditional independence. A conjugate starting point is $\lambda \sim \text{Gamma}(a, b)$ (shape–rate), giving

$$\lambda \mid y \sim \text{Gamma}\left(a + \sum_j y_j,\ b + \sum_j e_j\right).$$

The model implies conditional variance equal to conditional mean for each count and one common rate after exposure adjustment.
Stage 2 – compute the posterior. Use the closed-form update above if the common-rate model is retained. If a richer model is selected, use a suitable sampler or numerical integration and check computation separately from model fit.
Stage 3 – evaluate the model. Generate replicated counts Y_j^{rep} from the posterior predictive

distribution. Useful checks include: (i) compare $\mathrm{Var}(Y_j/e_j)$ or a dispersion statistic with replications to detect excess between-district variability; (ii) compare the maximum standardized district residual to reveal unusually high-risk districts; (iii) compare zero counts or upper-tail counts to assess the frequency of rare/extreme districts; and (iv) compare the distribution of pairwise or rank patterns if spatial structure is suspected.

Revision. A Gamma–Poisson random-effects model, $\lambda_j \mid k, m \sim \mathrm{Gamma}(k, k/m)$, produces negative-binomial marginal counts and allows district heterogeneity. Covariates or a spatial hierarchy may be appropriate if the extra variation has structure. Neither a larger sample nor more MCMC iterations repairs a variance assumption that is scientifically false.

Problem 5

Original question (verbatim).
A laboratory measures patients in four hospitals. An analyst assumes all observations are conditionally independent and identically distributed given one common mean, but hospital-level plots show persistent shifts. Specify two competing Bayesian models, one with a common parameter and one with hospital-specific parameters. For each model, describe the prior specification, the posterior target, and checks you would perform before interpreting the result. Explain how the three stages of analysis interact rather than functioning as a one-way pipeline.

Required theory (verbatim from the supplied lecture manual).
A Bayesian analysis is not just one calculation. It is a three-stage process, and all three stages matter. First, we construct the full probability model. This means stating what is random, what is unknown, how observations are generated, what is assumed independent, and what prior information is being used. A model that is mathematically convenient but scientifically unreasonable is still a poor model.

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Third, we evaluate the model and the consequences of the posterior. We ask whether replicated data from the model look plausible, whether the conclusions make scientific sense, and whether they change substantially under reasonable alternative prior assumptions. If not, we revise the model and repeat the process. I want to emphasise that a posterior can be calculated perfectly and still be misleading if the first-stage model is inappropriate. Computation does not repair a bad model automatically.

Detailed solution and reasoning.
Model A: one common mean. For observation Y_{hj} in hospital h, one possible model is $Y_{hj} \mid \mu, \sigma^2 \overset{ind}{\sim} N(\mu, \sigma^2)$ with a proper prior on (μ, σ^2), for example a Normal–Inverse-Gamma prior. Its posterior target is $p(\mu, \sigma^2 \mid y)$.

Model B: hospital-specific means. Use $Y_{hj} \mid \mu_h, \sigma^2 \sim N(\mu_h, \sigma^2)$ and $\mu_h \mid \mu_0, \tau^2 \sim N(\mu_0, \tau^2)$, with hyperpriors on $(\mu_0, \tau^2, \sigma^2)$. The target is the joint posterior of all hospital means and hyperparameters. This allows partial pooling rather than treating each hospital as unrelated.

Checks and comparison. For both models, inspect posterior predictive distributions of hospital means, within-hospital spread, and between-hospital spread. Check residuals, sensitivity to priors, and whether hospital labels or known case-mix variables explain systematic shifts. Compare predictive performance using a prespecified out-of-sample criterion when appropriate.

Why the stages interact. Model construction determines the posterior target; computation

approximates that target; evaluation may reveal a wrong likelihood or hierarchy. A failed predictive check sends the analyst back to model construction, after which posterior computation and checking are repeated. It is therefore an iterative process, not a one-way pipeline.

Problem 6

Original question (verbatim).
A posterior distribution has been computed exactly under a convenient conjugate model and is mathematically proper. The posterior predictive distribution nevertheless assigns negligible probability to a pattern that appears repeatedly in the observed data. Explain why posterior propriety and model adequacy are logically distinct. Design a model-checking procedure using replicated data, identify what would count as a serious discrepancy, and state how the analysis should change if the discrepancy is systematic rather than an isolated random fluctuation.

Required theory (verbatim from the supplied lecture manual).
A Bayesian analysis is not just one calculation. It is a three-stage process, and all three stages matter. First, we construct the full probability model. This means stating what is random, what is unknown, how observations are generated, what is assumed independent, and what prior information is being used. A model that is mathematically convenient but scientifically unreasonable is still a poor model.
Second, after the data have been observed, we condition on those data and calculate the posterior distribution. In simple conjugate problems we may obtain a closed-form expression. In a more complicated model we may need numerical integration or simulation. The computational method changes, but the target remains the posterior distribution.
Third, we evaluate the model and the consequences of the posterior. We ask whether replicated data from the model look plausible, whether the conclusions make scientific sense, and whether they change substantially under reasonable alternative prior assumptions. If not, we revise the model and repeat the process. I want to emphasise that a posterior can be calculated perfectly and still be misleading if the first-stage model is inappropriate. Computation does not repair a bad model automatically.

Detailed solution and reasoning.
Step 1: Separate propriety from fit. Posterior propriety means $\int p(\theta \mid y)d\theta = 1$ and the posterior is a valid probability distribution. Model adequacy asks whether the sampling model can generate data resembling the observed data. The first is a mathematical normalisation property; the second is a scientific/data-checking property. Neither implies the other.
Step 2: Simulate replicated data. Draw $\theta^{(m)} \sim p(\theta \mid y)$ and then $y^{rep,(m)} \sim p(y^{rep} \mid \theta^{(m)})$. Choose a discrepancy statistic before examining the replicated values, for example the number of extreme clusters, maximum cluster size, tail count, or a dispersion statistic. Compare $T(y)$ with the distribution of $T(y^{rep})$.
Step 3: Interpret the check. A repeated observed pattern lying far outside nearly all replications is evidence that the model systematically underpredicts that pattern. One isolated extreme replication is not conclusive because random fluctuations occur. Posterior predictive tail areas are diagnostic summaries, not automatically calibrated classical p-values.
Step 4: Revise the generative structure. For clustering, consider dependence, latent group effects, mixture components, temporal/spatial structure, or a heavier-tailed observation model, as scientifically justified. Then recompute the posterior and repeat the predictive checks. Exact conjugate calculation does not excuse model failure.

1.3 Parameters, data, likelihood, and predictive quantities

Problem 7

Original question (verbatim).
Suppose counts satisfy $Y_i \mid \lambda \sim \text{Poisson}(x_i\lambda)$, where the known exposures are $(1, 2, 4, 3)$ and the prior is $\lambda \sim \text{Gamma}(a, b)$ in the shape-rate convention. The observed counts are $(0, 3, 5, 2)$. Derive the likelihood and identify the data summaries through which it depends on λ. Obtain the posterior and derive the posterior predictive distribution for a new count with exposure 2. Explain why substituting the posterior mean of λ into a Poisson distribution does not generally give the correct predictive distribution.

Required theory (verbatim from the supplied lecture manual).
Let us set up the notation carefully because the same symbols will appear throughout the course. We use theta for an unknown parameter, y for the data already observed, and y-tilde for a future or otherwise unobserved quantity that could in principle be measured. A parameter is not the same thing as an observation. A future observation is also not the same thing as a parameter: it contains both the uncertainty about the parameter and the natural randomness of a new measurement.
If observations are conditionally independent given theta, their joint likelihood is the product of their individual sampling densities. The phrase 'conditionally independent' is important. It means independent after we know the shared parameter. If we average over uncertainty about that parameter, observations can become dependent marginally.
Before observing the current data, we can average the sampling distribution over the prior to obtain a prior predictive distribution. This asks: if our prior and model were true, what kind of data would we expect to see? After observing the data, we average over the posterior instead. That gives the posterior predictive distribution for a new observation. Do not simply substitute the posterior mean into the likelihood and call the result the predictive distribution. The correct predictive calculation integrates over all posterior parameter values, weighted by their posterior probabilities. **Board equations from the notes.**

$$p(y \mid \theta) = \prod_{i=1}^{n} p(y_i \mid \theta).$$

$$p(\tilde{y}) = \int p(\tilde{y} \mid \theta) p(\theta)\, \mathrm{d}\theta.$$

Detailed solution and reasoning.
Step 1: Form the likelihood. Here $\sum_i x_i = 1 + 2 + 4 + 3 = 10$ and $\sum_i y_i = 0 + 3 + 5 + 2 = 10$. Independence gives

$$L(\lambda; y) = \prod_i \frac{e^{-x_i\lambda}(x_i\lambda)^{y_i}}{y_i!} \propto \lambda^{10} e^{-10\lambda}.$$

Thus the likelihood depends on the observations through $\sum_i y_i$ and $\sum_i x_i$ (the exposure total is known); the remaining factors do not depend on λ.
Step 2: Update the Gamma prior. Multiplying by $\pi(\lambda) \propto \lambda^{a-1}e^{-b\lambda}$ gives

$$\pi(\lambda \mid y) \propto \lambda^{a+9} e^{-(b+10)\lambda}, \qquad \lambda \mid y \sim \text{Gamma}(A, B),$$

where $A = a + 10$ and $B = b + 10$.
Step 3: Integrate for a new count. For $Y_* \mid \lambda \sim \text{Poisson}(2\lambda)$,

$$\Pr(Y_* = k \mid y) = \int \frac{e^{-2\lambda}(2\lambda)^k}{k!} \frac{B^A}{\Gamma(A)} \lambda^{A-1} e^{-B\lambda}\, d\lambda$$

$$= \frac{\Gamma(A+k)}{\Gamma(A)k!}\left(\frac{B}{B+2}\right)^{A}\left(\frac{2}{B+2}\right)^{k}.$$

This is negative binomial (Gamma–Poisson mixture), with mean $2A/B$ and variance $2A/B+4A/B^2$. Substituting $E(\lambda \mid y) = A/B$ into a Poisson gives variance only $2A/B$ and omits the extra $4A/B^2$ from posterior uncertainty.

Problem 8

Original question (verbatim).
Let $Y_1, \ldots, Y_n \mid \mu$ be conditionally independent normal observations with known variance, and let μ have a normal prior. A future observation Y_{n+1} is to be predicted. Derive the posterior predictive distribution by integrating over the posterior of μ. Compare its variance with the variance of $Y_{n+1} \mid \mu$, and explain which part represents uncertainty about the unknown mean and which part represents irreducible observation noise. Give a numerical design in which the distinction is practically important.

Required theory (verbatim from the supplied lecture manual).
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Before observing the current data, we can average the sampling distribution over the prior to obtain a prior predictive distribution. This asks: if our prior and model were true, what kind of data would we expect to see? After observing the data, we average over the posterior instead. That gives the posterior predictive distribution for a new observation. Do not simply substitute the posterior mean into the likelihood and call the result the predictive distribution. The correct predictive calculation integrates over all posterior parameter values, weighted by their posterior probabilities. **Board equations from the notes.**

$$p(y \mid \theta) = \prod_{i=1}^{n} p(y_i \mid \theta).$$

$$p(\tilde{y}) = \int p(\tilde{y} \mid \theta)p(\theta)\,\mathrm{d}\theta.$$

Detailed solution and reasoning.
Step 1: Posterior for the mean. With $Y_i \mid \mu \sim N(\mu, \sigma^2)$ and $\mu \sim N(\mu_0, \tau_0^2)$,

$$\tau_n^2 = \left(\tau_0^{-2} + n\sigma^{-2}\right)^{-1}, \qquad \mu_n = \tau_n^2\left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2}\right).$$

Thus $\mu \mid y \sim N(\mu_n, \tau_n^2)$.
Step 2: Integrate the future observation. Since $Y_{n+1} \mid \mu \sim N(\mu, \sigma^2)$,

$$p(y_{n+1} \mid y) = \int p(y_{n+1} \mid \mu)p(\mu \mid y)d\mu$$

and the normal–normal convolution gives

$$Y_{n+1} \mid y \sim N(\mu_n, \sigma^2 + \tau_n^2).$$

The σ^2 term is irreducible measurement variation; τ_n^2 is uncertainty about the unknown population mean.
Step 3: Numerical illustration. Take $\sigma^2 = 4$, prior variance $\tau_0^2 = 9$, and $n = 2$. Then $\tau_n^2 = (1/9 + 2/4)^{-1} = 1.6364$, so the predictive variance is 5.6364, not 4. With many observations τ_n^2 becomes small, but the future observation variance remains σ^2.

Problem 9

Original question (verbatim).
Let $P \sim \text{Beta}(a, b)$ and, conditional on P, let Y_1, Y_2 be independent Bernoulli observations with success probability P. Derive the marginal probability that both observations are successes and the marginal covariance $\text{Cov}(Y_1, Y_2)$. Explain why conditional independence does not imply marginal independence. Then explain how the answer changes if the two observations have separate independent parameters rather than a shared uncertain parameter.

Required theory (verbatim from the supplied lecture manual).
Let us set up the notation carefully because the same symbols will appear throughout the course. We use theta for an unknown parameter, y for the data already observed, and y-tilde for a future or otherwise unobserved quantity that could in principle be measured. A parameter is not the same thing as an observation. A future observation is also not the same thing as a parameter: it contains both the uncertainty about the parameter and the natural randomness of a new measurement.
If observations are conditionally independent given theta, their joint likelihood is the product of their individual sampling densities. The phrase 'conditionally independent' is important. It means independent after we know the shared parameter. If we average over uncertainty about that parameter, observations can become dependent marginally.
Before observing the current data, we can average the sampling distribution over the prior to obtain a prior predictive distribution. This asks: if our prior and model were true, what kind of data would we expect to see? After observing the data, we average over the posterior instead. That gives the posterior predictive distribution for a new observation. Do not simply substitute the posterior mean into the likelihood and call the result the predictive distribution. The correct predictive calculation integrates over all posterior parameter values, weighted by their posterior probabilities. **Board equations from the notes.**

$$p(y \mid \theta) = \prod_{i=1}^{n} p(y_i \mid \theta).$$

$$p(\tilde{y}) = \int p(\tilde{y} \mid \theta) p(\theta) \, \mathrm{d}\theta.$$

Detailed solution and reasoning.
Step 1: Calculate the joint success probability. Conditional independence gives $\Pr(Y_1 = 1, Y_2 = 1 \mid P) = P^2$. Integrating over the Beta prior,

$$\Pr(Y_1 = 1, Y_2 = 1) = E(P^2) = \frac{a(a+1)}{(a+b)(a+b+1)}.$$

Also $E(Y_i) = E(P) = a/(a+b)$.

Step 2: Calculate covariance. By total expectation,

$$\mathrm{Cov}(Y_1, Y_2) = E(P^2) - E(P)^2 = \mathrm{Var}(P) = \frac{ab}{(a+b)^2(a+b+1)} > 0.$$

Thus they are independent conditional on P, but positively associated after the shared uncertain P is integrated out: observing a success suggests a larger P and therefore makes another success more likely.

Step 3: Change to independent parameters. If instead P_1 and P_2 are independent and $Y_i \mid P_i \sim \mathrm{Bernoulli}(P_i)$, then $E(P_1P_2) = E(P_1)E(P_2)$ and the marginal covariance is zero. The shared random parameter is the source of marginal dependence.

1.4 Posterior odds and the likelihood ratio

Problem 10

Original question (verbatim).
A disease has prior prevalence 0.004. A test has sensitivity 0.98 and false-positive probability 0.025. A patient tests positive. Derive the prior odds, likelihood ratio, posterior odds, and posterior probability of disease. A second test is then performed and is positive. Determine when multiplying the two likelihood ratios is valid, and show how the calculation changes if the tests are conditionally dependent given disease status. Explain why multiplying posterior probabilities directly is invalid.

Required theory (verbatim from the supplied lecture manual).
For two competing hypotheses, probability can be expressed as odds. Odds for H-one against H-zero are the probability of H-one divided by the probability of H-zero. Bayes' rule has a particularly simple form in odds: posterior odds equal prior odds multiplied by the likelihood ratio. In a simple-hypothesis comparison, the likelihood ratio is the probability of the observed data under H-one divided by the probability of the same data under H-zero.
This is a very useful interpretation. The prior odds tell us where we started. The likelihood ratio tells us how strongly the observed data distinguish the two hypotheses. The posterior odds tell us where we finish. The data do not multiply a prior probability directly; they multiply the prior odds. After obtaining posterior odds, convert them back into probabilities by dividing the odds numerator by the sum of the numerator and denominator. For composite hypotheses, the corresponding comparison uses marginal likelihoods and is called a Bayes factor. **Board equations from the notes.**

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = \frac{\Pr(H_1)}{\Pr(H_0)} \times \frac{p(y \mid H_1)}{p(y \mid H_0)}.$$

Detailed solution and reasoning.
Step 1: Prior odds and first likelihood ratio.

$$O_0 = \frac{0.004}{0.996} = 0.00401606, \qquad LR_1 = \frac{\Pr(+ \mid D)}{\Pr(+ \mid D^c)} = \frac{0.98}{0.025} = 39.2.$$

Therefore

$$O_1 = O_0 LR_1 = 0.157430, \qquad \Pr(D \mid +) = \frac{O_1}{1+O_1} = 0.136017.$$

The positive test moves disease probability from 0.4% to about 13.60%.

Step 2: Add a second positive result. If the second test has the same sensitivity and false-positive probability and the two tests are conditionally independent given disease status, $LR_2 = 39.2$, so

$$O_2 = O_0(39.2)^2 = 6.171245, \qquad \Pr(D \mid +,+) = \frac{O_2}{1+O_2} = 0.860554.$$

Equivalently, update the first posterior odds by multiplying by LR_2.
Step 3: Handle dependence correctly. If test results are dependent conditional on D or D^c, use

$$LR_{12} = \frac{\Pr(+,+ \mid D)}{\Pr(+,+ \mid D^c)}, \qquad O_2 = O_0 LR_{12}.$$

The product $LR_1 LR_2$ is valid only when the conditional joint probabilities factor under both disease states. Posterior probabilities are not likelihood factors; multiplying them double-counts the prior and ignores the correct normalisation.

Problem 11

Original question (verbatim).
Two hypotheses concern the failure mechanism of a component: H_0 says a failure probability is 0.02, while H_1 says it is 0.08. Prior model probabilities are 0.7 and 0.3. In 20 independent trials, 4 failures are observed. Calculate the posterior odds and posterior probability of H_1. Then repeat the reasoning after a second independent batch of observations. Explain how the direction of the likelihood ratio determines whether the data favour the numerator or denominator hypothesis.

Required theory (verbatim from the supplied lecture manual).
For two competing hypotheses, probability can be expressed as odds. Odds for H-one against H-zero are the probability of H-one divided by the probability of H-zero. Bayes' rule has a particularly simple form in odds: posterior odds equal prior odds multiplied by the likelihood ratio. In a simple-hypothesis comparison, the likelihood ratio is the probability of the observed data under H-one divided by the probability of the same data under H-zero.
This is a very useful interpretation. The prior odds tell us where we started. The likelihood ratio tells us how strongly the observed data distinguish the two hypotheses. The posterior odds tell us where we finish. The data do not multiply a prior probability directly; they multiply the prior odds. After obtaining posterior odds, convert them back into probabilities by dividing the odds numerator by the sum of the numerator and denominator. For composite hypotheses, the corresponding comparison uses marginal likelihoods and is called a Bayes factor. **Board equations from the notes.**

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = \frac{\Pr(H_1)}{\Pr(H_0)} \times \frac{p(y \mid H_1)}{p(y \mid H_0)}.$$

Detailed solution and reasoning.
Step 1: Form the likelihood ratio for the first batch. With $k = 4$ failures in $n = 20$ trials, the binomial coefficient cancels:

$$BF_{10}^{(1)} = \frac{0.08^4(0.92)^{16}}{0.02^4(0.98)^{16}} = \left(\frac{0.08}{0.02}\right)^4 \left(\frac{0.92}{0.98}\right)^{16} \approx 93.1597.$$

Step 2: Update prior odds. The prior odds for H_1 against H_0 are $0.3/0.7 = 3/7$. Hence

$$O_{10|y} = \frac{3}{7}(93.1597) = 39.9256, \qquad \Pr(H_1 \mid y) = \frac{39.9256}{1 + 39.9256} = 0.975565.$$

The observed batch strongly favours the 0.08 failure-probability hypothesis.
Step 3: Add a second independent batch. The prompt does not specify its size or failure count, so a second numerical result is not determined. If it has n_2 trials and k_2 failures, its likelihood ratio is

$$BF_{10}^{(2)} = \left(\frac{0.08}{0.02}\right)^{k_2} \left(\frac{0.92}{0.98}\right)^{n_2 - k_2},$$

and the new odds are $O_{10|y_1,y_2} = O_{10|y_1} BF_{10}^{(2)}$. A ratio above one favours the numerator hypothesis H_1; below one favours H_0.

Problem 12

Original question (verbatim).
A hospital must decide whether to administer an additional diagnostic procedure. Let the posterior odds of disease be O, the cost of treating a healthy patient be C_{FP}, and the cost of failing to treat a diseased patient be C_{FN}. Derive the posterior-probability or posterior-odds threshold at which treatment minimises posterior expected loss. Explain how a likelihood ratio from a new test changes the decision, and distinguish evidence updating from the final decision rule.

Required theory (verbatim from the supplied lecture manual).
For two competing hypotheses, probability can be expressed as odds. Odds for H-one against H-zero are the probability of H-one divided by the probability of H-zero. Bayes' rule has a particularly simple form in odds: posterior odds equal prior odds multiplied by the likelihood ratio. In a simple-hypothesis comparison, the likelihood ratio is the probability of the observed data under H-one divided by the probability of the same data under H-zero.
This is a very useful interpretation. The prior odds tell us where we started. The likelihood ratio tells us how strongly the observed data distinguish the two hypotheses. The posterior odds tell us where we finish. The data do not multiply a prior probability directly; they multiply the prior odds. After obtaining posterior odds, convert them back into probabilities by dividing the odds numerator by the sum of the numerator and denominator. For composite hypotheses, the corresponding comparison uses marginal likelihoods and is called a Bayes factor. **Board equations from the notes.**

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = \frac{\Pr(H_1)}{\Pr(H_0)} \times \frac{p(y \mid H_1)}{p(y \mid H_0)}.$$

Detailed solution and reasoning.
Let $q = \Pr(D \mid y)$ and $O = q/(1-q)$. Under the two available actions:

$$\text{Treat: } R_T = C_{FP}(1-q), \qquad \text{Do not treat: } R_N = C_{FN} q.$$

Treat when $R_T \le R_N$. Solving,

$$C_{FP}(1-q) \le C_{FN} q \iff q \ge \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

In odds form this is

$$O \ge \frac{C_{FP}}{C_{FN}}.$$

A new test updates the evidence first: $O_{new} = O_{old} \times LR$ (or the correct joint likelihood ratio if tests are dependent). Only after updating do we apply the loss-based threshold. Evidence updating is a probability calculation; treatment is a decision under specified costs. Changing costs changes the decision threshold but not the posterior evidence.

1.5 Exchangeability and conditional independence

Problem 13

Original question (verbatim).
A researcher observes n Bernoulli outcomes that are conditionally independent given a common unknown success probability $P \sim \text{Beta}(a, b)$. Derive the joint marginal probability of an ordered sequence containing s successes, and show that it depends on the sequence only through its success count. Calculate the marginal covariance between two distinct outcomes. Explain in precise terms why the sequence is exchangeable but not generally independent.

Required theory (verbatim from the supplied lecture manual).
Exchangeability means that the joint distribution of the observations does not change when we relabel their order. If I permute the observation labels, the probability model remains the same. This is a statement about symmetry, not a claim that every observed value is identical.
A common practical representation is to assume that observations are independent and identically distributed conditional on a shared unknown parameter. For example, each Bernoulli trial has the same success probability p once p is known. But when p itself is uncertain, learning from one observation changes our distribution for p, and that affects what we predict for another observation. Thus conditional independence need not imply marginal independence. This idea becomes central in hierarchical modelling, where observations within a group share group-level parameters and groups share population-level hyperparameters. **Board equations from the notes.**

$$Y_i \mid \theta \overset{\text{iid}}{\sim} p(y \mid \theta), \qquad \theta \sim p(\theta).$$

Detailed solution and reasoning.
For an ordered sequence $(y_1, \ldots, y_n)$ with $s = \sum_i y_i$ successes and $n - s$ failures,

$$\Pr(y_1, \ldots, y_n) = \int_0^1 p^s (1-p)^{n-s} \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)} dp = \frac{B(a+s, b+n-s)}{B(a,b)}.$$

The result depends on the order only through s, so every permutation has the same probability: the sequence is exchangeable.
For distinct $i \neq j$, $E(Y_i) = a/(a+b)$ and $E(Y_iY_j) = E(P^2)$. Therefore

$$\text{Cov}(Y_i, Y_j) = \text{Var}(P) = \frac{ab}{(a+b)^2(a+b+1)}.$$

Unless the prior is degenerate, this is positive, so the observations are not marginally independent. Exchangeability means invariance under permutations; it does not mean factorisation of the joint distribution.

Problem 14

Original question (verbatim).
A hospital selects 3 patient files uniformly without replacement from a finite collection containing exactly 8 positive and 12 negative cases. Let Y_i indicate whether the i-th selected file is positive. Show that the sequence is exchangeable but that the indicators are dependent. Compare this mechanism with conditionally iid Bernoulli sampling given an unknown probability. Explain why exchangeability is weaker than conditional iid sampling under a parameter model.

Required theory (verbatim from the supplied lecture manual).
Exchangeability means that the joint distribution of the observations does not change when we relabel their order. If I permute the observation labels, the probability model remains the same. This is a statement about symmetry, not a claim that every observed value is identical.
A common practical representation is to assume that observations are independent and identically distributed conditional on a shared unknown parameter. For example, each Bernoulli trial has the same success probability p once p is known. But when p itself is uncertain, learning from one observation changes our distribution for p, and that affects what we predict for another observation. Thus conditional independence need not imply marginal independence. This idea becomes central in hierarchical modelling, where observations within a group share group-level parameters and groups share population-level hyperparameters. **Board equations from the notes.**

$$Y_i \mid \theta \overset{\text{iid}}{\sim} p(y \mid \theta), \qquad \theta \sim p(\theta).$$

Detailed solution and reasoning.
Let $N = 20$, $K = 8$ positives and draw $m = 3$ files without replacement. Any ordered sequence with the same number of positives has the same probability, because the probability depends only on its positive/negative composition. Thus the indicators are exchangeable.
For two draws,

$$\Pr(Y_1 = 1) = \frac{8}{20} = 0.4, \qquad \Pr(Y_1 = 1, Y_2 = 1) = \frac{8}{20}\frac{7}{19} = \frac{14}{95}.$$

Hence

$$\operatorname{Cov}(Y_1, Y_2) = \frac{14}{95} - (0.4)^2 = -\frac{0.4(0.6)}{19} = -0.012632.$$

The negative covariance comes from depletion: after drawing a positive, only seven positives remain among nineteen files.
By contrast, conditionally iid Bernoulli sampling given a shared unknown probability gives positive marginal dependence from uncertainty in that probability. Exchangeability alone only requires permutation invariance; it does not guarantee a representation as conditionally iid Bernoulli observations under a particular parameter model.

Problem 15

Original question (verbatim).
Three sensors monitor the same machine. Their outputs are independent conditional on an unobserved machine state Z, but the state is uncertain and has two possible values. Specify a joint model and derive the marginal joint probability of two sensor alarms. Give conditions under which the alarms become positively associated after integrating out Z. Explain how an analyst who assumes marginal independence could overstate the strength of combined evidence.

Required theory (verbatim from the supplied lecture manual).
Exchangeability means that the joint distribution of the observations does not change when we relabel their order. If I permute the observation labels, the probability model remains the same. This is a statement about symmetry, not a claim that every observed value is identical.
A common practical representation is to assume that observations are independent and identically distributed conditional on a shared unknown parameter. For example, each Bernoulli trial has the same success probability p once p is known. But when p itself is uncertain, learning from one observation changes our distribution for p, and that affects what we predict for another observation. Thus conditional independence need not imply marginal independence. This idea becomes central in hierarchical modelling, where observations within a group share group-level parameters and groups share population-level hyperparameters. **Board equations from the notes.**

$$Y_i \mid \theta \overset{\text{iid}}{\sim} p(y \mid \theta), \qquad \theta \sim p(\theta).$$

Detailed solution and reasoning.
Let $Z \in \{0, 1\}$ denote machine state, with $\Pr(Z = 1) = \pi$. Let A_i denote alarm from sensor i and set $q_i(z) = \Pr(A_i = 1 \mid Z = z)$. Conditional independence gives

$$\Pr(A_1 = 1, A_2 = 1) = \pi q_1(1)q_2(1) + (1 - \pi)q_1(0)q_2(0).$$

The marginal alarm probabilities are $m_i = \pi q_i(1) + (1 - \pi)q_i(0)$. Subtracting their product yields

$$\mathrm{Cov}(1_{A_1}, 1_{A_2}) = \pi(1 - \pi)[q_1(1) - q_1(0)][q_2(1) - q_2(0)].$$

Thus alarms are positively associated if both sensors become more likely to alarm in the same state (or both become less likely in the same state). If the state shifts their alarm probabilities in opposite directions, association can be negative. Treating positively associated alarms as marginally independent counts the shared state evidence twice and can exaggerate the combined likelihood ratio.

1.6 Worked examples: understanding Bayes' rule

Problem 16

Original question (verbatim).
A rare condition affects 0.5% of a population. A first test has sensitivity 0.9 and specificity 0.97. A confirmatory test has sensitivity 0.95 and specificity 0.99, and the tests are conditionally independent given the true condition status. Derive the probability of disease after (i) a positive first test and (ii) two positive tests. Calculate the probability of two positive results under each true status, and explain why the second positive result does not simply multiply the first posterior probability by 0.95.

Required theory (verbatim from the supplied lecture manual).
I will use the diagnostic-test example to show why we must write the conditioning direction carefully. We are given the prevalence, the sensitivity, and the false-positive rate. The question asks for disease probability after a positive test. So the denominator must be the probability of a positive test from either route: a diseased person testing positive, or a non-diseased person testing positive. The numerator is only the first route. This gives the posterior probability. It is not equal to sensitivity because sensitivity conditions on disease, while the question conditions on the test

result.
In the odds example, the arithmetic is even shorter. First convert the prior probabilities into prior odds. Then multiply by the likelihood ratio. Finally convert the posterior odds into a probability. I want students to learn the order, not just the numerical answer. If we keep the order straight, the calculation becomes almost mechanical. **Board equations from the notes.**

$$\Pr(D,+) = 0.02(0.95) = 0.019, \qquad \Pr(D^c,+) = 0.98(0.10) = 0.098.$$
$$\Pr(+) = 0.019 + 0.098 = 0.117.$$

Detailed solution and reasoning.
Let D be disease, with $\Pr(D) = 0.005$. For the first test, sensitivity is 0.9 and false-positive probability is 0.03. Thus

$$\Pr(D \mid T_1+) = \frac{0.005(0.9)}{0.005(0.9) + 0.995(0.03)} = 0.131004.$$

For two positive results, conditional independence gives

$$\Pr(T_1+, T_2+ \mid D) = 0.9(0.95) = 0.855,$$

$$\Pr(T_1+, T_2+ \mid D^c) = 0.03(0.01) = 0.0003.$$

Therefore

$$\Pr(D \mid T_1+, T_2+) = \frac{0.005(0.855)}{0.005(0.855) + 0.995(0.0003)} = 0.934733.$$

The second positive changes odds by its likelihood ratio $0.95/0.01 = 95$ conditional on disease status and the first result. It does not multiply the first posterior probability by 0.95: 0.95 is $\Pr(T_2+ \mid D)$, not $\Pr(D \mid T_2+)$. The relevant false-positive probability and updated base rate must also enter.

Problem 17

Original question (verbatim).
A genetic marker can arise from one of three mutually exclusive genetic states G_1, G_2, G_3, with prior probabilities $(0.2, 0.5, 0.3)$. A laboratory observation E has probabilities $(0.8, 0.2, 0.1)$ under the three states. Derive the posterior distribution over the states and the posterior odds for every pair. Now suppose the observation is reported only as E or as a coarsened event $E \cup F$. Explain how the likelihoods and posterior conclusions change under the coarsening.

Required theory (verbatim from the supplied lecture manual).
I will use the diagnostic-test example to show why we must write the conditioning direction carefully. We are given the prevalence, the sensitivity, and the false-positive rate. The question asks for disease probability after a positive test. So the denominator must be the probability of a positive test from either route: a diseased person testing positive, or a non-diseased person testing positive. The numerator is only the first route. This gives the posterior probability. It is not equal to sensitivity because sensitivity conditions on disease, while the question conditions on the test result.
In the odds example, the arithmetic is even shorter. First convert the prior probabilities into prior odds. Then multiply by the likelihood ratio. Finally convert the posterior odds into a probability. I want students to learn the order, not just the numerical answer. If we keep the order straight,

the calculation becomes almost mechanical. **Board equations from the notes.**

$$\Pr(D,+) = 0.02(0.95) = 0.019, \qquad \Pr(D^c,+) = 0.98(0.10) = 0.098.$$
$$\Pr(+) = 0.019 + 0.098 = 0.117.$$

Detailed solution and reasoning.
Bayes' rule gives unnormalised posterior weights

$$(0.2)(0.8) = 0.16, \quad (0.5)(0.2) = 0.10, \quad (0.3)(0.1) = 0.03.$$

Their sum is 0.29, so

$$\Pr(G_1 \mid E) = 0.16/0.29 = 0.551724, \quad \Pr(G_2 \mid E) = 0.344828, \quad \Pr(G_3 \mid E) = 0.103448.$$

Posterior odds are the ratios of these probabilities: $G_1 : G_2 = 1.6 : 1$, $G_1 : G_3 = 5.333 : 1$, and $G_2 : G_3 = 3.333 : 1$.
If the observed report is the coarsened event $E \cup F$, replace each likelihood by

$$\Pr(E \cup F \mid G_i) = \Pr(E \mid G_i) + \Pr(F \mid G_i) - \Pr(E \cap F \mid G_i).$$

The problem supplies no probabilities for F or the overlap, so a numerical coarsened posterior cannot be calculated. Coarsening generally changes the likelihood ratios and therefore the posterior; it loses the distinction between outcomes E and F.

Problem 18

Original question (verbatim).
A classifier is evaluated under two possible population models. Under model M_0, a word is correctly classified with probability 0.85; under model M_1, that probability is 0.95. The prior probability of M_1 is 0.25. In 10 independent classifications, 9 are correct. Calculate the posterior model probability and compare it with the maximum-likelihood choice of model. Explain what the Bayesian calculation adds and why the result is not determined by the larger likelihood alone.

Required theory (verbatim from the supplied lecture manual).
I will use the diagnostic-test example to show why we must write the conditioning direction carefully. We are given the prevalence, the sensitivity, and the false-positive rate. The question asks for disease probability after a positive test. So the denominator must be the probability of a positive test from either route: a diseased person testing positive, or a non-diseased person testing positive. The numerator is only the first route. This gives the posterior probability. It is not equal to sensitivity because sensitivity conditions on disease, while the question conditions on the test result.
In the odds example, the arithmetic is even shorter. First convert the prior probabilities into prior odds. Then multiply by the likelihood ratio. Finally convert the posterior odds into a probability. I want students to learn the order, not just the numerical answer. If we keep the order straight, the calculation becomes almost mechanical. **Board equations from the notes.**

$$\Pr(D,+) = 0.02(0.95) = 0.019, \qquad \Pr(D^c,+) = 0.98(0.10) = 0.098.$$
$$\Pr(+) = 0.019 + 0.098 = 0.117.$$

Detailed solution and reasoning.
The binomial coefficients cancel in the model likelihood ratio. For nine correct and one incorrect classification,

$$BF_{10} = \frac{0.95^9(0.05)}{0.85^9(0.15)} = \left(\frac{0.95}{0.85}\right)^9 \frac{0.05}{0.15} \approx 0.907028.$$

With prior model probabilities $\Pr(M_1) = 0.25$ and $\Pr(M_0) = 0.75$, prior odds are $1/3$. Thus

$$O_{10|y} = \frac{1}{3}(0.907028) = 0.302343, \quad \Pr(M_1 \mid y) = \frac{0.302343}{1 + 0.302343} = 0.232153.$$

The maximum-likelihood choice is M_0, since its likelihood is slightly larger for this observed 9/10 outcome. The Bayesian result adds prior model odds and averages the alternative evidence as specified by the model; it does not select a model solely because its success probability is numerically larger. Here the data are slightly closer to 0.85 than 0.95 in likelihood terms, and the prior also favours M_0.

1.7 Common conceptual distinctions

Problem 19

Original question (verbatim).
An analyst makes the following claims: (i) the likelihood is a probability distribution over the unknown parameter; (ii) a 95% posterior predictive interval means the parameter lies in it with probability 0.95; (iii) a proper posterior proves that the model fits; and (iv) conditional independence guarantees marginal independence. For each claim, determine whether it is correct, give a counterexample where it is not, and replace it with a mathematically accurate statement. Your answer must distinguish prior, likelihood, posterior, parameter, and future observation.

Required theory (verbatim from the supplied lecture manual).
Before moving on, let us separate five ideas that students often blend together. A prior is not necessarily an arbitrary guess; it may represent earlier evidence, a scientific constraint, or a population distribution. A likelihood is not a probability distribution over parameter values. A posterior is the updated distribution after conditioning on the observed data. A credible interval describes posterior probability for a parameter, whereas a predictive interval describes uncertainty about a future observation. A predictive interval is generally wider because it includes both uncertainty about the parameter and new observation-level randomness.
Finally, a proper posterior does not automatically mean a model fits well. Posterior calculation and model checking answer different questions. We can have a mathematically valid distribution under assumptions that are nevertheless inconsistent with the data. Keep this distinction in mind whenever you interpret a result.

Detailed solution and reasoning.
(i) False. The likelihood $L(\theta; y) = p(y \mid \theta)$ is a function of θ with the data fixed. It is not generally normalised over θ. A prior is a probability distribution over θ; multiplying prior by likelihood and normalising gives the posterior.
(ii) False. A 95% posterior predictive interval concerns a future observation $\widetilde{Y}$, so its statement is $\Pr(\widetilde{Y} \in I \mid y) = 0.95$. A parameter credible interval C instead satisfies $\Pr(\theta \in C \mid y) = 0.95$. Prediction includes observation noise as well as parameter uncertainty.
(iii) False. Posterior propriety only says the posterior integrates to one. A proper posterior can

come from a likelihood that systematically misses overdispersion, clustering, or tail behaviour. Posterior predictive checks assess adequacy.
(iv) False in general. Conditional independence given a shared random parameter does not imply marginal independence. For example, if $Y_1, Y_2 \mid P$ are iid Bernoulli(P), then $\mathrm{Cov}(Y_1, Y_2) = \mathrm{Var}(P) > 0$ after integrating out P.
The correct distinctions are: prior $p(\theta)$; likelihood $p(y \mid \theta)$; posterior $p(\theta \mid y)$; parameter θ; future observation $\tilde{y}$ with predictive density $\int p(\tilde{y} \mid \theta)p(\theta \mid y)d\theta$.

Problem 20

Original question (verbatim).
A posterior distribution for a defect probability is sharply concentrated, but posterior predictive replications almost never reproduce the number of extreme defect clusters observed in the actual factory. Explain how a posterior can be concentrated and proper while the model remains inadequate. Design a posterior predictive discrepancy statistic for clustering and explain what conclusions can and cannot be drawn from the observed discrepancy.

Required theory (verbatim from the supplied lecture manual).
Before moving on, let us separate five ideas that students often blend together. A prior is not necessarily an arbitrary guess; it may represent earlier evidence, a scientific constraint, or a population distribution. A likelihood is not a probability distribution over parameter values. A posterior is the updated distribution after conditioning on the observed data. A credible interval describes posterior probability for a parameter, whereas a predictive interval describes uncertainty about a future observation. A predictive interval is generally wider because it includes both uncertainty about the parameter and new observation-level randomness.
Finally, a proper posterior does not automatically mean a model fits well. Posterior calculation and model checking answer different questions. We can have a mathematically valid distribution under assumptions that are nevertheless inconsistent with the data. Keep this distinction in mind whenever you interpret a result.

Detailed solution and reasoning.
A concentrated posterior for the defect probability describes uncertainty about a single parameter under the assumed likelihood. It does not guarantee that the likelihood reproduces the factory's dependence structure. If the model is iid Bernoulli, it may fit the marginal defect rate but fail to reproduce clusters caused by machine, shift, lot, or time effects.
Define, for example, $T(y)$ as the number of adjacent defect pairs, the maximum run length of defects, or the number of batches with defect fraction above a chosen threshold. Draw posterior parameters $p^{(m)}$ and replicated factory sequences $y^{rep,(m)} \sim p(y^{rep} \mid p^{(m)})$. Compare $T(y)$ with $T(y^{rep,(m)})$. If the observed clustering statistic is consistently beyond the replicated range, the iid assumption is inadequate.
This discrepancy supports the claim that the current model cannot reproduce the observed clustering; it does not by itself identify the unique cause or prove a specific alternative. A revised model could include batch-specific probabilities, a beta-binomial/random-effects structure, a Markov dependence model, or measured covariates. The posterior can be precise and proper while its predictive mechanism is wrong.

Problem 21

Original question (verbatim).
A report gives a narrow 95% credible interval for a population mean and a much wider 95% predictive interval for an individual future measurement. A manager says that the two intervals are inconsistent because they concern the same population. Explain the distinction by decomposing predictive uncertainty into parameter uncertainty and observation-level randomness. Construct a normal example in which the credible interval becomes arbitrarily narrow with increasing sample size while the predictive interval retains a nonzero width.

Required theory (verbatim from the supplied lecture manual).
Before moving on, let us separate five ideas that students often blend together. A prior is not necessarily an arbitrary guess; it may represent earlier evidence, a scientific constraint, or a population distribution. A likelihood is not a probability distribution over parameter values. A posterior is the updated distribution after conditioning on the observed data. A credible interval describes posterior probability for a parameter, whereas a predictive interval describes uncertainty about a future observation. A predictive interval is generally wider because it includes both uncertainty about the parameter and new observation-level randomness.
Finally, a proper posterior does not automatically mean a model fits well. Posterior calculation and model checking answer different questions. We can have a mathematically valid distribution under assumptions that are nevertheless inconsistent with the data. Keep this distinction in mind whenever you interpret a result.

Detailed solution and reasoning.
A credible interval for μ describes uncertainty about the population mean. A predictive interval for one future Y_{n+1} describes both uncertainty about μ and individual measurement noise. Under $Y_i \mid \mu \sim N(\mu, \sigma^2)$ and a normal prior, if $\mu \mid y \sim N(\mu_n, \tau_n^2)$, then

$$\mu \mid y: \quad \operatorname{Var}(\mu \mid y) = \tau_n^2, \qquad Y_{n+1} \mid y: \quad \operatorname{Var}(Y_{n+1} \mid y) = \sigma^2 + \tau_n^2.$$

For a numerical design, take known $\sigma^2 = 9$ and a regular prior with fixed variance. As $n \to \infty$, $\tau_n^2 = (\tau_0^{-2} + n/9)^{-1} \to 0$, so the credible interval for μ shrinks to zero width. The predictive variance tends to 9, so an individual prediction interval retains standard deviation 3. There is no inconsistency: the mean becomes known while a new individual observation remains noisy.

Chapter 2

Exponential families and conjugate prior distributions

2.1 Why exponential families are useful

Problem 22

Original question (verbatim).
Write the Bernoulli, Poisson, and normal-with-known-variance likelihoods in the canonical exponential-family form $h(y)\exp\{\eta T(y) - A(\eta)\}$. For each model, identify the natural parameter, sufficient statistic, log-partition function, and base measure. Then show how the joint likelihood of an iid sample reduces to a function of a low-dimensional summary. Explain why this representation is useful for conjugate updating, without claiming that membership in an exponential family makes every prior scientifically appropriate.

Required theory (verbatim from the supplied lecture manual).
The reason we study exponential families is that they reveal a common algebraic structure behind many familiar distributions. Rather than treating Bernoulli, Poisson, exponential, and normal models as unrelated formulas, we write them in a shared form involving a natural parameter and a sufficient statistic. For independent observations, the data enter through a sum of those sufficient statistics. That is why a complicated-looking dataset can sometimes be reduced to a small number of summary quantities without losing information about the model parameter.
Conjugacy takes advantage of this structure. We choose a prior whose algebraic kernel has the same form as the likelihood kernel. After multiplication, the posterior is in the same family as the prior, with updated hyperparameters. This gives us a closed-form answer and also makes the role of prior information visible. But remember: conjugacy is a mathematical convenience. It does not prove that the model or prior is scientifically appropriate. **Board equations from the notes.**

$$p(y \mid \eta) = h(y)\exp\{\eta T(y) - A(\eta)\}.$$

$$p(y_1, \ldots, y_n \mid \eta) = \left\{\prod_{i=1}^{n} h(y_i)\right\} \exp\left\{\eta \sum_{i=1}^{n} T(y_i) - nA(\eta)\right\}.$$

Detailed solution and reasoning.
For a single Bernoulli observation $y \in \{0, 1\}$,

$$p(y \mid p) = p^y(1-p)^{1-y} = \exp\{y \log[p/(1-p)] + \log(1-p)\}.$$

Thus $\eta = \log\{p/(1-p)\}$, $T(y) = y$, $A(\eta) = \log(1+e^{\eta})$, and $h(y) = 1$. For a Poisson observation,

$$p(y \mid \lambda) = \frac{e^{-\lambda}\lambda^{y}}{y!} = \frac{1}{y!}\exp\{y\log\lambda - \lambda\},$$

so $\eta = \log\lambda$, $T(y) = y$, $A(\eta) = e^{\eta}$, and $h(y) = 1/y!$.
For $Y \sim N(\mu, \sigma^2)$ with known variance,

$$p(y \mid \mu) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{\frac{y\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} - \frac{y^2}{2\sigma^2}\right\}.$$

Here $\eta = \mu/\sigma^2$, $T(y) = y$, $A(\eta) = \sigma^2\eta^2/2$, and $h(y) = (2\pi\sigma^2)^{-1/2}e^{-y^2/(2\sigma^2)}$.
For iid samples, the natural statistic becomes $\sum_i T(y_i)$, so the likelihood depends on a low-dimensional summary (success count, total count, or sample sum). This factorisation exposes sufficient statistics and makes it easier to select a conjugate prior. Exponential-family membership is an algebraic convenience, not a guarantee that a prior is scientifically suitable.

Problem 23

Original question (verbatim).
Two independent datasets are observed under the same Poisson-rate model. Dataset A contains 10 counts with total 24, and Dataset B contains 40 counts with total 24; exposures differ but have the same total in both datasets. Determine whether the two datasets produce the same likelihood kernel for the rate. Extend the argument to a Gamma prior and identify exactly which summaries must match for the posterior distributions to be identical. Explain what information is discarded by those summaries and why it is irrelevant to this particular parameter under this model.

Required theory (verbatim from the supplied lecture manual).
The reason we study exponential families is that they reveal a common algebraic structure behind many familiar distributions. Rather than treating Bernoulli, Poisson, exponential, and normal models as unrelated formulas, we write them in a shared form involving a natural parameter and a sufficient statistic. For independent observations, the data enter through a sum of those sufficient statistics. That is why a complicated-looking dataset can sometimes be reduced to a small number of summary quantities without losing information about the model parameter.
Conjugacy takes advantage of this structure. We choose a prior whose algebraic kernel has the same form as the likelihood kernel. After multiplication, the posterior is in the same family as the prior, with updated hyperparameters. This gives us a closed-form answer and also makes the role of prior information visible. But remember: conjugacy is a mathematical convenience. It does not prove that the model or prior is scientifically appropriate. **Board equations from the notes.**

$$p(y \mid \eta) = h(y)\exp\{\eta T(y) - A(\eta)\}.$$

$$p(y_1, \ldots, y_n \mid \eta) = \left\{\prod_{i=1}^{n} h(y_i)\right\}\exp\left\{\eta\sum_{i=1}^{n} T(y_i) - nA(\eta)\right\}.$$

Detailed solution and reasoning.
For $Y_i \mid \lambda \sim \text{Poisson}(x_i\lambda)$, the likelihood kernel is

$$L(\lambda) \propto \lambda^{\sum_i y_i}\exp\{-\lambda\sum_i x_i\}.$$

Both datasets have $\sum y_i = 24$ and, by the question, the same total exposure $\sum x_i$. Their likelihood kernels are therefore identical for λ, even though one has 10 counts and the other 40 counts. The number of observations alone is not the sufficient summary for the common rate when exposures are known and may differ.
With prior $\lambda \sim \text{Gamma}(a, b)$, the posterior is

$$\lambda \mid y \sim \text{Gamma}\left(a + 24, b + \sum_i x_i\right).$$

Therefore same total count and same total exposure imply the same posterior. The discarded information is the allocation of counts among observation units and the number/order of rows; under this common-rate independent Poisson model it contributes only factors independent of λ. If rates vary by unit, counts are dependent, or covariates/time patterns are introduced, the allocation may matter.

Problem 24

Original question (verbatim).
Consider a normal sample with known variance and unknown mean. Starting from the joint density, isolate the part involving the unknown mean and identify a sufficient statistic. Now suppose one observation is replaced by two values whose sum is unchanged but whose individual values are very different. Determine whether the likelihood for the mean changes. Explain why sufficiency is model- and parameter-specific, and identify which aspects of the data would matter if the variance were also unknown.

Required theory (verbatim from the supplied lecture manual).
The reason we study exponential families is that they reveal a common algebraic structure behind many familiar distributions. Rather than treating Bernoulli, Poisson, exponential, and normal models as unrelated formulas, we write them in a shared form involving a natural parameter and a sufficient statistic. For independent observations, the data enter through a sum of those sufficient statistics. That is why a complicated-looking dataset can sometimes be reduced to a small number of summary quantities without losing information about the model parameter.
Conjugacy takes advantage of this structure. We choose a prior whose algebraic kernel has the same form as the likelihood kernel. After multiplication, the posterior is in the same family as the prior, with updated hyperparameters. This gives us a closed-form answer and also makes the role of prior information visible. But remember: conjugacy is a mathematical convenience. It does not prove that the model or prior is scientifically appropriate. **Board equations from the notes.**

$$p(y \mid \eta) = h(y) \exp\{\eta T(y) - A(\eta)\}.$$

$$p(y_1, \ldots, y_n \mid \eta) = \left\{\prod_{i=1}^{n} h(y_i)\right\} \exp\left\{\eta \sum_{i=1}^{n} T(y_i) - nA(\eta)\right\}.$$

Detailed solution and reasoning.
The joint density is

$$p(y_1, \ldots, y_n \mid \mu) \propto \exp\left[-\frac{1}{2\sigma^2} \sum_i (y_i - \mu)^2\right].$$

Use the identity

$$\sum_i (y_i - \mu)^2 = \sum_i (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2.$$

The first term does not involve μ, so for a fixed sample size and known σ^2, the likelihood for μ depends on the sample only through $\bar{y}$ (equivalently $\sum_i y_i$). Thus $\bar{Y}$ is sufficient for μ.
Interpret the wording carefully. If it means that two entries in a fixed-size sample are replaced by two different values with the same pairwise sum, then n and $\sum_i y_i$ are unchanged, so the likelihood for μ is unchanged. If it literally means one observation is replaced by two new observations, the sample size changes from n to $n + 1$; even if the total sum is preserved, the sufficient statistic includes the changed n and the likelihood does change. If σ^2 is also unknown, the likelihood depends on $\bar{y}$ and $S = \sum_i (y_i - \bar{y})^2$; a change in within-sample spread affects inference about variance.

2.2 Beta-Bernoulli and Beta-Binomial models

Problem 25

Original question (verbatim).
A reliability engineer assigns $P \sim \text{Beta}(2, 5)$ to the probability of successful operation. The first 20 trials give 14 successes. Derive the posterior distribution, posterior mean, and probability that the next 5 trials contain at least 4 successes. Do not treat the future trials as having a fixed success probability equal to the posterior mean: derive the predictive probability by integrating over P, and explain the extra variability created by uncertainty about P.

Required theory (verbatim from the supplied lecture manual).
Here the unknown quantity is a success probability p, so it must lie between zero and one. The Bernoulli likelihood contributes one factor of p for every success and one factor of one-minus-p for every failure. If we have s successes and f failures, the likelihood kernel is p to the power s times one-minus-p to the power f.
The Beta prior has exactly the same two factors, with exponents a-minus-one and b-minus-one. When we multiply prior and likelihood, we add the exponents. That is why the posterior is Beta with parameters a plus s and b plus f. The posterior mean is the updated success-like count divided by the total updated count. The predictive probability of success on the next trial equals that posterior mean because we average the conditional success probability p over the posterior distribution. This is the simplest example of conjugate updating and should become automatic.
Board equations from the notes.

$$p(y \mid p) \propto p^s (1-p)^f.$$

$$p \sim \text{Beta}(a, b), \qquad \pi(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a, b)}, \quad 0 < p < 1.$$

Detailed solution and reasoning.
After 14 successes and 6 failures, Beta conjugacy gives

$$P \mid y \sim \text{Beta}(2 + 14, 5 + 6) = \text{Beta}(16, 11), \qquad E(P \mid y) = \frac{16}{27} = 0.592593.$$

Let K be the number of successes in the next five trials. Integrating the binomial probability over

the posterior,

$$\Pr(K = k \mid y) = \binom{5}{k} \frac{B(16+k, 11+5-k)}{B(16,11)}.$$

Hence

$$\Pr(K \geq 4 \mid y) = \sum_{k=4}^{5} \binom{5}{k} \frac{B(16+k, 16-k)}{B(16,11)} \approx 0.342179.$$

The predictive law is Beta–Binomial, not Binomial$(5, 16/27)$. The latter conditions on a fixed plug-in value and ignores posterior variation in P, which creates extra predictive dispersion.

Problem 26

Original question (verbatim).
A prior is to be chosen for a Bernoulli success probability. The expert states that the prior mean must be 0.3 and the prior effective sample size $a + b$ must be 12. Determine the Beta hyperparameters. After observing 2 successes in 5 trials, derive the posterior and compare the posterior mean with the sample proportion. Then discuss whether the pseudo-count interpretation means the prior is literally equivalent to 12 additional observed trials in every inferential respect.

Required theory (verbatim from the supplied lecture manual).
Here the unknown quantity is a success probability p, so it must lie between zero and one. The Bernoulli likelihood contributes one factor of p for every success and one factor of one-minus-p for every failure. If we have s successes and f failures, the likelihood kernel is p to the power s times one-minus-p to the power f.
The Beta prior has exactly the same two factors, with exponents a-minus-one and b-minus-one. When we multiply prior and likelihood, we add the exponents. That is why the posterior is Beta with parameters a plus s and b plus f. The posterior mean is the updated success-like count divided by the total updated count. The predictive probability of success on the next trial equals that posterior mean because we average the conditional success probability p over the posterior distribution. This is the simplest example of conjugate updating and should become automatic.
Board equations from the notes.

$$p(y \mid p) \propto p^s (1-p)^f.$$

$$p \sim \text{Beta}(a,b), \qquad \pi(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}, \quad 0 < p < 1.$$

Detailed solution and reasoning.
For $P \sim \text{Beta}(a,b)$, the prior mean is $a/(a+b)$ and the effective sample size is $a+b$. The constraints give

$$a + b = 12, \qquad a/12 = 0.3 \quad \Rightarrow \quad a = 3.6,\ b = 8.4.$$

After 2 successes and 3 failures,

$$P \mid y \sim \text{Beta}(5.6, 11.4), \qquad E(P \mid y) = \frac{5.6}{17} = 0.329412.$$

The sample proportion is $2/5 = 0.4$. The posterior mean lies between the prior mean 0.3 and the sample proportion 0.4, weighted by prior and data concentration.
The pseudo-count interpretation is an algebraic analogy: a and b act like prior success/failure counts in the update. Here they are fractional. The prior is not literally 12 observed Bernoulli

trials for every purpose: it is a distribution over p, can be elicited from substantive information, and produces predictive and inferential implications that need not match a particular historical sample or sampling design.

Problem 27

Original question (verbatim).
Two laboratories observe the same number of successes but in different batches and in different orders. One analyst updates a Beta prior after each observation; another uses the final success and failure totals in one batch. Prove that the final posterior is the same under the iid Bernoulli model. Then derive the predictive probability of the next two outcomes in the two possible orders (success-failure and failure-success), and explain how exchangeability is reflected in the result.

Required theory (verbatim from the supplied lecture manual).
Here the unknown quantity is a success probability p, so it must lie between zero and one. The Bernoulli likelihood contributes one factor of p for every success and one factor of one-minus-p for every failure. If we have s successes and f failures, the likelihood kernel is p to the power s times one-minus-p to the power f.
The Beta prior has exactly the same two factors, with exponents a-minus-one and b-minus-one. When we multiply prior and likelihood, we add the exponents. That is why the posterior is Beta with parameters a plus s and b plus f. The posterior mean is the updated success-like count divided by the total updated count. The predictive probability of success on the next trial equals that posterior mean because we average the conditional success probability p over the posterior distribution. This is the simplest example of conjugate updating and should become automatic.
Board equations from the notes.

$$p(y \mid p) \propto p^s (1-p)^f.$$

$$p \sim \mathrm{Beta}(a,b), \qquad \pi(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}, \quad 0 < p < 1.$$

Detailed solution and reasoning.
If the first n observations contain s successes, repeated sequential updates give

$$\mathrm{Beta}(a,b) \to \mathrm{Beta}(a+y_1, b+1-y_1) \to \cdots \to \mathrm{Beta}(a+s, b+n-s).$$

Multiplication is commutative, so the final posterior depends only on total successes and failures, not the order or batching.
Let the current posterior be $\mathrm{Beta}(A,B)$. The predictive probability of success then failure is

$$\Pr(S,F \mid y) = \frac{A}{A+B}\frac{B}{A+B+1} = \frac{AB}{(A+B)(A+B+1)}.$$

For failure then success,

$$\Pr(F,S \mid y) = \frac{B}{A+B}\frac{A}{A+B+1} = \frac{AB}{(A+B)(A+B+1)}.$$

They are equal, demonstrating exchangeability of the next pair. Conditional probabilities change after the first outcome, but the joint probability of the two orders is equal.

2.3 Gamma-Poisson model, including exposure

Problem 28

Original question (verbatim).
A clinic records counts $Y_i \mid \lambda \sim \text{Poisson}(x_i\lambda)$ with exposures $x = (0.5, 1, 2, 4)$, counts $y = (0, 2, 3, 7)$, and prior $\lambda \sim \text{Gamma}(3, 2)$ in the shape-rate convention. Derive the posterior and calculate its mean and variance. Then derive the posterior predictive distribution for a new observation with exposure 1.5 and calculate its probability of zero events. Explain why using the number of clinics rather than total exposure in the posterior rate is incorrect.

Required theory (verbatim from the supplied lecture manual).
The Poisson model is natural for counts, but be careful about what the rate means. If an observation has exposure x, its expected count is x times lambda. The exposure may represent time, population size, area, or another amount of opportunity for events. The joint likelihood contains lambda raised to the total observed count, multiplied by an exponential term involving lambda times the total exposure.
A Gamma prior in the shape-rate parameterisation has a power of lambda and an exponential decay term. Multiplying the prior and likelihood adds the total count to the shape and adds the total exposure to the rate. The important exam trap is to update the rate with exposure, not merely with the number of observations. When each exposure equals one, the total exposure is n and we recover the familiar simpler formula. Integrating the Poisson likelihood over a Gamma prior gives a negative-binomial predictive distribution. **Board equations from the notes.**

$$L(\lambda; y) \propto \lambda^{\sum_i y_i} \exp\left(-\lambda \sum_i x_i\right).$$

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + \sum_i y_i,\ b + \sum_i x_i\right).}$$

Detailed solution and reasoning.
The total exposure is $0.5 + 1 + 2 + 4 = 7.5$ and the total count is $0 + 2 + 3 + 7 = 12$. Multiplying the likelihood by the prior $\text{Gamma}(3, 2)$ gives

$$\lambda \mid y \sim \text{Gamma}(3 + 12, 2 + 7.5) = \text{Gamma}(15, 9.5).$$

Thus the posterior mean and variance are

$$E(\lambda \mid y) = 15/9.5 = 1.578947, \qquad \text{Var}(\lambda \mid y) = 15/9.5^2 = 0.166205.$$

For exposure $x_* = 1.5$, the predictive count is negative binomial:

$$\Pr(Y_* = k \mid y) = \frac{\Gamma(15 + k)}{\Gamma(15)k!} \left(\frac{9.5}{11}\right)^{15} \left(\frac{1.5}{11}\right)^{k}.$$

In particular,

$$\Pr(Y_* = 0 \mid y) = \left(\frac{9.5}{11}\right)^{15} = 0.110908.$$

The posterior rate adds total exposure, not the number of clinics: the exponential likelihood factor is $e^{-\lambda \sum_i x_i}$. Using 4 instead of 7.5 would misstate the amount of time at risk.

Problem 29

Original question (verbatim).
Two regions have observed counts 12 and 18 over exposures 3 and 9, respectively. Under a common rate model with a Gamma prior, derive the posterior. Then consider a model with separate regional rates and independent Gamma priors. Determine which information is lost by imposing a common rate, and describe a posterior predictive comparison that could reveal whether the common-rate assumption is unreasonable. Your answer must distinguish evidence about the common rate from evidence about the adequacy of the common-rate model.

Required theory (verbatim from the supplied lecture manual).
The Poisson model is natural for counts, but be careful about what the rate means. If an observation has exposure x, its expected count is x times lambda. The exposure may represent time, population size, area, or another amount of opportunity for events. The joint likelihood contains lambda raised to the total observed count, multiplied by an exponential term involving lambda times the total exposure.
A Gamma prior in the shape-rate parameterisation has a power of lambda and an exponential decay term. Multiplying the prior and likelihood adds the total count to the shape and adds the total exposure to the rate. The important exam trap is to update the rate with exposure, not merely with the number of observations. When each exposure equals one, the total exposure is n and we recover the familiar simpler formula. Integrating the Poisson likelihood over a Gamma prior gives a negative-binomial predictive distribution. **Board equations from the notes.**

$$L(\lambda; y) \propto \lambda^{\sum_i y_i} \exp\left(-\lambda \sum_i x_i\right).$$

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + \sum_i y_i,\ b + \sum_i x_i\right).}$$

Detailed solution and reasoning.
Under a common rate and prior $\lambda \sim \text{Gamma}(a, b)$, total count is $12 + 18 = 30$ and total exposure is $3 + 9 = 12$. Therefore

$$\lambda \mid y \sim \text{Gamma}(a + 30, b + 12).$$

Under separate regional rates with independent $\lambda_r \sim \text{Gamma}(a_r, b_r)$,

$$\lambda_1 \mid y \sim \text{Gamma}(a_1 + 12, b_1 + 3), \quad \lambda_2 \mid y \sim \text{Gamma}(a_2 + 18, b_2 + 9).$$

The common-rate model discards regional rate differences and pools evidence; it assumes the two regions share the same event rate after exposure adjustment.
A predictive check can simulate replicated regional counts under the common posterior and compare a discrepancy such as $|Y_1/3 - Y_2/9|$, the maximum regional standardized residual, or the ratio of regional rates. A posterior for a common λ answers uncertainty conditional on the common-rate assumption; a check of regional differences addresses whether that assumption is defensible. These are distinct questions.

Problem 30

Original question (verbatim).
A data analyst reports that the posterior predictive number of cases over exposure 10 is Poisson with mean equal to 10 times the posterior mean of λ. Derive the correct posterior predictive distribution under a Gamma-Poisson model and compare its variance with the plug-in Poisson variance. Identify the parameter regime in which the discrepancy is largest, and explain how posterior uncertainty produces overdispersion relative to a fixed-rate Poisson model.

Required theory (verbatim from the supplied lecture manual).
The Poisson model is natural for counts, but be careful about what the rate means. If an observation has exposure x, its expected count is x times lambda. The exposure may represent time, population size, area, or another amount of opportunity for events. The joint likelihood contains lambda raised to the total observed count, multiplied by an exponential term involving lambda times the total exposure.
A Gamma prior in the shape-rate parameterisation has a power of lambda and an exponential decay term. Multiplying the prior and likelihood adds the total count to the shape and adds the total exposure to the rate. The important exam trap is to update the rate with exposure, not merely with the number of observations. When each exposure equals one, the total exposure is n and we recover the familiar simpler formula. Integrating the Poisson likelihood over a Gamma prior gives a negative-binomial predictive distribution. **Board equations from the notes.**

$$L(\lambda; y) \propto \lambda^{\sum_i y_i} \exp\left(-\lambda \sum_i x_i\right).$$

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + \sum_i y_i,\ b + \sum_i x_i\right).}$$

Detailed solution and reasoning.
If $\lambda \mid y \sim \text{Gamma}(A, B)$ in shape–rate form and $Y_* \mid \lambda \sim \text{Poisson}(10\lambda)$, integration yields

$$\Pr(Y_* = k \mid y) = \frac{\Gamma(A+k)}{\Gamma(A)k!}\left(\frac{B}{B+10}\right)^A\left(\frac{10}{B+10}\right)^k.$$

Its mean is $10A/B$ and variance is

$$\text{Var}(Y_* \mid y) = 10A/B + 100A/B^2.$$

A plug-in Poisson with mean $10A/B$ has variance only $10A/B$. The extra term is $10^2 \,\text{Var}(\lambda \mid y)$. The discrepancy is largest when the posterior relative uncertainty in λ is large, for example when prior shape plus observed count is small or exposure/data information is weak. The mixture is overdispersed relative to a fixed-rate Poisson because different posterior draws of λ produce different conditional means.

2.4 Gamma-Exponential model

Problem 31

Original question (verbatim).
A reliability study records exact failure times 1.2, 2.1, and 0.7, together with right-censoring times 1.5 and 3.0 for two units that have not failed. Assume an exponential rate λ and prior Gamma(a, b). Derive the likelihood contribution of each exact and censored observation, obtain the posterior, and explain why the shape increases by the number of observed failures while the rate increases by total time at risk. State what would go wrong if the censored units were incorrectly treated as failures at their censoring times.

Required theory (verbatim from the supplied lecture manual).
The exponential distribution models positive waiting times, and here lambda is a rate. A larger rate means shorter expected waiting times because the mean is one over lambda. For n independent waiting times, the likelihood contains lambda to the power n and an exponential term involving minus lambda times the sum of the waiting times.
The Gamma prior is conjugate, so the posterior shape increases by the number of observed events and the posterior rate increases by the total waiting time. Please distinguish exact event times from right-censored observations. An exact waiting time contributes an event and its duration. A right-censored observation tells us only that the unit survived to a known time; it contributes a survival probability, which increases the total exposure but does not count an observed event. If we transform from the rate lambda to the mean waiting time, which is one over lambda, we must transform the density using the Jacobian. **Board equations from the notes.**

$$p(y \mid \lambda) \propto \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} y_i\right).$$

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + n, b + \sum_{i=1}^{n} y_i\right).}$$

Detailed solution and reasoning.
For an exact event time t, the exponential density contributes $\lambda e^{-\lambda t}$. For a right-censored unit observed up to c, the contribution is the survival probability $\Pr(T > c \mid \lambda) = e^{-\lambda c}$. Thus the three failures and two censored units contribute

$$L(\lambda) \propto \lambda^3 \exp\{-\lambda(1.2 + 2.1 + 0.7 + 1.5 + 3.0)\} = \lambda^3 e^{-8.5\lambda}.$$

Multiplying by Gamma(a, b) gives

$$\lambda \mid \mathit{data} \sim \text{Gamma}(a + 3, b + 8.5).$$

The posterior shape increases by the number of observed failures because each exact failure supplies one factor of λ. The rate increases by total time at risk because every unit contributes survival exposure until failure or censoring.
Treating censored units as failures would incorrectly add two powers of λ, yielding shape $a + 5$, and would claim events occurred when they were not observed. This biases the inferred rate upward and confuses survival information with event information.

Problem 32

Original question (verbatim).
Suppose $\lambda \mid y \sim \text{Gamma}(a_n, b_n)$ in the shape-rate convention, but the scientific quantity of interest is the mean waiting time $\tau = 1/\lambda$. Derive the posterior density of τ using the Jacobian. Determine its mode and state the conditions under which its mean exists. Explain why simply replacing λ by $1/\tau$ in the Gamma density gives an incorrect transformed density.

Required theory (verbatim from the supplied lecture manual).
The exponential distribution models positive waiting times, and here lambda is a rate. A larger rate means shorter expected waiting times because the mean is one over lambda. For n independent waiting times, the likelihood contains lambda to the power n and an exponential term involving minus lambda times the sum of the waiting times.
The Gamma prior is conjugate, so the posterior shape increases by the number of observed events and the posterior rate increases by the total waiting time. Please distinguish exact event times from right-censored observations. An exact waiting time contributes an event and its duration. A right-censored observation tells us only that the unit survived to a known time; it contributes a survival probability, which increases the total exposure but does not count an observed event. If we transform from the rate lambda to the mean waiting time, which is one over lambda, we must transform the density using the Jacobian. **Board equations from the notes.**

$$p(y \mid \lambda) \propto \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} y_i\right).$$

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + n, b + \sum_{i=1}^{n} y_i\right).}$$

Detailed solution and reasoning.
Let $\tau = 1/\lambda$. If $\lambda \sim \text{Gamma}(A, B)$, then

$$f_\lambda(\lambda) = \frac{B^A}{\Gamma(A)} \lambda^{A-1} e^{-B\lambda}, \quad \lambda > 0.$$

The inverse transformation has $|d\lambda/d\tau| = 1/\tau^2$, so

$$f_\tau(\tau) = f_\lambda(1/\tau)\frac{1}{\tau^2} = \frac{B^A}{\Gamma(A)} \tau^{-A-1} e^{-B/\tau}, \quad \tau > 0.$$

This is inverse-Gamma with shape A and kernel $\tau^{-A-1}e^{-B/\tau}$. For $A > 1$, its mode is $B/(A+1)$; for $A > 2$, its mean exists and equals $B/(A-1)$. The mean condition follows by checking the tail integral $\int^\infty \tau f_\tau(\tau) d\tau$.
Simply substituting $1/\tau$ into the Gamma formula omits the Jacobian factor $1/\tau^2$ and therefore does not define the transformed density. A density must transform with both the substituted argument and the derivative of the transformation.

Problem 33

Original question (verbatim).
A survival study has three exact event times and four right-censored observations. Two analysts use the same total observed time, but one records the number of failures incorrectly as seven. Derive the correct posterior and compare it with the incorrect one. Explain which part of the likelihood carries information about the event rate and how the event indicator and total exposure play different roles. Give a general expression for the likelihood with event indicators.

Required theory (verbatim from the supplied lecture manual).
The exponential distribution models positive waiting times, and here lambda is a rate. A larger rate means shorter expected waiting times because the mean is one over lambda. For n independent waiting times, the likelihood contains lambda to the power n and an exponential term involving minus lambda times the sum of the waiting times.
The Gamma prior is conjugate, so the posterior shape increases by the number of observed events and the posterior rate increases by the total waiting time. Please distinguish exact event times from right-censored observations. An exact waiting time contributes an event and its duration. A right-censored observation tells us only that the unit survived to a known time; it contributes a survival probability, which increases the total exposure but does not count an observed event. If we transform from the rate lambda to the mean waiting time, which is one over lambda, we must transform the density using the Jacobian. **Board equations from the notes.**

$$p(y \mid \lambda) \propto \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} y_i\right).$$

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a+n, b+\sum_{i=1}^{n} y_i\right).}$$

Detailed solution and reasoning.
Let $D_i = 1$ for an exact event and $D_i = 0$ for right censoring, with observed time t_i. The likelihood is

$$L(\lambda) \propto \prod_i [\lambda e^{-\lambda t_i}]^{D_i} [e^{-\lambda t_i}]^{1-D_i} = \lambda^{\sum_i D_i} e^{-\lambda \sum_i t_i}.$$

Here $\sum_i D_i = 3$ and the total observed time is the common value reported by the analysts, say T. The correct posterior is Gamma$(a+3, b+T)$, whereas recording all seven observations as failures gives the incorrect Gamma$(a+7, b+T)$.
The event indicators determine the power/shape update; total time at risk determines the rate update. Equal total time does not make the event count irrelevant. Censoring contributes exposure but no event factor. The incorrect analyst overstates the number of failures and hence shifts the posterior toward higher rates.

2.5 Dirichlet-Multinomial model

Problem 34

Original question (verbatim).
A website has four possible landing-page outcomes with probability vector $\boldsymbol{p} \sim \text{Dirichlet}(1, 2, 1, 3)$. In 100 visits, the observed counts are $(10, 20, 15, 55)$. Derive the posterior distribution, posterior

mean vector, and posterior predictive probability that the next 10 visits contain exactly 2 outcomes of type 1 and 5 of type 4 (with the remaining 3 visits split between types 2 and 3 in any way). Show how the multinomial coefficient and Dirichlet normalising constants enter the calculation.

Required theory (verbatim from the supplied lecture manual).
The Beta distribution handles one probability, while the Dirichlet distribution handles a vector of category probabilities. The components must be nonnegative and add to one, so the parameter space is a simplex. In a multinomial sample, the likelihood kernel is the product of each category probability raised to its observed count.
The Dirichlet prior has the same product structure, with each exponent equal to its concentration parameter minus one. Multiplying adds the observed category count to the corresponding prior concentration parameter. The posterior is therefore Dirichlet with updated component-wise parameters. The posterior mean for category k is its updated concentration divided by the sum of all updated concentrations. The predictive probability that the next observation belongs to category k is exactly this posterior mean. If there are only two categories, the Dirichlet model reduces to the Beta model. **Board equations from the notes.**

$$p(y \mid \boldsymbol{p}) \propto \prod_{k=1}^{K} p_k^{n_k}.$$

$$\boxed{\boldsymbol{p} \mid y \sim \text{Dirichlet}(\alpha_1 + n_1, \ldots, \alpha_K + n_K).}$$

Detailed solution and reasoning.
For counts $n = (10, 20, 15, 55)$ and prior Dirichlet$(1, 2, 1, 3)$, conjugacy gives

$$\boldsymbol{p} \mid y \sim \text{Dirichlet}(11, 22, 16, 58),$$

with concentration total 107. The posterior mean is

$$E(\boldsymbol{p} \mid y) = \frac{1}{107}(11, 22, 16, 58) = (0.102804, 0.205607, 0.149533, 0.542056).$$

The probability of next-10 count vector $(2, n_2, n_3, 5)$ with $n_2 + n_3 = 3$ is Dirichlet–Multinomial. Summing over all allocations of those remaining three visits,

$$\Pr(N_1 = 2, N_4 = 5, N_2 + N_3 = 3 \mid y) = \sum_{n_2=0}^{3} \frac{10!}{2! n_2! (3 - n_2)! 5!} \frac{B_4(\alpha' + (2, n_2, 3 - n_2, 5))}{B_4(\alpha')},$$

where $\alpha' = (11, 22, 16, 58)$ and $B_4(\alpha) = \prod_k \Gamma(\alpha_k) / \Gamma(\sum_k \alpha_k)$. Evaluating the four terms gives approximately 0.0517283. This sum includes the multinomial coefficient for each allocation and the Dirichlet normalising-constant ratio; the wording asks for the aggregate event over categories 2 and 3.

Problem 35

Original question (verbatim).
Let $\boldsymbol{p} \sim \text{Dirichlet}(\alpha_1, \alpha_2, \alpha_3)$. Researchers decide to combine categories 1 and 2 into a single category after collecting data. Derive the marginal posterior distribution for $p_1 + p_2$, and compare it with the posterior obtained by aggregating the data before updating. Explain why the Beta marginal property is useful and state the conditions under which the result holds.

Required theory (verbatim from the supplied lecture manual).
The Beta distribution handles one probability, while the Dirichlet distribution handles a vector of category probabilities. The components must be nonnegative and add to one, so the parameter space is a simplex. In a multinomial sample, the likelihood kernel is the product of each category probability raised to its observed count.
The Dirichlet prior has the same product structure, with each exponent equal to its concentration parameter minus one. Multiplying adds the observed category count to the corresponding prior concentration parameter. The posterior is therefore Dirichlet with updated component-wise parameters. The posterior mean for category k is its updated concentration divided by the sum of all updated concentrations. The predictive probability that the next observation belongs to category k is exactly this posterior mean. If there are only two categories, the Dirichlet model reduces to the Beta model. **Board equations from the notes.**

$$p(y \mid \boldsymbol{p}) \propto \prod_{k=1}^{K} p_k^{n_k}.$$

$$\boxed{\boldsymbol{p} \mid y \sim \text{Dirichlet}(\alpha_1 + n_1, \ldots, \alpha_K + n_K).}$$

Detailed solution and reasoning.
The Dirichlet aggregation property states that if $\boldsymbol{p} \sim \text{Dirichlet}(\alpha_1, \alpha_2, \alpha_3)$, then

$$(p_1 + p_2, p_3) \sim \text{Dirichlet}(\alpha_1 + \alpha_2, \alpha_3),$$

which is equivalently $p_1 + p_2 \sim \text{Beta}(\alpha_1 + \alpha_2, \alpha_3)$. After data counts (n_1, n_2, n_3), the posterior is $\text{Dirichlet}(\alpha_1 + n_1, \alpha_2 + n_2, \alpha_3 + n_3)$, so

$$p_1 + p_2 \mid y \sim \text{Beta}(\alpha_1 + \alpha_2 + n_1 + n_2, \alpha_3 + n_3).$$

If categories 1 and 2 are aggregated before updating, the aggregated prior is $\text{Beta}(\alpha_1 + \alpha_2, \alpha_3)$ and the aggregated counts are $(n_1 + n_2, n_3)$, producing exactly the same posterior. The property holds for a Dirichlet prior and multinomial sampling with category definitions fixed; it does not mean all arbitrary prior families are aggregation-consistent.

Problem 36

Original question (verbatim).
For a three-category multinomial model with a Dirichlet prior, derive the covariance between two distinct category probabilities under the posterior. Explain why the components of the probability vector are negatively dependent even though the data are conditionally iid given the vector. Then derive the predictive probability that the next two observations belong to different categories and compare it with the product of the one-step predictive probabilities.

Required theory (verbatim from the supplied lecture manual).
The Beta distribution handles one probability, while the Dirichlet distribution handles a vector of category probabilities. The components must be nonnegative and add to one, so the parameter space is a simplex. In a multinomial sample, the likelihood kernel is the product of each category probability raised to its observed count.
The Dirichlet prior has the same product structure, with each exponent equal to its concentration parameter minus one. Multiplying adds the observed category count to the corresponding prior concentration parameter. The posterior is therefore Dirichlet with updated component-wise

parameters. The posterior mean for category k is its updated concentration divided by the sum of all updated concentrations. The predictive probability that the next observation belongs to category k is exactly this posterior mean. If there are only two categories, the Dirichlet model reduces to the Beta model. **Board equations from the notes.**

$$p(y \mid \boldsymbol{p}) \propto \prod_{k=1}^{K} p_k^{n_k}.$$

$$\boxed{\boldsymbol{p} \mid y \sim \text{Dirichlet}(\alpha_1 + n_1, \ldots, \alpha_K + n_K).}$$

Detailed solution and reasoning.
Let $\alpha_0 = \sum_k \alpha'_k$, where $\alpha'_k = \alpha_k + n_k$ are the posterior Dirichlet parameters. For $i \neq j$,

$$\text{Cov}(p_i, p_j \mid y) = -\frac{\alpha'_i \alpha'_j}{\alpha_0^2(\alpha_0 + 1)} < 0.$$

The components must sum to one: increasing one category probability leaves less probability for the others. Conditional iid sampling given $\boldsymbol{p}$ therefore does not imply marginal independence of the category probabilities or of repeated observations after integrating out the shared vector.
For the next two observations to be different,

$$\Pr(Y_1 \neq Y_2 \mid y) = \sum_{i \neq j} E(p_i p_j \mid y) = 1 - E\left(\sum_i p_i^2 \mid y\right).$$

Since $E(p_i p_j) = \alpha'_i \alpha'_j / [\alpha_0(\alpha_0 + 1)]$, this is

$$\frac{\alpha_0^2 - \sum_i (\alpha'_i)^2}{\alpha_0(\alpha_0 + 1)}.$$

It is not obtained by treating the two outcomes as independent with fixed probabilities equal to posterior means; the shared uncertain vector creates predictive dependence.

2.6 Normal likelihood with a conjugate Normal prior: known variance

Problem 37

Original question (verbatim).
A sensor reports $Y_i \mid \mu \sim N(\mu, 9)$, with prior $\mu \sim N(10, 4)$. A sample of size 5 has mean 13. Derive the posterior distribution by completing the square and identify the weights assigned to the prior mean and sample mean. Determine the smallest sample size, with sample mean fixed at 13, for which the posterior mean lies within 0.5 of 13. Interpret this as a competition between prior precision and data precision.

Required theory (verbatim from the supplied lecture manual).
In this model, the sampling variance is known, and the unknown parameter is the population mean. The sample mean contains the relevant information about the mean. We assign a normal prior to the unknown mean. The likelihood precision is n divided by sigma-squared, while the prior precision is one divided by the prior variance.

When we multiply the two normal kernels and complete the square, the posterior is normal. Its precision is the sum of prior precision and data precision. Its mean is a weighted average of the prior mean and the sample mean, with weights proportional to precision. This is why we say the posterior is a compromise between prior information and data. Strong prior precision gives greater influence to the prior mean; a large sample or small known sampling variance gives greater influence to the sample mean. **Board equations from the notes.**

$$\mu \mid y \sim \mathcal{N}(\mu_n, \tau_n^2),$$

$$\boxed{\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1}, \qquad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2}\right).}$$

Detailed solution and reasoning.
The prior is $\mu \sim N(10, 4)$ and the sampling variance is 9. With $n = 5$, $\bar{y} = 13$,

$$\tau_n^2 = \left(\frac{1}{4} + \frac{5}{9}\right)^{-1} = \frac{36}{29} = 1.241379,$$

$$\mu_n = \frac{10/4 + 5(13)/9}{1/4 + 5/9} = 12.068966.$$

More transparently, the posterior mean is the precision-weighted average

$$\mu_n = \frac{10/4 + 5(13)/9}{1/4 + 5/9} = \frac{155}{13} = 11.923077.$$

The prior weight is $\frac{1/4}{1/4+n/9}$ and the data weight is $\frac{n/9}{1/4+n/9}$.
For general n with sample mean fixed at 13,

$$13 - \mu_n = \frac{(13-10)/4}{1/4 + n/9} = \frac{0.75}{0.25 + n/9}.$$

Require this to be at most 0.5: $0.75 \leq 0.5(0.25 + n/9)$, so $n \geq 12$. At $n = 12$, $\mu_n = 12.526316$, within 0.5 of 13. This is a precision competition: increasing n increases data precision and pulls the posterior mean toward the sample mean.

Problem 38

Original question (verbatim).
An experiment is performed in two stages. The first stage has sample size 8 and mean 4.2; the second has sample size 12 and mean 5.1. The observation variance is known to be 4 and the prior on the unknown mean is normal. Derive the posterior after stage one and then update it using stage two. Prove that the final posterior is identical to the posterior obtained by analysing all 20 observations at once. Explain which summary is sufficient for sequential updating.

Required theory (verbatim from the supplied lecture manual).
In this model, the sampling variance is known, and the unknown parameter is the population mean. The sample mean contains the relevant information about the mean. We assign a normal prior to the unknown mean. The likelihood precision is n divided by sigma-squared, while the prior precision is one divided by the prior variance.
When we multiply the two normal kernels and complete the square, the posterior is normal. Its

precision is the sum of prior precision and data precision. Its mean is a weighted average of the prior mean and the sample mean, with weights proportional to precision. This is why we say the posterior is a compromise between prior information and data. Strong prior precision gives greater influence to the prior mean; a large sample or small known sampling variance gives greater influence to the sample mean. **Board equations from the notes.**

$$\mu \mid y \sim \mathcal{N}(\mu_n, \tau_n^2),$$

$$\boxed{\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1}, \qquad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2}\right).}$$

Detailed solution and reasoning.
Let the prior be $\mu \sim N(\mu_0, \tau_0^2)$ and known observation variance be 4. After stage one ($n_1 = 8$, $\bar{y}_1 = 4.2$),

$$\tau_1^{-2} = \tau_0^{-2} + 8/4, \qquad \mu_1 = \tau_1^2 \left(\mu_0/\tau_0^2 + 8(4.2)/4\right).$$

The second stage has $n_2 = 12$, $\bar{y}_2 = 5.1$. Updating the normal posterior again gives

$$\tau_2^{-2} = \tau_1^{-2} + 12/4 = \tau_0^{-2} + 20/4,$$

$$\mu_2 = \tau_2^2 \left(\mu_1/\tau_1^2 + 12(5.1)/4\right) = \tau_2^2 \left(\mu_0/\tau_0^2 + \frac{8(4.2) + 12(5.1)}{4}\right).$$

The combined sample mean is $\bar{y} = (8(4.2) + 12(5.1))/20 = 4.74$. A one-stage update uses exactly the same precision and numerator, so the final posterior is identical.
The sufficient information for this known-variance normal mean is total sample size and sum (or equivalently sample mean); the stage labels/order are irrelevant under the iid model.

Problem 39

Original question (verbatim).
A scientist claims that a normal prior $N(\mu_0, \tau_0^2)$ is 'weak' because its variance is numerically large. The measurement unit is then changed from metres to millimetres. Show how the prior parameters and likelihood variance transform, and demonstrate that the posterior for the physical mean is unchanged when the transformation is handled correctly. Explain why judging prior informativeness from a bare numerical variance without reference to the measurement scale is misleading.

Required theory (verbatim from the supplied lecture manual).
In this model, the sampling variance is known, and the unknown parameter is the population mean. The sample mean contains the relevant information about the mean. We assign a normal prior to the unknown mean. The likelihood precision is n divided by sigma-squared, while the prior precision is one divided by the prior variance.
When we multiply the two normal kernels and complete the square, the posterior is normal. Its precision is the sum of prior precision and data precision. Its mean is a weighted average of the prior mean and the sample mean, with weights proportional to precision. This is why we say the posterior is a compromise between prior information and data. Strong prior precision gives greater influence to the prior mean; a large sample or small known sampling variance gives greater influence to the sample mean. **Board equations from the notes.**

$$\mu \mid y \sim \mathcal{N}(\mu_n, \tau_n^2),$$

$$\boxed{\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1}, \qquad \mu_n = \tau_n^2\left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2}\right).}$$

Detailed solution and reasoning.
Let the physical measurement be X in metres and define $X^* = 1000X$ in millimetres. Then $\mu^* = 1000\mu$, $\mu_0^* = 1000\mu_0$, and variances transform quadratically: $\tau_0^{*2} = 10^6\tau_0^2$ and $\sigma^{*2} = 10^6\sigma^2$. The sample mean also becomes $\bar{x}^* = 1000\bar{x}$.
The normal posterior formula in millimetres is

$$\tau_n^{*2} = \left(\frac{1}{10^6\tau_0^2} + \frac{n}{10^6\sigma^2}\right)^{-1} = 10^6\tau_n^2,$$

$$\mu_n^* = 1000\mu_n.$$

Converting back to metres gives the same physical posterior. A numerical variance that looks large in metres may look enormous in millimetres because its units are squared. Informativeness is relative to the measurement scale and to the likelihood variance, not to the bare number written in a chosen unit.

2.7 Normal-Gamma and Normal-Inverse-Gamma conjugacy

Problem 40

Original question (verbatim).
Let $Y_1, \ldots, Y_n \mid \mu, \sigma^2$ be iid normal, with $\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0)$ and $\sigma^2 \sim \text{InvGamma}(a_0, b_0)$. Derive the joint posterior and all updated hyperparameters in terms of n, $\bar{Y}$, and $S = \sum_i (Y_i - \bar{Y})^2$. Your derivation must show where the between-centre term involving $(\bar{Y} - \mu_0)^2$ enters and explain its role in the variance update.

Required theory (verbatim from the supplied lecture manual).
Now both the mean and variance are unknown, so the prior has to describe their relationship as well. In the Normal–Gamma version, we work with the precision tau, which is one over sigma-squared. Conditional on tau, the mean is normal, and the precision itself has a Gamma prior. In the Normal–Inverse-Gamma version, we use sigma-squared directly and give it an inverse-Gamma prior.
The posterior remains in the same family. The effective prior sample size parameter kappa increases by n. The posterior mean is the weighted average of the prior mean and sample mean. The Gamma shape or inverse-Gamma shape increases by n over two. The updated scale parameter includes the within-sample sum of squares and a correction for the difference between the sample mean and prior mean. That last term matters: it records the disagreement between the data location and the prior location. In an exam, state whether your Gamma distribution uses rate or scale and whether the prior is on variance or precision. **Board equations from the notes.**

$$\tau \sim \text{Gamma}(a_0, b_0), \qquad \mu \mid \tau \sim \mathcal{N}\left(\mu_0, \frac{1}{\kappa_0\tau}\right), \qquad Y_i \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^{-1}).$$

$$\tau \mid y \sim \text{Gamma}(a_n, b_n), \qquad \mu \mid \tau, y \sim \mathcal{N}\left(\mu_n, \frac{1}{\kappa_n\tau}\right),$$

Detailed solution and reasoning.
The likelihood kernel is

$$p(y \mid \mu, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}\sum_i (y_i - \mu)^2\right].$$

The prior kernel is $(\sigma^2)^{-a_0-1}\exp(-b_0/\sigma^2)$ times the conditional normal kernel for μ. Complete the square using

$$\sum_i (y_i - \mu)^2 + \kappa_0(\mu - \mu_0)^2 = S + \frac{\kappa_0 n}{\kappa_0 + n}(\bar{y} - \mu_0)^2 + (\kappa_0 + n)(\mu - \mu_n)^2,$$

where

$$\kappa_n = \kappa_0 + n, \quad \mu_n = \frac{\kappa_0\mu_0 + n\bar{y}}{\kappa_n}.$$

Thus

$$\mu \mid \sigma^2, y \sim N(\mu_n, \sigma^2/\kappa_n),$$
$$\sigma^2 \mid y \sim \text{InvGamma}(a_n, b_n), \quad a_n = a_0 + n/2,$$
$$b_n = b_0 + \frac{1}{2}S + \frac{1}{2}\frac{\kappa_0 n}{\kappa_n}(\bar{y} - \mu_0)^2.$$

The final between-centre term records disagreement between prior centre and data mean. Greater disagreement increases posterior uncertainty about variance; it must not be dropped.

Problem 41

Original question (verbatim).
A statistician uses a Normal-Gamma prior on precision $\tau = 1/\sigma^2$, while a colleague writes an apparently similar Normal-Inverse-Gamma prior on variance. Translate one parameterisation into the other, including the density transformation. Identify the Gamma shape-rate convention and the inverse-Gamma kernel explicitly. Explain how a missing Jacobian or a confusion between rate and scale would alter the posterior.

Required theory (verbatim from the supplied lecture manual).
Now both the mean and variance are unknown, so the prior has to describe their relationship as well. In the Normal–Gamma version, we work with the precision tau, which is one over sigma-squared. Conditional on tau, the mean is normal, and the precision itself has a Gamma prior. In the Normal–Inverse-Gamma version, we use sigma-squared directly and give it an inverse-Gamma prior.
The posterior remains in the same family. The effective prior sample size parameter kappa increases by n. The posterior mean is the weighted average of the prior mean and sample mean. The Gamma shape or inverse-Gamma shape increases by n over two. The updated scale parameter includes the within-sample sum of squares and a correction for the difference between the sample mean and prior mean. That last term matters: it records the disagreement between the data location and the prior location. In an exam, state whether your Gamma distribution uses rate or scale and whether the prior is on variance or precision. **Board equations from the notes.**

$$\tau \sim \text{Gamma}(a_0, b_0), \qquad \mu \mid \tau \sim \mathcal{N}\left(\mu_0, \frac{1}{\kappa_0\tau}\right), \qquad Y_i \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^{-1}).$$

$$\tau \mid y \sim \text{Gamma}(a_n, b_n), \qquad \mu \mid \tau, y \sim \mathcal{N}\left(\mu_n, \frac{1}{\kappa_n\tau}\right),$$

Detailed solution and reasoning.
Let $v = \sigma^2$ and $\tau = 1/v$. If $v \sim \text{InvGamma}(a, b)$ with density

$$f_V(v) = \frac{b^a}{\Gamma(a)} v^{-a-1} e^{-b/v}, \quad v > 0,$$

then $v = 1/\tau$ and $|dv/d\tau| = \tau^{-2}$. Hence

$$f_{\mathrm{T}}(\tau) = f_V(1/\tau)\tau^{-2} = \frac{b^a}{\Gamma(a)} \tau^{a-1} e^{-b\tau},$$

so $\tau \sim \text{Gamma}(a, b)$ in shape–rate form. Conversely, a Gamma prior on precision transforms to an inverse-Gamma prior on variance.
Under the Normal–Gamma version, $\mu \mid \tau \sim N(\mu_0, 1/(\kappa_0\tau))$. The Gamma parameter b is a rate, not a scale. A missing Jacobian changes the power of the transformed variable and therefore changes the prior; confusing rate and scale changes both the mean and the posterior update. Always state which parameter is being modelled and use its density transformation.

Problem 42

Original question (verbatim).
Under a Normal-Inverse-Gamma prior, derive the posterior predictive distribution of one future observation after integrating out both the regression mean and variance. State its location, scale, and degrees of freedom in terms of the updated hyperparameters. Explain why the predictive distribution has heavier tails than a normal distribution with the posterior mean and posterior expected variance plugged in.

Required theory (verbatim from the supplied lecture manual).
Now both the mean and variance are unknown, so the prior has to describe their relationship as well. In the Normal–Gamma version, we work with the precision tau, which is one over sigma-squared. Conditional on tau, the mean is normal, and the precision itself has a Gamma prior. In the Normal–Inverse-Gamma version, we use sigma-squared directly and give it an inverse-Gamma prior.
The posterior remains in the same family. The effective prior sample size parameter kappa increases by n. The posterior mean is the weighted average of the prior mean and sample mean. The Gamma shape or inverse-Gamma shape increases by n over two. The updated scale parameter includes the within-sample sum of squares and a correction for the difference between the sample mean and prior mean. That last term matters: it records the disagreement between the data location and the prior location. In an exam, state whether your Gamma distribution uses rate or scale and whether the prior is on variance or precision. **Board equations from the notes.**

$$\tau \sim \text{Gamma}(a_0, b_0), \qquad \mu \mid \tau \sim \mathcal{N}\left(\mu_0, \frac{1}{\kappa_0\tau}\right), \qquad Y_i \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^{-1}).$$

$$\tau \mid y \sim \text{Gamma}(a_n, b_n), \qquad \mu \mid \tau, y \sim \mathcal{N}\left(\mu_n, \frac{1}{\kappa_n\tau}\right),$$

Detailed solution and reasoning.
For the scalar Normal–Inverse-Gamma model, after updating let

$$\mu \mid \sigma^2, y \sim N(\mu_n, \sigma^2/\kappa_n), \qquad \sigma^2 \mid y \sim \text{InvGamma}(a_n, b_n).$$

A future observation satisfies $Y_* \mid \mu, \sigma^2 \sim N(\mu, \sigma^2)$. Integrating out μ first gives $Y_* \mid \sigma^2, y \sim N(\mu_n, \sigma^2(1 + 1/\kappa_n))$. Integrating out σ^2 yields

$$Y_* \mid y \sim t_{2a_n}\left(\mu_n,\, s_n\right), \qquad s_n^2 = \frac{b_n(\kappa_n + 1)}{a_n \kappa_n},$$

where s_n is the Student-t scale and degrees of freedom are $2a_n$.
The Student-t predictive has heavier tails because it averages over uncertainty in variance as well as mean. A normal distribution using posterior mean and posterior expected variance plugs in one variance value and suppresses this tail uncertainty. If $a_n \leq 1$, the posterior mean of σ^2 is not finite, but the Student-t predictive remains well-defined.

2.8 Exponential-family summary table

Problem 43

Original question (verbatim).
A study contains four data streams: (i) binary treatment responses, (ii) event counts observed over unequal person-time exposures, (iii) exact and right-censored waiting times, and (iv) counts in five mutually exclusive outcome categories. For each stream, identify a likelihood family and a conjugate prior from the chapter. Derive the likelihood kernel and posterior update symbolically, carefully stating which data summary updates each hyperparameter. Explain why the same 'add the observations to the prior' slogan is not adequate for all four models.

Required theory (verbatim from the supplied lecture manual).
This table is intended to be used as a quick recognition exercise. When you see successes and failures, think Beta–Binomial. When you see counts and exposure, think Gamma–Poisson. When you see waiting times and a rate, think Gamma–Exponential. When you see category counts, think Dirichlet–Multinomial. When you see normal observations with a known variance and unknown mean, think Normal–Normal. When both the mean and variance are unknown, think Normal–Gamma or Normal–Inverse-Gamma.
Do not memorise only the final update. Train yourself to recognise the likelihood kernel. Then write the prior kernel and multiply. If you can do those steps, you can reconstruct the posterior even if you forget a formula.

Detailed solution and reasoning.
(i) Binary responses. For s successes and f failures, $L(p) \propto p^s(1-p)^f$. With $p \sim \text{Beta}(a, b)$, the posterior is $\text{Beta}(a + s, b + f)$; the summaries are success and failure counts.
(ii) Unequal person-time counts. For $Y_i \sim \text{Poisson}(x_i\lambda)$, $L(\lambda) \propto \lambda^{\sum y_i} e^{-\lambda \sum x_i}$. A $\text{Gamma}(a, b)$ prior gives $\text{Gamma}(a + \sum y_i, b + \sum x_i)$; total count updates shape and total exposure updates rate.
(iii) Exact and censored waiting times. With event indicators d_i, $L(\lambda) \propto \lambda^{\sum d_i} e^{-\lambda \sum t_i}$. A Gamma prior gives $\text{Gamma}(a + \sum d_i, b + \sum t_i)$; event count and total time at risk have different roles.
(iv) Five categories. For counts $n_1, \ldots, n_5$, $L(\boldsymbol{p}) \propto \prod_k p_k^{n_k}$. A Dirichlet prior updates componentwise to $\text{Dirichlet}(\alpha_k + n_k)$.
The slogan 'add the observations' is incomplete: different models have different sufficient summaries and those summaries update different prior parameters. One must derive the kernel to see what is being added.

Problem 44

Original question (verbatim).
A researcher has access only to the following summaries: number of successes and failures; total count and total exposure; number of exact events and total time at risk; and category counts. For each summary, determine whether it is sufficient for the corresponding unknown parameter under the model in the notes. Construct two raw datasets with identical sufficient summaries but visibly different observation patterns, and explain why they nevertheless yield the same posterior under the specified model. State one plausible model extension under which the discarded pattern could become relevant.

Required theory (verbatim from the supplied lecture manual).
This table is intended to be used as a quick recognition exercise. When you see successes and failures, think Beta–Binomial. When you see counts and exposure, think Gamma–Poisson. When you see waiting times and a rate, think Gamma–Exponential. When you see category counts, think Dirichlet–Multinomial. When you see normal observations with a known variance and unknown mean, think Normal–Normal. When both the mean and variance are unknown, think Normal–Gamma or Normal–Inverse-Gamma.
Do not memorise only the final update. Train yourself to recognise the likelihood kernel. Then write the prior kernel and multiply. If you can do those steps, you can reconstruct the posterior even if you forget a formula.

Detailed solution and reasoning.
Under the stated models, the sufficient summaries are: (i) Bernoulli: number of successes and failures; (ii) Poisson with known exposures: total event count and total exposure; (iii) exponential with right censoring: number of observed events and total time at risk; (iv) multinomial: category-count vector. The factorisation theorem shows each likelihood can be written as a function of the listed summary and parameter times a factor free of the parameter.
Examples of raw data with the same summaries include binary sequences $1, 1, 0, 0$ and $0, 1, 0, 1$ (two successes, two failures); Poisson count rows $(0, 4)$ and $(2, 2)$ with equal total exposure; survival records with the same number of failures and same total time but different allocations of time across subjects; and categorical sequences with identical category counts in a different order. Under the specified likelihoods, their posteriors are identical.
The discarded pattern may become relevant with a time trend, covariates, subject-specific rates, serial dependence, overdispersion, non-constant hazard, or a model where censoring is informative. Sufficiency is always relative to a model and parameter, not a universal statement about the data.

Problem 45

Original question (verbatim).
A company wants to predict a future batch of binary outcomes, a future count over a known exposure, a future waiting time, and the next categorical outcome. For each case, derive or identify the posterior predictive distribution implied by the conjugate model. Compare the source of predictive uncertainty and explain why a plug-in estimate generally understates uncertainty. Your answer should connect the likelihood kernel, prior kernel, posterior family, and predictive family rather than merely listing distribution names.

Required theory (verbatim from the supplied lecture manual).
This table is intended to be used as a quick recognition exercise. When you see successes and

failures, think Beta–Binomial. When you see counts and exposure, think Gamma–Poisson. When you see waiting times and a rate, think Gamma–Exponential. When you see category counts, think Dirichlet–Multinomial. When you see normal observations with a known variance and unknown mean, think Normal–Normal. When both the mean and variance are unknown, think Normal–Gamma or Normal–Inverse-Gamma.
Do not memorise only the final update. Train yourself to recognise the likelihood kernel. Then write the prior kernel and multiply. If you can do those steps, you can reconstruct the posterior even if you forget a formula.

Detailed solution and reasoning.
Binary future batch. If $p \mid y \sim \text{Beta}(A, B)$, then a future count in m trials is Beta–Binomial:

$$\Pr(K = k \mid y) = \binom{m}{k} \frac{B(A + k, B + m - k)}{B(A, B)}.$$

Future count with exposure. If $\lambda \mid y \sim \text{Gamma}(A, B)$ and $Y_* \mid \lambda \sim \text{Poisson}(x\lambda)$, then

$$\Pr(Y_* = k \mid y) = \frac{\Gamma(A + k)}{\Gamma(A)k!} \left(\frac{B}{B + x}\right)^A \left(\frac{x}{B + x}\right)^k.$$

Future waiting time. With $\lambda \mid y \sim \text{Gamma}(A, B)$ and $T_* \mid \lambda \sim \text{Exp}(\lambda)$,

$$\Pr(T_* > t \mid y) = E(e^{-\lambda t} \mid y) = \left(\frac{B}{B + t}\right)^A,$$

with density $AB^A/(B + t)^{A+1}$ (a Lomax/Pareto-II form). **Next category.** Under a Dirichlet posterior $\boldsymbol{p} \mid y \sim \text{Dirichlet}(\alpha'_1, \ldots, \alpha'_K)$, the next outcome is categorical with probabilities α'_k/α'_0; a future batch has a Dirichlet–Multinomial distribution.
In every case prediction integrates the conditional sampling distribution against the posterior. Plug-in prediction suppresses variation in the unknown probability/rate/vector and therefore generally understates predictive uncertainty.

Chapter 3

Bayesian estimation, MAP, credible intervals, and prior choice

3.1 Posterior distributions contain more than a point estimate

Problem 46

Original question (verbatim).
A posterior distribution for a treatment effect is a mixture: with probability 0.65 it is $N(0.2, 0.1^2)$, and with probability 0.35 it is $N(1.1, 0.25^2)$. Determine the posterior mean and identify the posterior median and mode conceptually (numerical computation may be used for the median and mode). For squared-error, absolute-error, and zero-one neighbourhood loss, explain which summary is the relevant Bayes estimator. Discuss why reporting only the posterior mean can conceal a scientifically important second mode.

Required theory (verbatim from the supplied lecture manual).
After the data have been observed, the posterior distribution is our complete representation of uncertainty under the model. A point estimate is a summary, and the choice of summary depends on the question or loss function. Under squared-error loss, the posterior mean is optimal. Under absolute-error loss, a posterior median is optimal. The posterior mode, or MAP estimate, is the value at which the posterior density is highest.
The MAP estimate is obtained by maximising the log likelihood plus the log prior, because the normalising constant does not depend on the parameter. If the prior is flat on the chosen parameter scale, the posterior mode and maximum-likelihood estimate coincide, assuming a maximum exists. But a density mode can change after a nonlinear transformation because the transformed density includes a Jacobian. Therefore, do not treat mean, median, and mode as interchangeable, and do not assume MAP is invariant to parameterisation. **Board equations from the notes.**

$$\hat{\theta}_{\text{Bayes}} = \mathbb{E}(\theta \mid y).$$

$$\boxed{\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid y) = \arg\max_{\theta} \{\log p(y \mid \theta) + \log p(\theta)\}.}$$

Detailed solution and reasoning.
The posterior is a two-component mixture. Its mean is the mixture of component means:

$$E(\theta \mid y) = 0.65(0.2) + 0.35(1.1) = 0.515.$$

For squared-error loss, the Bayes estimator is the posterior mean, so it is 0.515.
The median is the value m solving

$$0.65\Phi\left(\frac{m-0.2}{0.1}\right)+0.35\Phi\left(\frac{m-1.1}{0.25}\right)=0.5,$$

which is approximately $m=0.27355$. The mode maximises the mixture density, not the weighted average of component modes; numerical maximisation gives approximately 0.20005 (the first component dominates the density peak).
Under absolute-error loss, a posterior median minimises expected loss. Under zero-one neighbourhood loss, the relevant estimate is a point in the neighbourhood with greatest posterior mass/density, which is mode-like and depends on the neighbourhood width. The mean 0.515 lies between two separated components and hides the fact that substantial posterior mass is near both 0.2 and 1.1. Report the mixture or a density plot in addition to a point summary.

Problem 47

Original question (verbatim).
Let $\theta>0$ have a posterior density that is Gamma with shape 3 and rate 2. Find the posterior mode of θ, and then transform to $\phi=\log\theta$. Derive the density of ϕ and find its mode. Explain why the transformed mode is not generally the logarithm of the original mode, and state which posterior summaries are invariant under a one-to-one transformation and which are not.

Required theory (verbatim from the supplied lecture manual).
After the data have been observed, the posterior distribution is our complete representation of uncertainty under the model. A point estimate is a summary, and the choice of summary depends on the question or loss function. Under squared-error loss, the posterior mean is optimal. Under absolute-error loss, a posterior median is optimal. The posterior mode, or MAP estimate, is the value at which the posterior density is highest.
The MAP estimate is obtained by maximising the log likelihood plus the log prior, because the normalising constant does not depend on the parameter. If the prior is flat on the chosen parameter scale, the posterior mode and maximum-likelihood estimate coincide, assuming a maximum exists. But a density mode can change after a nonlinear transformation because the transformed density includes a Jacobian. Therefore, do not treat mean, median, and mode as interchangeable, and do not assume MAP is invariant to parameterisation. **Board equations from the notes.**

$$\hat{\theta}_{\text{Bayes}}=\mathbb{E}(\theta\mid y).$$

$$\boxed{\hat{\theta}_{\text{MAP}}=\arg\max_{\theta}p(\theta\mid y)=\arg\max_{\theta}\{\log p(y\mid\theta)+\log p(\theta)\}.}$$

Detailed solution and reasoning.
For $\theta\sim\text{Gamma}(3,2)$ (shape–rate),

$$f_\theta(\theta)\propto\theta^2e^{-2\theta},\quad\theta>0.$$

The mode is $(3-1)/2=1$.
Let $\phi=\log\theta$, so $\theta=e^\phi$ and $|d\theta/d\phi|=e^\phi$. The transformed density is

$$f_\phi(\phi)=f_\theta(e^\phi)e^\phi\propto e^{3\phi}e^{-2e^\phi}.$$

Its log-density derivative is $3 - 2e^{\phi}$, so the transformed mode satisfies $e^{\phi} = 3/2$ and is $\phi_{mode} = \log(1.5)$, not $\log(1) = 0$.
A monotone one-to-one transformation preserves probability statements and quantiles: the median and equal-tailed interval transform by applying the function to their endpoints. A density mode does not generally transform this way because the Jacobian changes density height. Means also do not commute with nonlinear transformations: $E(\log\theta) \neq \log E(\theta)$ in general.

Problem 48

Original question (verbatim).
A decision maker must choose an action a and incurs loss $L(a, \theta) = (a - \theta)^2 + 4|a - \theta|$. Derive the posterior expected loss and characterise the Bayes action. Compare it with the posterior mean and median. Explain why the optimal point estimate is a property of both the posterior distribution and the loss function, rather than a universal feature of the posterior.

Required theory (verbatim from the supplied lecture manual).
After the data have been observed, the posterior distribution is our complete representation of uncertainty under the model. A point estimate is a summary, and the choice of summary depends on the question or loss function. Under squared-error loss, the posterior mean is optimal. Under absolute-error loss, a posterior median is optimal. The posterior mode, or MAP estimate, is the value at which the posterior density is highest.
The MAP estimate is obtained by maximising the log likelihood plus the log prior, because the normalising constant does not depend on the parameter. If the prior is flat on the chosen parameter scale, the posterior mode and maximum-likelihood estimate coincide, assuming a maximum exists. But a density mode can change after a nonlinear transformation because the transformed density includes a Jacobian. Therefore, do not treat mean, median, and mode as interchangeable, and do not assume MAP is invariant to parameterisation. **Board equations from the notes.**

$$\hat{\theta}_{\text{Bayes}} = \mathbb{E}(\theta \mid y).$$

$$\boxed{\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid y) = \arg\max_{\theta} \{\log p(y \mid \theta) + \log p(\theta)\}.}$$

Detailed solution and reasoning.
The Bayes action minimises posterior expected loss

$$R(a \mid y) = E\{(a - \theta)^2 + 4|a - \theta| \mid y\}.$$

For a continuous posterior, differentiate with respect to a:

$$R'(a) = 2\{a - E(\theta \mid y)\} + 4\{2F(a \mid y) - 1\}.$$

A minimiser therefore solves

$$2(a - E(\theta \mid y)) + 4(2F(a \mid y) - 1) = 0.$$

Because the quadratic component makes the risk strictly convex (and the absolute-value component convex), the minimiser is unique under ordinary regularity conditions. Its exact numerical value cannot be determined without the posterior distribution.
For squared-error loss alone the derivative gives $a = E(\theta \mid y)$. For absolute-error loss alone any posterior median minimises risk. The combined loss balances both effects, so the action is not generally equal to either summary. A Bayes estimator is defined jointly by the posterior and the loss function, not by a universal rule.

3.2 Posterior variance and the value of data

Problem 49

Original question (verbatim).
Let θ have prior variance 16. A future observation Y can take values 0 or 1, and the posterior means are known to be 2 when $Y = 0$ and 6 when $Y = 1$, with prior predictive probabilities 0.75 and 0.25. Use the law of total expectation and law of total variance to determine the prior mean and the expected posterior variance. Explain the role of $\mathrm{Var}\{E(\theta \mid Y)\}$ as the amount of prior uncertainty resolved by the data.

Required theory (verbatim from the supplied lecture manual).
The law of total variance tells us how prior uncertainty is divided. One part is the expected posterior variance, which represents uncertainty remaining after observing data. The other part is the variance of the posterior mean across possible datasets, which represents how much the data can move our estimate. Their sum is the prior variance.
This identity implies that posterior variance is no greater than prior variance on average over the prior predictive distribution. It does not say that every particular dataset must reduce uncertainty. If the observed data conflict with a very concentrated prior, the posterior can actually be more spread out than the prior. So when you explain this theorem, use the phrase 'on average over possible data' rather than saying 'the posterior variance is always smaller'. **Board equations from the notes.**

$$\mathbb{E}(\theta) = \mathbb{E}\{\mathbb{E}(\theta \mid Y)\},$$

$$\boxed{\mathrm{Var}(\theta) = \mathbb{E}\{\mathrm{Var}(\theta \mid Y)\} + \mathrm{Var}\{\mathbb{E}(\theta \mid Y)\}.}$$

Detailed solution and reasoning.
By the law of total expectation,

$$E(\theta) = E\{E(\theta \mid Y)\} = 0.75(2) + 0.25(6) = 3.$$

The variance of the posterior mean is

$$\mathrm{Var}\{E(\theta \mid Y)\} = 0.75(2-3)^2 + 0.25(6-3)^2 = 3.$$

The law of total variance states

$$\mathrm{Var}(\theta) = E\{\mathrm{Var}(\theta \mid Y)\} + \mathrm{Var}\{E(\theta \mid Y)\}.$$

Given prior variance 16,

$$E\{\mathrm{Var}(\theta \mid Y)\} = 16 - 3 = 13.$$

Thus the data are expected to resolve three units of variance, on average. The second term measures how much the posterior mean varies across possible data outcomes; it is the expected reduction in squared-error uncertainty due to observing Y.

Problem 50

Original question (verbatim).
A very concentrated prior is centred at 0, but the observed data strongly favour values near 5. Construct a conjugate normal-normal example in which the posterior variance for this particular dataset is larger than the prior variance, or prove that such an example is impossible under the specified known-variance normal model. Reconcile your conclusion with the law of total variance. Your answer must distinguish a statement about every realised dataset from a statement averaged over the prior predictive distribution.

Required theory (verbatim from the supplied lecture manual).
The law of total variance tells us how prior uncertainty is divided. One part is the expected posterior variance, which represents uncertainty remaining after observing data. The other part is the variance of the posterior mean across possible datasets, which represents how much the data can move our estimate. Their sum is the prior variance.
This identity implies that posterior variance is no greater than prior variance on average over the prior predictive distribution. It does not say that every particular dataset must reduce uncertainty. If the observed data conflict with a very concentrated prior, the posterior can actually be more spread out than the prior. So when you explain this theorem, use the phrase 'on average over possible data' rather than saying 'the posterior variance is always smaller'. **Board equations from the notes.**

$$\mathbb{E}(\theta) = \mathbb{E}\{\mathbb{E}(\theta \mid Y)\},$$

$$\boxed{\mathrm{Var}(\theta) = \mathbb{E}\{\mathrm{Var}(\theta \mid Y)\} + \mathrm{Var}\{\mathbb{E}(\theta \mid Y)\}.}$$

Detailed solution and reasoning.
It is impossible in the specified known-variance Normal–Normal model. If $Y_i \mid \mu \sim N(\mu, \sigma^2)$ and $\mu \sim N(\mu_0, \tau_0^2)$, then for every realised sample

$$\mathrm{Var}(\mu \mid y) = \left(\tau_0^{-2} + n\sigma^{-2}\right)^{-1} < \tau_0^2.$$

The posterior variance does not depend on $\bar{y}$, so even data with mean near 5 cannot make the posterior variance larger than the prior variance in this model. The posterior mean may shift far from zero, but a shifted centre is not the same as increased variance.
The law of total variance is an average identity:

$$\mathrm{Var}(\mu) = E_Y\{\mathrm{Var}(\mu \mid Y)\} + \mathrm{Var}_Y\{E(\mu \mid Y)\}.$$

The expected posterior variance is no greater than the prior variance, with the remainder represented by variation in posterior means. This example is stronger: posterior variance decreases for every dataset because known-variance conjugacy makes it deterministic given n. In more general models, an individual realised dataset can increase posterior variance even though the prior-predictive average satisfies the identity.

Problem 51

Original question (verbatim).
A laboratory can collect either 10 additional measurements from an existing group or 10 measurements from a new group. Formulate a Bayesian design argument using expected posterior variance to compare the two plans in a hierarchical setting. Identify which unknown quantity each plan is expected to inform and explain why the value of data cannot be assessed solely by counting the number of additional observations.

Required theory (verbatim from the supplied lecture manual).
The law of total variance tells us how prior uncertainty is divided. One part is the expected posterior variance, which represents uncertainty remaining after observing data. The other part is the variance of the posterior mean across possible datasets, which represents how much the data can move our estimate. Their sum is the prior variance.
This identity implies that posterior variance is no greater than prior variance on average over the prior predictive distribution. It does not say that every particular dataset must reduce uncertainty. If the observed data conflict with a very concentrated prior, the posterior can actually be more spread out than the prior. So when you explain this theorem, use the phrase 'on average over possible data' rather than saying 'the posterior variance is always smaller'. **Board equations from the notes.**

$$\mathbb{E}(\theta) = \mathbb{E}\{\mathbb{E}(\theta \mid Y)\},$$

$$\boxed{\mathrm{Var}(\theta) = \mathbb{E}\{\mathrm{Var}(\theta \mid Y)\} + \mathrm{Var}\{\mathbb{E}(\theta \mid Y)\}.}$$

Detailed solution and reasoning.
The design target must be specified first: estimating the existing group's effect θ_j, the population mean μ, the between-group variance τ^2, or a future group's effect. For the existing group, ten more measurements reduce its sampling variance s_j^2 and usually sharpen its conditional posterior. They provide limited direct information about how groups vary in the population if the number of groups remains unchanged.
Ten measurements from a new group add a new group-level estimate. In a hierarchical model $\theta_j \sim N(\mu, \tau^2)$, this can inform μ and especially τ^2 because it adds another draw from the group distribution. The value of each design can be compared by preposterior expected posterior variance, for example

$$\mathrm{EVSI} = \mathrm{Var}(\psi \mid y) - E_{Y_{new}|y}\{\mathrm{Var}(\psi \mid y, Y_{new})\},$$

for the chosen target ψ (or by expected loss reduction).
Which plan is better depends on the current number and precision of groups, the hierarchy, and the target. Counting ten observations in each design is not sufficient: repeated observations in one group and a new independent group inform different levels of the model.

3.3 Credible intervals

Problem 52

Original question (verbatim).
After observing data, a parameter has posterior distribution Beta$(2, 8)$. Define the equal-tailed 90% credible interval and the 90% highest posterior density (HPD) interval. Explain how each

is constructed and why they need not have the same endpoints. Without relying on a normal approximation, describe how you would compute both intervals numerically and verify their posterior probability content.

Required theory (verbatim from the supplied lecture manual).
A credible interval is a set of parameter values that contains a specified amount of posterior probability. For a central 95 percent interval, we take the 2.5th and 97.5th posterior percentiles. The interpretation is direct: conditional on the observed data and the model, the posterior probability that the parameter lies in that interval is 95 percent.
A highest posterior density region also contains 95 percent posterior mass, but it gathers the points with greatest density first. For a symmetric unimodal posterior, the central interval and HPD interval agree. For a skewed or multimodal posterior, they may differ, and the HPD set may even consist of separate pieces. The central interval is often easier to calculate from posterior simulations and has a clear quantile definition. The HPD region depends on the parameter scale, so it generally does not stay the same under nonlinear transformations. **Board equations from the notes.**

$$\Pr\{\theta \in C(y) \mid y\} = 1 - \alpha.$$

$$\left[q_{\alpha/2}, q_{1-\alpha/2}\right],$$

Detailed solution and reasoning.
For a continuous posterior CDF F of Beta$(2, 8)$, the equal-tailed 90% interval is

$$[F^{-1}(0.05), F^{-1}(0.95)] \approx [0.04102, 0.42914].$$

The HPD interval is the shortest (for a unimodal continuous density, highest-density) interval with posterior mass 0.90. Its endpoints satisfy

$$F(u) - F(l) = 0.90, \qquad f(l) = f(u),$$

with the endpoints chosen so that density inside is at least as high as density outside. Numerical solution gives approximately $[0.01543, 0.37611]$.
To calculate either interval without a normal approximation, evaluate the Beta CDF and invert it for equal tails. For HPD, numerically solve the two equations above (or find the density cutoff c such that $\int_{f(p)\geq c} f(p)dp = 0.90$). Verify probability content by evaluating $F(u) - F(l)$ for an interval, or integrating over the HPD set. The endpoints differ because the Beta posterior is asymmetric.

Problem 53

Original question (verbatim).
A posterior density is bimodal, with 60% of its probability near -2 and 40% near 3, and very low density around zero. Construct a 90% central credible interval and describe the possible shape of a 90% HPD region. Explain why a single interval may include a low-density region and why the HPD set may be disconnected. Discuss what each representation communicates to a scientist.

Required theory (verbatim from the supplied lecture manual).
A credible interval is a set of parameter values that contains a specified amount of posterior

probability. For a central 95 percent interval, we take the 2.5th and 97.5th posterior percentiles. The interpretation is direct: conditional on the observed data and the model, the posterior probability that the parameter lies in that interval is 95 percent.
A highest posterior density region also contains 95 percent posterior mass, but it gathers the points with greatest density first. For a symmetric unimodal posterior, the central interval and HPD interval agree. For a skewed or multimodal posterior, they may differ, and the HPD set may even consist of separate pieces. The central interval is often easier to calculate from posterior simulations and has a clear quantile definition. The HPD region depends on the parameter scale, so it generally does not stay the same under nonlinear transformations. **Board equations from the notes.**

$$\Pr\{\theta \in C(y) \mid y\} = 1 - \alpha.$$

$$\left[q_{\alpha/2}, q_{1-\alpha/2}\right],$$

Detailed solution and reasoning.
A 90% central interval is $[F^{-1}(0.05), F^{-1}(0.95)]$. It places 5% posterior probability in each tail. Since 60% of the mass is near -2 and 40% is near 3, this interval will generally extend from the left component into the right component and include the low-density region around zero.
An HPD region contains the points with the largest posterior density until 90% mass is accumulated. For a separated bimodal density, it can be a union of two disjoint intervals: one around each mode. Since the right mode contains 40% total mass, the remaining 50% needed may be taken from the denser part near the left mode. Exact endpoints require the actual component densities, which are not supplied.
The central interval communicates equal tail probability and is always a single interval in one dimension; the HPD set communicates where posterior density is largest and can be disconnected. Reporting only a central interval can visually suggest substantial support in a valley where the density is actually low.

Problem 54

Original question (verbatim).
Let θ have a continuous posterior and let $\phi = \exp(\theta)$. A 95% equal-tailed interval for θ is $[l, u]$. Derive the corresponding interval for ϕ, and explain why it remains an equal-tailed interval. Then explain why transforming a 95% HPD interval for θ does not generally produce the 95% HPD interval for ϕ. Identify the role of the Jacobian in this difference.

Required theory (verbatim from the supplied lecture manual).
A credible interval is a set of parameter values that contains a specified amount of posterior probability. For a central 95 percent interval, we take the 2.5th and 97.5th posterior percentiles. The interpretation is direct: conditional on the observed data and the model, the posterior probability that the parameter lies in that interval is 95 percent.
A highest posterior density region also contains 95 percent posterior mass, but it gathers the points with greatest density first. For a symmetric unimodal posterior, the central interval and HPD interval agree. For a skewed or multimodal posterior, they may differ, and the HPD set may even consist of separate pieces. The central interval is often easier to calculate from posterior simulations and has a clear quantile definition. The HPD region depends on the parameter scale, so it generally does not stay the same under nonlinear transformations. **Board equations from the notes.**

$$\Pr\{\theta \in C(y) \mid y\} = 1 - \alpha.$$

$$\left[q_{\alpha/2}, q_{1-\alpha/2}\right],$$

Detailed solution and reasoning.
For a monotone increasing transformation $\phi = e^{\theta}$, the event $l \le \theta \le u$ is exactly equivalent to $e^{l} \le \phi \le e^{u}$. Therefore the transformed interval

$$[e^{l}, e^{u}]$$

retains 2.5% probability in each tail and is an equal-tailed 95% interval for ϕ.
An HPD interval is defined by density height on the chosen scale. The transformed density is

$$f_{\phi}(\phi) = f_{\theta}(\log \phi)\frac{1}{\phi},$$

so the Jacobian $1/\phi$ changes the relative density heights. A set with highest density in θ need not be highest density in ϕ, even though its posterior probability is unchanged. Probability content is transformation-invariant; HPD ordering and shortest-interval status generally are not.

3.4 Informative, weakly informative, and noninformative priors

Problem 55

Original question (verbatim).
An expert believes a failure probability is usually below 0.05 but cannot rule out values as high as 0.2. Propose two proper prior distributions that express different degrees of information while respecting this belief. Compare their prior predictive implications for a batch of 20 trials. Explain why a prior can be weakly informative on the probability scale but highly informative on the log-odds scale, and state how you would communicate the elicitation assumptions.

Required theory (verbatim from the supplied lecture manual).
A prior can be informative when it contains substantial prior knowledge about the quantity, weakly informative when it rules out implausible values but still allows a broad range, or noninformative in an attempt to let the likelihood dominate. These labels are not absolute. A prior that is weak on one scale may be highly informative on another scale.
A flat prior is not automatically neutral. Flatness depends on parameterisation, and an improper prior has an arbitrary multiplicative constant because it cannot be normalised to integrate to one. Such priors can sometimes produce proper posteriors for parameter estimation, but they require care. The practical lesson is to choose priors with the actual scale and scientific meaning in mind, check the prior predictive implications, and examine sensitivity of conclusions to reasonable alternatives.

Detailed solution and reasoning.
One possible pair is $p \sim \text{Beta}(1, 19)$ (broader) and $p \sim \text{Beta}(2, 38)$ (more concentrated); both have mean 0.05, while the second has twice the concentration. They place most prior mass at small failure probabilities but retain nonzero tail probability above 0.2. For 20 trials, the prior predictive

distribution is Beta–Binomial:

$$\Pr(K = k) = \binom{20}{k} \frac{B(a + k, b + 20 - k)}{B(a, b)}.$$

The two priors yield different probabilities of zero failures and of unusually large failure counts; these predictive implications should be shown to the expert before observing the current batch. Prior informativeness depends on parameterisation. A density that appears broad in p may be concentrated or skewed in $\eta = \log\{p/(1-p)\}$ because the Jacobian changes the density. Elicitation should state the scale, plausible quantiles, prior mean, concentration/effective sample size, and prior predictive checks. The Beta choices are examples, not uniquely determined by the verbal statement.

Problem 56

Original question (verbatim).
A colleague proposes a flat prior on a positive rate λ and calls it noninformative. Another proposes a flat prior on the mean waiting time $\tau = 1/\lambda$. Derive the induced density under each parameterisation and show whether the two priors express the same beliefs. Discuss propriety and explain why 'flat' is not a parameterisation-free concept.

Required theory (verbatim from the supplied lecture manual).
A prior can be informative when it contains substantial prior knowledge about the quantity, weakly informative when it rules out implausible values but still allows a broad range, or noninformative in an attempt to let the likelihood dominate. These labels are not absolute. A prior that is weak on one scale may be highly informative on another scale.
A flat prior is not automatically neutral. Flatness depends on parameterisation, and an improper prior has an arbitrary multiplicative constant because it cannot be normalised to integrate to one. Such priors can sometimes produce proper posteriors for parameter estimation, but they require care. The practical lesson is to choose priors with the actual scale and scientific meaning in mind, check the prior predictive implications, and examine sensitivity of conclusions to reasonable alternatives.

Detailed solution and reasoning.
A flat prior on the rate is $\pi_\lambda(\lambda) \propto 1$, $\lambda > 0$. Under $\tau = 1/\lambda$,

$$\pi_\tau(\tau) = \pi_\lambda(1/\tau) \left| \frac{d(1/\tau)}{d\tau} \right| \propto \frac{1}{\tau^2}.$$

A flat prior on the mean waiting time is instead $\pi_\tau(\tau) \propto 1$; transforming it back gives

$$\pi_\lambda(\lambda) \propto \frac{1}{\lambda^2}.$$

These are different priors and express different relative beliefs. Neither flat prior is a proper probability distribution on $(0, \infty)$ because its integral diverges.
The word 'flat' is not invariant to reparameterisation: densities transform with a Jacobian. One must choose a meaningful parameter scale and assess induced prior predictions rather than calling a constant density universally noninformative.

Problem 57

Original question (verbatim).
Two reasonable priors lead to posterior means 0.04 and 0.07 for a risk parameter, while the posterior probability that risk exceeds 0.1 is 0.02 under one prior and 0.18 under the other. Design a prior-sensitivity analysis that goes beyond comparing posterior means. Specify which prior predictive checks, posterior summaries, and scientific conclusions should be compared, and explain what result would make you describe the conclusion as prior-sensitive.

Required theory (verbatim from the supplied lecture manual).
A prior can be informative when it contains substantial prior knowledge about the quantity, weakly informative when it rules out implausible values but still allows a broad range, or noninformative in an attempt to let the likelihood dominate. These labels are not absolute. A prior that is weak on one scale may be highly informative on another scale.
A flat prior is not automatically neutral. Flatness depends on parameterisation, and an improper prior has an arbitrary multiplicative constant because it cannot be normalised to integrate to one. Such priors can sometimes produce proper posteriors for parameter estimation, but they require care. The practical lesson is to choose priors with the actual scale and scientific meaning in mind, check the prior predictive implications, and examine sensitivity of conclusions to reasonable alternatives.

Detailed solution and reasoning.
Use a prespecified set of scientifically defensible priors rather than selecting one based on the preferred conclusion. For each prior, first examine prior predictive distributions (typical risks, tail risks, and plausible event counts). Then fit the same likelihood and record posterior mean/median, credible interval endpoints, and decision-relevant probabilities such as $\Pr(p > 0.1 \mid y)$.
A useful sensitivity table contains rows for each prior and columns for prior mean/concentration, prior predictive event probabilities, posterior mean, 95% interval, $\Pr(p > 0.1 \mid y)$, and the resulting scientific decision. In this example, means 0.04 versus 0.07 and tail probabilities 0.02 versus 0.18 show that the conclusion about exceeding 0.1 is highly prior-sensitive, even if the means look fairly close.
Sensitivity is important when plausible prior choices change a substantive decision or uncertainty statement. It is not a search for a prior that produces a desired answer. If data are highly informative, reasonable priors should lead to more similar conclusions; if they do not, the prior dependence must be reported.

3.5 Jeffreys' prior

Problem 58

Original question (verbatim).
For $X \sim \text{Binomial}(n, p)$, reparameterise using $\eta = \log\{p/(1-p)\}$. Write the likelihood in terms of η, derive the Fisher information for η, and obtain Jeffreys' prior directly on the log-odds scale. Transform it back to the probability scale and explain how the Jacobian reconciles the two expressions. Compare the result with a prior that is uniform in p.

Required theory (verbatim from the supplied lecture manual).
Jeffreys' prior is constructed from the Fisher information. For a scalar parameter, we calculate the expected negative second derivative of the log likelihood, and then take its square root. The result

is used as a prior density up to proportionality. Its central motivation is invariance under smooth one-to-one reparameterisation.
For the binomial probability p, the Fisher information is n divided by p times one-minus-p. Taking the square root gives a density proportional to p to the minus one-half times one-minus-p to the minus one-half. That is a Beta one-half, one-half prior. It differs from the uniform Beta one, one prior. Jeffreys' prior is not a magic answer for all models: in multiparameter problems it can be controversial, and the resulting posterior still has to be checked for propriety. **Board equations from the notes.**

$$I(\theta) = \mathbb{E}\left[\left\{\frac{\partial}{\partial\theta}\log p(Y \mid \theta)\right\}^2 \middle| \theta\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log p(Y \mid \theta) \middle| \theta\right],$$

$$\boxed{\pi_J(\theta) \propto \sqrt{I(\theta)}.}$$

Detailed solution and reasoning.
Let $\eta = \log\{p/(1-p)\}$, so $p = e^\eta/(1+e^\eta)$. For x successes,

$$L(\eta) \propto p^x(1-p)^{n-x} = \frac{e^{x\eta}}{(1+e^\eta)^n}.$$

The log-likelihood is $x\eta - n\log(1+e^\eta)$. The Fisher information is

$$I_\eta = np(1-p),$$

so Jeffreys' density with respect to $d\eta$ is

$$\pi_J(\eta) \propto \sqrt{np(1-p)}.$$

Now $d\eta/dp = 1/[p(1-p)]$, so the induced density with respect to dp is

$$\pi_J(p) \propto \sqrt{p(1-p)}\frac{1}{p(1-p)} \propto p^{-1/2}(1-p)^{-1/2},$$

which is Beta$(1/2, 1/2)$. A prior uniform in p is Beta$(1,1)$ and is not the same. The Jacobian reconciles the coordinate-specific density expressions.

Problem 59

Original question (verbatim).
Let $Y_1, \ldots, Y_n$ be iid exponential observations with unknown rate $\lambda > 0$. Derive the Fisher information for λ and Jeffreys' prior. Then transform to the mean waiting time $\tau = 1/\lambda$ and derive the induced prior. Explain how this example illustrates the invariance motivation for Jeffreys' construction, and discuss whether the resulting prior is proper.

Required theory (verbatim from the supplied lecture manual).
Jeffreys' prior is constructed from the Fisher information. For a scalar parameter, we calculate the expected negative second derivative of the log likelihood, and then take its square root. The result is used as a prior density up to proportionality. Its central motivation is invariance under smooth one-to-one reparameterisation.
For the binomial probability p, the Fisher information is n divided by p times one-minus-p. Taking

the square root gives a density proportional to p to the minus one-half times one-minus-p to the minus one-half. That is a Beta one-half, one-half prior. It differs from the uniform Beta one, one prior. Jeffreys' prior is not a magic answer for all models: in multiparameter problems it can be controversial, and the resulting posterior still has to be checked for propriety. **Board equations from the notes.**

$$I(\theta) = \mathbb{E}\left[\left\{\frac{\partial}{\partial\theta}\log p(Y \mid \theta)\right\}^2 \middle| \theta\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log p(Y \mid \theta) \middle| \theta\right],$$

$$\boxed{\pi_J(\theta) \propto \sqrt{I(\theta)}.}$$

Detailed solution and reasoning.
For iid exponential observations with rate λ, the log-likelihood is

$$\ell(\lambda) = n\log\lambda - \lambda\sum_i y_i.$$

The score is $n/\lambda - \sum_i y_i$ and the negative expected second derivative is

$$I(\lambda) = n/\lambda^2.$$

Thus Jeffreys' prior is $\pi_J(\lambda) \propto \sqrt{I(\lambda)} \propto 1/\lambda$. For $\tau = 1/\lambda$, the induced density is

$$\pi_J(\tau) = \pi_J(1/\tau)\frac{1}{\tau^2} \propto \tau\frac{1}{\tau^2} = 1/\tau.$$

It has the same functional form in the mean parameterisation, illustrating the reparameterisation motivation of Jeffreys' construction. The prior is improper because $\int_0^\infty d\lambda/\lambda$ diverges (at both boundaries); it can sometimes yield a proper posterior for estimation but cannot automatically be used for Bayes factors.

Problem 60

Original question (verbatim).
A model has density $f(y \mid \theta)$ and Fisher information $I(\theta)$. A student computes $I(\theta)$ from the observed sample rather than taking its expectation, then uses its square root as a prior. Explain why this is not the standard Jeffreys prior described in the notes. Give a complete derivation template for the scalar case and discuss at least two cautions that remain even after Jeffreys' prior has been derived correctly.

Required theory (verbatim from the supplied lecture manual).
Jeffreys' prior is constructed from the Fisher information. For a scalar parameter, we calculate the expected negative second derivative of the log likelihood, and then take its square root. The result is used as a prior density up to proportionality. Its central motivation is invariance under smooth one-to-one reparameterisation.
For the binomial probability p, the Fisher information is n divided by p times one-minus-p. Taking the square root gives a density proportional to p to the minus one-half times one-minus-p to the minus one-half. That is a Beta one-half, one-half prior. It differs from the uniform Beta one, one prior. Jeffreys' prior is not a magic answer for all models: in multiparameter problems it can be controversial, and the resulting posterior still has to be checked for propriety. **Board equations**

from the notes.

$$I(\theta) = \mathbb{E}\left[\left\{\frac{\partial}{\partial\theta}\log p(Y \mid \theta)\right\}^2 \middle| \theta\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log p(Y \mid \theta)\middle|\theta\right],$$

$$\boxed{\pi_J(\theta) \propto \sqrt{I(\theta)}.}$$

Detailed solution and reasoning.
The standard scalar Jeffreys prior is based on expected Fisher information under the sampling model:

$$I(\theta) = E_\theta\left[-\frac{\partial^2}{\partial\theta^2}\log f(Y \mid \theta)\right] = E_\theta\left[\left\{\frac{\partial}{\partial\theta}\log f(Y \mid \theta)\right\}^2\right],$$

when the regularity conditions make these expressions equal. Then $\pi_J(\theta) \propto \sqrt{I(\theta)}$. The expectation is over repeated data generated at fixed θ; using the observed sample's Hessian is an observed-information construction, not the standard definition in the notes.
A sound derivation: (1) write $\ell(\theta; y) = \log f(y \mid \theta)$; (2) differentiate twice; (3) take expectation under $f(y \mid \theta)$; (4) take the square root; (5) specify support and normalisation/propriety; (6) transform with a Jacobian if changing parameterisation. Even a correctly derived Jeffreys prior may be improper, may behave problematically near boundaries, and may require care with nuisance parameters or non-regular models. It is not automatically scientifically appropriate in every application.

3.6 Prior sensitivity and transformations

Problem 61

Original question (verbatim).
Let p have a Beta$(2, 3)$ prior and define the odds $\omega = p/(1-p)$. Derive the density of ω with the correct Jacobian and identify its support. Compare this induced prior with a prior that is uniform on ω. Explain why choosing a prior before deciding which parameter scale has substantive meaning can change the analysis.

Required theory (verbatim from the supplied lecture manual).
Prior sensitivity asks whether substantive conclusions change when we replace the chosen prior by other reasonable prior distributions. This is not a search for a prior that gives the answer we prefer. It is a way of measuring how strongly the conclusion depends on assumptions that are not determined by the current data alone.
When transforming a parameter, transform the density, not just the variable name. If eta is a function of theta, the density of eta contains the absolute derivative of theta with respect to eta. This Jacobian can make a flat prior on one scale non-flat on another. The same issue affects modes and HPD intervals. Whenever a question asks for an induced prior, write the transformation, calculate the inverse transformation, differentiate it, and multiply by the absolute Jacobian. **Board equations from the notes.**

$$p_\phi(\phi) = p_\theta(h^{-1}(\phi))\left|\frac{\mathrm{d}h^{-1}(\phi)}{\mathrm{d}\phi}\right|.$$

Detailed solution and reasoning.
For $p \sim \text{Beta}(2,3)$,

$$f_P(p) = \frac{p^1(1-p)^2}{B(2,3)}, \quad 0 < p < 1, \qquad B(2,3) = 1/12.$$

Let $\omega = p/(1-p)$, so $p = \omega/(1+\omega)$, $1-p = 1/(1+\omega)$ and $dp/d\omega = (1+\omega)^{-2}$. Therefore

$$f_\Omega(\omega) = 12\frac{\omega}{(1+\omega)^3}\frac{1}{(1+\omega)^2} = \frac{12\omega}{(1+\omega)^5}, \quad \omega > 0.$$

A uniform prior on ω would have constant density on $(0,\infty)$ and is improper; it is not this induced prior. Choosing a prior flat or convenient on odds rather than probability scale changes the implied beliefs on p. The scientifically meaningful scale should be chosen deliberately and prior predictive implications should be checked.

Problem 62

Original question (verbatim).
A scientific conclusion is stated as $\Pr(\theta > c \mid y) > 0.95$. Repeat the analysis under three reasonable priors that differ in concentration and tail behaviour. Design a sensitivity table that records posterior means, interval endpoints, and the probability of the scientific event. Explain why sensitivity should focus on the decision-relevant conclusion rather than searching for the prior that produces a preferred answer.

Required theory (verbatim from the supplied lecture manual).
Prior sensitivity asks whether substantive conclusions change when we replace the chosen prior by other reasonable prior distributions. This is not a search for a prior that gives the answer we prefer. It is a way of measuring how strongly the conclusion depends on assumptions that are not determined by the current data alone.
When transforming a parameter, transform the density, not just the variable name. If eta is a function of theta, the density of eta contains the absolute derivative of theta with respect to eta. This Jacobian can make a flat prior on one scale non-flat on another. The same issue affects modes and HPD intervals. Whenever a question asks for an induced prior, write the transformation, calculate the inverse transformation, differentiate it, and multiply by the absolute Jacobian. **Board equations from the notes.**

$$p_\phi(\phi) = p_\theta(h^{-1}(\phi)) \left| \frac{\mathrm{d}h^{-1}(\phi)}{\mathrm{d}\phi} \right|.$$

Detailed solution and reasoning.
Construct a sensitivity analysis with three priors that span plausible concentration and tail behaviour, for example a moderately concentrated Beta prior, a broader Beta prior with similar centre, and a prior allowing heavier upper-tail mass. Keep the likelihood and data unchanged. For each fit compute

$$E(\theta \mid y), \quad [q_{0.025}, q_{0.975}], \quad \Pr(\theta > c \mid y).$$

Also show prior predictive distributions and the posterior probability relevant to the actual scientific decision.
A clear table should list the prior family/parameters, prior mean and spread, posterior mean, interval endpoints, and $\Pr(\theta > c \mid y)$. If the scientific conclusion changes across reasonable priors

(for example, the probability crosses the prespecified 0.95 decision threshold), state that the conclusion is prior-sensitive. Do not select the prior because it yields the desired result; explain which prior assumptions are defensible before fitting.

Problem 63

Original question (verbatim).
An analyst reports that the posterior mode and the 95% HPD interval are unchanged after replacing a parameter by a nonlinear one-to-one transformation. Determine whether these invariance claims are generally valid. Use a concrete transformation to show the effect of the Jacobian on the density and explain which probability statements are preserved even when modes and HPD sets change.

Required theory (verbatim from the supplied lecture manual).
Prior sensitivity asks whether substantive conclusions change when we replace the chosen prior by other reasonable prior distributions. This is not a search for a prior that gives the answer we prefer. It is a way of measuring how strongly the conclusion depends on assumptions that are not determined by the current data alone.
When transforming a parameter, transform the density, not just the variable name. If eta is a function of theta, the density of eta contains the absolute derivative of theta with respect to eta. This Jacobian can make a flat prior on one scale non-flat on another. The same issue affects modes and HPD intervals. Whenever a question asks for an induced prior, write the transformation, calculate the inverse transformation, differentiate it, and multiply by the absolute Jacobian. **Board equations from the notes.**

$$p_\phi(\phi) = p_\theta(h^{-1}(\phi)) \left| \frac{\mathrm{d}h^{-1}(\phi)}{\mathrm{d}\phi} \right|.$$

Detailed solution and reasoning.
The claims are not generally valid. Under a nonlinear transformation $\phi = g(\theta)$, the density becomes

$$f_\phi(\phi) = f_\theta(g^{-1}(\phi)) \left| \frac{dg^{-1}(\phi)}{d\phi} \right|.$$

The Jacobian changes density heights, so the mode need not transform as $g(\theta_{mode})$, and an HPD set need not map to an HPD set. For example, if $\theta \sim \text{Gamma}(3, 2)$, its mode is 1, but for $\phi = \log\theta$ the transformed mode is $\log(3/2)$.
Probability statements are preserved under one-to-one transformations: $\Pr(\theta \in C \mid y) = \Pr(\phi \in g(C) \mid y)$. Quantiles/equal-tailed intervals transform appropriately when g is monotone. Modes and HPD intervals are density-based and therefore scale-dependent. A report should state the parameterisation on which those summaries were computed.

Chapter 4

Bayesian hypothesis testing and Bayes factors

4.1 Simple and composite hypotheses

Problem 64

Original question (verbatim).
A coin is tested under $H_0 : p = 0.5$ and $H_1 : p$ unknown with $p \sim \text{Beta}(2, 2)$. In 12 tosses, 10 heads are observed. Derive the likelihood under H_0, the marginal likelihood under H_1, and the Bayes factor BF_{10}. Explain why evaluating the alternative likelihood at its maximum-likelihood estimate is not the same as computing its marginal likelihood.

Required theory (verbatim from the supplied lecture manual).
In Bayesian hypothesis testing, a hypothesis is treated as a model with a prior probability. A simple hypothesis specifies the parameter completely. A composite hypothesis leaves some parameter values unknown and assigns a prior distribution over them. For a simple hypothesis, the probability of the data under that hypothesis is obtained directly from the sampling model. For a composite hypothesis, we average the likelihood across the parameter prior to obtain a marginal likelihood. This distinction explains why the Bayes factor is not always the ratio of two likelihood values evaluated at best-fitting parameters. It compares the probability of the observed data under the complete models, including their prior distributions. We must specify prior model probabilities as well as parameter priors before the posterior model probabilities can be calculated. **Board equations from the notes.**

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = \frac{\Pr(H_1)}{\Pr(H_0)} BF_{10},$$

$$\boxed{BF_{10} = \frac{p(y \mid H_1)}{p(y \mid H_0)}.}$$

Detailed solution and reasoning.
For $y = 10$ heads in $n = 12$, the likelihood under the point null is

$$p(y \mid H_0) = \binom{12}{10}(0.5)^{10}(0.5)^2 = \binom{12}{10}0.5^{12}.$$

Under H_1, $p \sim \text{Beta}(2,2)$, so integrate over the prior:

$$p(y \mid H_1) = \binom{12}{10} \int_0^1 p^{10}(1-p)^2 \frac{p(1-p)}{B(2,2)} dp = \binom{12}{10} \frac{B(12,4)}{B(2,2)}.$$

The Bayes factor is

$$BF_{10} = \frac{B(12,4)}{B(2,2)\,0.5^{12}} \approx 4.50110.$$

The binomial coefficient cancels. This is a marginal likelihood, averaging likelihood over the entire Beta prior. Evaluating the alternative at its MLE $\hat{p} = 10/12$ gives the maximum likelihood, not the prior-weighted average and not a Bayes factor. The prior must be specified before the alternative's evidence is defined.

Problem 65

Original question (verbatim).
A measurement Y is normal with known variance 1. Under H_0, the mean is fixed at zero; under H_1, the mean has prior $N(0, \tau^2)$. For an observed sample mean $\bar{y}$ from n measurements, derive the two marginal likelihoods and the Bayes factor. Identify the integral over the alternative parameter and explain which model quantities must be specified before posterior model probabilities can be reported.

Required theory (verbatim from the supplied lecture manual).
In Bayesian hypothesis testing, a hypothesis is treated as a model with a prior probability. A simple hypothesis specifies the parameter completely. A composite hypothesis leaves some parameter values unknown and assigns a prior distribution over them. For a simple hypothesis, the probability of the data under that hypothesis is obtained directly from the sampling model. For a composite hypothesis, we average the likelihood across the parameter prior to obtain a marginal likelihood. This distinction explains why the Bayes factor is not always the ratio of two likelihood values evaluated at best-fitting parameters. It compares the probability of the observed data under the complete models, including their prior distributions. We must specify prior model probabilities as well as parameter priors before the posterior model probabilities can be calculated. **Board equations from the notes.**

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = \frac{\Pr(H_1)}{\Pr(H_0)} BF_{10},$$

$$\boxed{BF_{10} = \frac{p(y \mid H_1)}{p(y \mid H_0)}.}$$

Detailed solution and reasoning.
The sample mean has sampling distribution $\bar{Y} \mid \theta \sim N(\theta, 1/n)$. Under H_0, its marginal density is $N(\bar{y}; 0, 1/n)$. Under H_1, with $\theta \sim N(0, \tau^2)$,

$$p(\bar{y} \mid H_1) = \int N(\bar{y}; \theta, 1/n) N(\theta; 0, \tau^2) d\theta = N(\bar{y}; 0, 1/n + \tau^2).$$

Residual information about within-sample variation is common to both models and cancels from

the ratio. Hence

$$BF_{10} = \frac{\phi(\bar{y}; 0, 1/n + \tau^2)}{\phi(\bar{y}; 0, 1/n)} = \sqrt{\frac{1/n}{1/n + \tau^2}} \exp\left[\frac{\bar{y}^2}{2}\left\{n - \frac{1}{1/n + \tau^2}\right\}\right].$$

To report posterior model probabilities, one must specify τ^2 and prior model probabilities, as well as the sampling variance and the hypotheses. The integral under H_1 is essential.

Problem 66

Original question (verbatim).
Two candidate models describe the same count data. Model A fixes the event rate at a specified value. Model B assigns a Gamma prior to an unknown event rate. Formulate the marginal likelihood under each model and derive the Bayes factor symbolically. Explain how the prior distribution under Model B affects its evidence even when Model B can fit the observed count better at its maximum-likelihood estimate.

Required theory (verbatim from the supplied lecture manual).
In Bayesian hypothesis testing, a hypothesis is treated as a model with a prior probability. A simple hypothesis specifies the parameter completely. A composite hypothesis leaves some parameter values unknown and assigns a prior distribution over them. For a simple hypothesis, the probability of the data under that hypothesis is obtained directly from the sampling model. For a composite hypothesis, we average the likelihood across the parameter prior to obtain a marginal likelihood. This distinction explains why the Bayes factor is not always the ratio of two likelihood values evaluated at best-fitting parameters. It compares the probability of the observed data under the complete models, including their prior distributions. We must specify prior model probabilities as well as parameter priors before the posterior model probabilities can be calculated. **Board equations from the notes.**

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = \frac{\Pr(H_1)}{\Pr(H_0)} BF_{10},$$

$$\boxed{BF_{10} = \frac{p(y \mid H_1)}{p(y \mid H_0)}.}$$

Detailed solution and reasoning.
Let total count be y over known exposure e. Under Model A, the rate is fixed at λ_0, so

$$p(y \mid M_A) = \frac{e^{-e\lambda_0}(e\lambda_0)^y}{y!}.$$

Under Model B, $\lambda \sim \text{Gamma}(a, b)$ and

$$p(y \mid M_B) = \int \frac{e^{-e\lambda}(e\lambda)^y}{y!} \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} d\lambda = \frac{\Gamma(a+y)}{\Gamma(a) y!} \left(\frac{b}{b+e}\right)^a \left(\frac{e}{b+e}\right)^y.$$

Thus $BF_{BA} = p(y \mid M_B)/p(y \mid M_A)$.
Model B may attain a higher likelihood at its MLE, but marginal likelihood averages over all rates according to the Gamma prior. If the prior spreads mass over many rates that predict the observed count poorly, that dilution lowers its integrated evidence. Prior location and concentration therefore matter to model evidence.

4.2 Interpreting a Bayes factor

Problem 67

Original question (verbatim).
A study reports $BF_{10} = 12$, but the prior probability of H_1 is only 0.01. Calculate the posterior probability of H_1. A second analyst incorrectly reports 12% as the posterior probability. Explain the error and determine what Bayes factor would be required to make the posterior probability of H_1 exceed 0.5 under the same prior model probability.

Required theory (verbatim from the supplied lecture manual).
The Bayes factor for model one against model zero is the marginal likelihood under model one divided by the marginal likelihood under model zero. It tells us how the observed data change the odds between the models. Posterior odds equal prior odds multiplied by the Bayes factor.
A Bayes factor greater than one favours the numerator model relative to the denominator model, while a value less than one favours the denominator model. But the strength of evidence is conditional on the models and their priors. A Bayes factor does not by itself give the posterior probability of a hypothesis. To obtain that probability, we also need the prior model odds.

Detailed solution and reasoning.
Bayes' rule in odds form gives

$$O_{10|y} = BF_{10}O_{10} = 12 \times \frac{0.01}{0.99} = 0.121212.$$

Therefore

$$\Pr(H_1 \mid y) = \frac{0.121212}{1 + 0.121212} = 0.108108.$$

The value 12 is a likelihood-evidence ratio, not a posterior percentage. It multiplies prior odds; it is not itself a posterior probability.
For posterior probability above 0.5, posterior odds must exceed one. Thus $BF_{10} > 1/O_{10} = 0.99/0.01 = 99$. A Bayes factor of exactly 99 gives posterior probability exactly 0.5; a larger value is required to exceed 0.5.

Problem 68

Original question (verbatim).
Three independent research teams collect evidence y_1, y_2, y_3 under the same two fully specified hypotheses. Show when the combined Bayes factor equals the product of the three study-specific Bayes factors. Construct a situation in which this multiplication fails because the studies share an unknown nuisance quantity or their data are conditionally dependent. Explain what joint model is needed in that case.

Required theory (verbatim from the supplied lecture manual).
The Bayes factor for model one against model zero is the marginal likelihood under model one divided by the marginal likelihood under model zero. It tells us how the observed data change the odds between the models. Posterior odds equal prior odds multiplied by the Bayes factor.
A Bayes factor greater than one favours the numerator model relative to the denominator model, while a value less than one favours the denominator model. But the strength of evidence is conditional on the models and their priors. A Bayes factor does not by itself give the posterior

probability of a hypothesis. To obtain that probability, we also need the prior model odds.

Detailed solution and reasoning.
If the studies are conditionally independent given each fully specified hypothesis, then

$$p(y_1, y_2, y_3 \mid H_k) = \prod_{r=1}^{3} p(y_r \mid H_k), \quad k = 0, 1.$$

Consequently,

$$BF_{10}(y_1, y_2, y_3) = \frac{\prod_r p(y_r \mid H_1)}{\prod_r p(y_r \mid H_0)} = \prod_r BF_{10}(y_r).$$

For composite hypotheses with a shared unknown parameter θ, the correct alternative marginal is

$$p(y_1, y_2, y_3 \mid H_1) = \int \left\{ \prod_r p(y_r \mid \theta, H_1) \right\} \pi(\theta \mid H_1) d\theta,$$

which is generally not equal to $\prod_r \int p(y_r \mid \theta, H_1)\pi(\theta)d\theta$. The shared parameter couples studies after integration. Likewise, dependent data do not factor. Use one joint model with the common nuisance quantity and dependence structure; multiplying separate marginal Bayes factors would double-count or mis-handle shared uncertainty.

Problem 69

Original question (verbatim).
A Bayes factor of 0.08 is reported for H_1 against H_0. One report calls this strong evidence for H_1, while another calls it evidence for H_0. Resolve the disagreement by fixing the numerator and denominator convention, convert the factor to the reciprocal comparison, and show how prior odds determine the final posterior odds.

Required theory (verbatim from the supplied lecture manual).
The Bayes factor for model one against model zero is the marginal likelihood under model one divided by the marginal likelihood under model zero. It tells us how the observed data change the odds between the models. Posterior odds equal prior odds multiplied by the Bayes factor.
A Bayes factor greater than one favours the numerator model relative to the denominator model, while a value less than one favours the denominator model. But the strength of evidence is conditional on the models and their priors. A Bayes factor does not by itself give the posterior probability of a hypothesis. To obtain that probability, we also need the prior model odds.

Detailed solution and reasoning.
The convention must be stated: $BF_{10} = p(y \mid H_1)/p(y \mid H_0) = 0.08$. Since it is below one, the data favour H_0 relative to H_1. The reciprocal comparison is

$$BF_{01} = 1/0.08 = 12.5,$$

which is evidence for H_0 against H_1.
Posterior odds still depend on prior odds:

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = 0.08 \frac{\Pr(H_1)}{\Pr(H_0)}.$$

A Bayes factor is not an absolute posterior statement. Both reports are resolved by specifying

numerator and denominator: 0.08 favours the denominator model H_0, equivalently 12.5 favours H_0 over H_1.

4.3 Point null hypotheses and continuous alternatives

Problem 70

Original question (verbatim).
Let $Y_1, \ldots, Y_n \mid \theta \sim N(\theta, \sigma^2)$ with known σ^2. Compare $H_0 : \theta = 0$ against $H_1 : \theta \sim N(0, \tau^2)$. Derive the Bayes factor in terms of $n, \bar{y}, \sigma^2, \tau^2$. Analyse its behaviour as $\tau^2 \to 0$ and $\tau^2 \to \infty$, and explain why a very diffuse alternative can receive a low marginal likelihood even when it contains values that fit the data extremely well.

Required theory (verbatim from the supplied lecture manual).
A point null hypothesis says that a parameter has exactly one specified value, such as theta equals zero. A continuous alternative assigns probability over a range of values. Under the null, we evaluate the likelihood at the fixed value. Under the alternative, we integrate the likelihood over the prior distribution for theta.
This integral is the key difference. The alternative is rewarded when a substantial portion of its prior distribution predicts data like those observed. A very flexible alternative may fit the data extremely well at one narrow point but assign most prior mass to values that predict the data poorly. Its marginal likelihood can therefore be smaller than expected. This is one reason Bayes factors naturally penalise diffuse alternatives.

Detailed solution and reasoning.
The sufficient statistic is $\bar{Y}$, with $\bar{Y} \mid \theta \sim N(\theta, v)$, $v = \sigma^2/n$. Under H_0, $\bar{Y} \sim N(0, v)$. Under H_1, integrating $\theta \sim N(0, \tau^2)$ gives $\bar{Y} \sim N(0, v + \tau^2)$. Therefore

$$BF_{10} = \sqrt{\frac{v}{v+\tau^2}} \exp\left\{ \frac{\bar{y}^2}{2} \left(\frac{1}{v} - \frac{1}{v+\tau^2} \right) \right\}.$$

As $\tau^2 \to 0$, the alternative collapses to the null and $BF_{10} \to 1$. As $\tau^2 \to \infty$, the leading factor tends to zero while the exponential factor tends to a finite value for fixed $\bar{y}$, so $BF_{10} \to 0$.
A diffuse alternative spreads prior mass across a huge range of effects. Although it contains values that fit the observed mean well, only a small fraction of its prior mass lies near the observed value; averaging over the whole prior can produce low marginal likelihood. This is the prior predictive dilution behind the Jeffreys–Lindley phenomenon.

Problem 71

Original question (verbatim).
For binomial data $Y \sim \text{Binomial}(n, p)$, compare the point null $H_0 : p = 0.4$ with a continuous alternative $H_1 : p \sim \text{Beta}(a, b)$. Derive the Bayes factor after observing y successes. Explain why the null receives positive prior probability only because it is represented as a separate model, and why a continuous Beta prior alone assigns probability zero to the event $p = 0.4$.

Required theory (verbatim from the supplied lecture manual).
A point null hypothesis says that a parameter has exactly one specified value, such as theta equals zero. A continuous alternative assigns probability over a range of values. Under the null, we evaluate the likelihood at the fixed value. Under the alternative, we integrate the likelihood over the prior distribution for theta.
This integral is the key difference. The alternative is rewarded when a substantial portion of its prior distribution predicts data like those observed. A very flexible alternative may fit the data extremely well at one narrow point but assign most prior mass to values that predict the data poorly. Its marginal likelihood can therefore be smaller than expected. This is one reason Bayes factors naturally penalise diffuse alternatives.

Detailed solution and reasoning.
Under $H_0 : p = 0.4$, the likelihood is

$$p(y \mid H_0) = \binom{n}{y} 0.4^y 0.6^{n-y}.$$

Under H_1 with $p \sim \text{Beta}(a, b)$,

$$p(y \mid H_1) = \binom{n}{y} \frac{B(a+y, b+n-y)}{B(a,b)}.$$

Thus

$$BF_{10} = \frac{B(a+y, b+n-y)}{B(a,b)\, 0.4^y 0.6^{n-y}}.$$

The binomial coefficient cancels.
The null has positive prior model probability because it is assigned a separate discrete model indicator (a point mass at $p = 0.4$). A continuous Beta distribution assigns probability zero to any single point, including $p = 0.4$. A test of the point null therefore requires an explicit point-null model plus a continuous alternative.

Problem 72

Original question (verbatim).
A very small p-value is obtained for testing $H_0 : \theta = 0$, yet the Bayes factor favours the null under a diffuse alternative. Construct a coherent normal-model explanation of how this can occur. Identify the role of sample size, effect size, prior scale, and the point mass at the null. Explain why a p-value is not the posterior probability of the null and why the two procedures answer different questions.

Required theory (verbatim from the supplied lecture manual).
A point null hypothesis says that a parameter has exactly one specified value, such as theta equals zero. A continuous alternative assigns probability over a range of values. Under the null, we evaluate the likelihood at the fixed value. Under the alternative, we integrate the likelihood over the prior distribution for theta.
This integral is the key difference. The alternative is rewarded when a substantial portion of its prior distribution predicts data like those observed. A very flexible alternative may fit the data extremely well at one narrow point but assign most prior mass to values that predict the data poorly. Its marginal likelihood can therefore be smaller than expected. This is one reason Bayes factors naturally penalise diffuse alternatives.

Detailed solution and reasoning.
A normal example makes the phenomenon clear. Let $\bar{Y} \mid \theta \sim N(\theta, \sigma^2/n)$, compare $H_0 : \theta = 0$ to $H_1 : \theta \sim N(0, \tau^2)$. A small frequentist p-value measures how extreme the observed statistic is under H_0; it does not assign probability to H_0. The Bayes factor compares the null density at the observation with the alternative's prior predictive density.
When τ is very large, the alternative spreads mass across a broad range of effects. Even if the sample provides a small p-value, the observed effect may occupy a low-density region of that diffuse prior predictive distribution, lowering BF_{10}. The sample size changes the sampling variance; effect size changes the observed mean; the prior scale changes how concentrated the alternative is near plausible effects. A point mass at zero is needed for H_0 to have positive prior probability. The two procedures answer different questions and need not agree.

4.4 Sensitivity, improper priors, and limitations

Problem 73

Original question (verbatim).
Suppose a model comparison uses an improper prior $\pi(\theta) = c\, g(\theta)$, where the arbitrary constant c is unknown. Show how the constant enters the marginal likelihood and explain why the Bayes factor may depend on an arbitrary choice. Give an example of why posterior propriety for parameter estimation does not automatically make an improper prior valid for model comparison. State what must be done before a Bayes factor can be interpreted.

Required theory (verbatim from the supplied lecture manual).
Bayes factors require marginal likelihoods, and marginal likelihoods require proper normalised priors or a carefully constructed alternative. With an improper prior, the arbitrary normalising constant does not cancel in general when comparing models. Consequently, the ordinary Bayes factor may be undefined.
Bayes factors can also be highly sensitive to prior scales under composite alternatives. Before interpreting a Bayes factor, report the model priors and the parameter priors, and consider how the answer changes under reasonable alternatives. A large Bayes factor is not a universal statement independent of modelling choices. It is evidence in favour of one fully specified model relative to another. **Board equations from the notes.**

$$BF_{10}(y_1, y_2) = BF_{10}(y_1) BF_{10}(y_2),$$

Detailed solution and reasoning.
If an improper prior is $\pi(\theta) = cg(\theta)$, its marginal likelihood is

$$p(y \mid M) = \int p(y \mid \theta, M) cg(\theta) d\theta = c \int p(y \mid \theta, M) g(\theta) d\theta.$$

The arbitrary normalising constant c therefore appears in the evidence. In a Bayes factor comparing two models with unrelated improper priors, the arbitrary constants generally do not cancel, so the numerical factor is undefined up to an arbitrary multiplier.
A posterior can be proper because the likelihood makes $p(\theta \mid y)$ integrable, even though the prior itself is improper. That does not determine the prior's arbitrary scale for model evidence. For example, a flat prior on a location parameter may yield a proper normal posterior within a model but leave its marginal likelihood dependent on the unspecified constant.
Before interpreting a Bayes factor, use proper priors for model-specific parameters or a principled

construction that makes the arbitrary constants cancel (with conditions justified). The prior predictive model must be well-defined.

Problem 74

Original question (verbatim).
For a composite alternative, calculate or analyse the Bayes factor under a family of prior scales $\tau \in \{0.1, 1, 10, 100\}$. Design a sensitivity analysis that distinguishes changes caused by the data from changes caused by the prior. Explain the role of prior predictive mass near the observed effect and why a single reported Bayes factor can create a misleading impression of prior-independent evidence.

Required theory (verbatim from the supplied lecture manual).
Bayes factors require marginal likelihoods, and marginal likelihoods require proper normalised priors or a carefully constructed alternative. With an improper prior, the arbitrary normalising constant does not cancel in general when comparing models. Consequently, the ordinary Bayes factor may be undefined.
Bayes factors can also be highly sensitive to prior scales under composite alternatives. Before interpreting a Bayes factor, report the model priors and the parameter priors, and consider how the answer changes under reasonable alternatives. A large Bayes factor is not a universal statement independent of modelling choices. It is evidence in favour of one fully specified model relative to another. **Board equations from the notes.**

$$BF_{10}(y_1, y_2) = BF_{10}(y_1)BF_{10}(y_2),$$

Detailed solution and reasoning.
For each prior scale $\tau \in \{0.1, 1, 10, 100\}$, compute

$$p(y \mid H_1, \tau) = \int p(y \mid \theta, H_1)\pi(\theta \mid \tau, H_1)d\theta, \quad BF_{10}(\tau) = \frac{p(y \mid H_1, \tau)}{p(y \mid H_0)}.$$

Hold the data and null model fixed. Alongside each factor, report prior predictive intervals or density near the observed effect, so the reader sees how much prior mass was assigned to values capable of generating the data.
The four scales represent different assumptions: a very narrow prior may exclude effects larger than 0.1, a moderate prior may place mass near the observed effect, and very diffuse scales distribute mass over many values that predict the data poorly. A changing Bayes factor across this grid is a prior-sensitivity result, not a change in the observed evidence itself. Report the range and explain which prior scales are scientifically credible rather than presenting one factor as prior-free.

Problem 75

Original question (verbatim).
Two independent datasets are analysed sequentially. A colleague claims that $BF_{10}(y_1, y_2) = BF_{10}(y_1)BF_{10}(y_2)$ always holds. State the conditions under which the identity is valid, derive the identity from joint marginal likelihoods, and give a counterexample involving a shared latent parameter or dependence between batches. Explain how the correct joint analysis avoids double-counting evidence.

Required theory (verbatim from the supplied lecture manual).
Bayes factors require marginal likelihoods, and marginal likelihoods require proper normalised priors or a carefully constructed alternative. With an improper prior, the arbitrary normalising constant does not cancel in general when comparing models. Consequently, the ordinary Bayes factor may be undefined.
Bayes factors can also be highly sensitive to prior scales under composite alternatives. Before interpreting a Bayes factor, report the model priors and the parameter priors, and consider how the answer changes under reasonable alternatives. A large Bayes factor is not a universal statement independent of modelling choices. It is evidence in favour of one fully specified model relative to another. **Board equations from the notes.**

$$BF_{10}(y_1, y_2) = BF_{10}(y_1)BF_{10}(y_2),$$

Detailed solution and reasoning.
The product identity is valid when each study is conditionally independent under each hypothesis and the hypothesis-specific predictive distributions factorise:

$$p(y_1, y_2 \mid H_k) = p(y_1 \mid H_k)p(y_2 \mid H_k), \quad k = 0, 1.$$

Then

$$BF_{10}(y_1, y_2) = \frac{p(y_1 \mid H_1)p(y_2 \mid H_1)}{p(y_1 \mid H_0)p(y_2 \mid H_0)} = BF_{10}(y_1)BF_{10}(y_2).$$

For shared unknown θ under H_1, however,

$$p(y_1, y_2 \mid H_1) = \int p(y_1 \mid \theta)p(y_2 \mid \theta)\pi(\theta)d\theta,$$

which is generally not the product of the two separately integrated marginal likelihoods. Conditional independence given θ does not imply marginal independence after θ is integrated out. A joint model retains the shared parameter and integrates it once, preventing double-counting.

Chapter 5

Posterior simulation: Gibbs sampling, Metropolis–Hastings, and diagnostics

5.1 Why MCMC is required

Problem 76

Original question (verbatim).
A logistic regression model has several coefficients with correlated posterior dependence and no closed-form normalising constant. Specify the posterior target and explain why direct sampling and exact integration may be impractical. Describe how MCMC draws can approximate a posterior mean, a tail probability, and a posterior predictive quantity. State the conditions that make the simulation output meaningful and explain why a long run does not by itself establish model correctness or convergence.

Required theory (verbatim from the supplied lecture manual).
In many Bayesian models, the posterior is known only up to a proportionality constant, or it is too complicated to integrate directly. Markov chain Monte Carlo allows us to generate dependent draws whose long-run distribution is the posterior target. We can then approximate expectations, quantiles, probabilities, and predictions using those draws.
We must remember what MCMC does and does not do. It does not make the model correct. It does not automatically guarantee that the algorithm has explored the posterior. It gives us a computational route to posterior summaries, provided the chain has reached the relevant stationary distribution and mixes adequately. That is why convergence diagnostics are part of the statistical analysis, not optional decoration. **Board equations from the notes.**

$$p(\theta^{(t)} \mid \theta^{(t-1)}, \dots, \theta^{(1)}) = p(\theta^{(t)} \mid \theta^{(t-1)}).$$

Detailed solution and reasoning.
For logistic regression with parameter vector β, data y and design matrix X, the posterior target is

$$\pi(\beta \mid y) \propto p(y \mid \beta, X)\pi(\beta).$$

For binary outcomes, $p(y \mid \beta, X) = \prod_i \text{Bernoulli}(y_i; \text{logit}^{-1}(x_i^T \beta))$. Correlated coefficients and nonlinear normalising integrals often prevent closed-form posterior summaries or direct independent sampling.
Given post-warm-up MCMC draws $\beta^{(1)}, \dots, \beta^{(M)}$, approximate a posterior mean by $M^{-1} \sum_m \beta^{(m)}$, a tail probability by $M^{-1} \sum_m 1\{g(\beta^{(m)}) > c\}$, and a predictive probability by simulating future

data from each draw and averaging. These are Monte Carlo approximations to posterior integrals. Meaningful output requires a correct sampler targeting the intended posterior, adequate exploration/mixing, multiple-chain and trace diagnostics, sufficient effective sample size, and acceptable Monte Carlo error. A long run does not prove convergence, and convergence does not prove the likelihood or prior is scientifically adequate.

Problem 77

Original question (verbatim).
Consider a posterior density known only up to proportionality, $\pi(\theta \mid y) \propto q(\theta)$, where direct sampling is unavailable but pointwise evaluation of q is possible. Explain why this is a natural setting for MCMC. Compare the computational target with numerical quadrature and independent Monte Carlo, and identify which features of a Markov chain are deliberately used to produce draws from the target distribution.

Required theory (verbatim from the supplied lecture manual).
In many Bayesian models, the posterior is known only up to a proportionality constant, or it is too complicated to integrate directly. Markov chain Monte Carlo allows us to generate dependent draws whose long-run distribution is the posterior target. We can then approximate expectations, quantiles, probabilities, and predictions using those draws.
We must remember what MCMC does and does not do. It does not make the model correct. It does not automatically guarantee that the algorithm has explored the posterior. It gives us a computational route to posterior summaries, provided the chain has reached the relevant stationary distribution and mixes adequately. That is why convergence diagnostics are part of the statistical analysis, not optional decoration. **Board equations from the notes.**

$$p(\theta^{(t)} \mid \theta^{(t-1)}, \ldots, \theta^{(1)}) = p(\theta^{(t)} \mid \theta^{(t-1)}).$$

Detailed solution and reasoning.
MCMC is useful when $\pi(\theta \mid y) \propto q(\theta)$ is known up to a constant and $q(\theta)$ can be evaluated, but direct independent draws or exact integration are unavailable. The unknown normalising constant cancels from Metropolis–Hastings ratios, which is a key advantage.
Numerical quadrature can be accurate in low dimensions but becomes expensive as dimension grows. Independent Monte Carlo requires a tractable proposal distribution from which independent draws can be generated and may have high variance if it poorly matches the target. MCMC deliberately constructs a Markov chain whose stationary distribution is the target; it uses local or conditional transitions so the unknown constant cancels. The draws are dependent, so Monte Carlo uncertainty must be assessed using effective sample size and Monte Carlo standard error.

Problem 78

Original question (verbatim).
A posterior has two separated modes and a sampler is started close to the first mode. After one million iterations, the trace appears stable and the posterior mean is reported. Explain why the run may still be inadequate. Design a multiple-chain strategy with dispersed initial values and propose diagnostics that would reveal failure to explore the second mode. Discuss how a failure of exploration affects posterior means and tail probabilities.

Required theory (verbatim from the supplied lecture manual).
In many Bayesian models, the posterior is known only up to a proportionality constant, or it is too complicated to integrate directly. Markov chain Monte Carlo allows us to generate dependent draws whose long-run distribution is the posterior target. We can then approximate expectations, quantiles, probabilities, and predictions using those draws.
We must remember what MCMC does and does not do. It does not make the model correct. It does not automatically guarantee that the algorithm has explored the posterior. It gives us a computational route to posterior summaries, provided the chain has reached the relevant stationary distribution and mixes adequately. That is why convergence diagnostics are part of the statistical analysis, not optional decoration. **Board equations from the notes.**

$$p(\theta^{(t)} \mid \theta^{(t-1)}, \ldots, \theta^{(1)}) = p(\theta^{(t)} \mid \theta^{(t-1)}).$$

Detailed solution and reasoning.
A stable trace near one mode does not establish global convergence. A chain started near the first mode may never cross a low-density valley to a second mode, even after one million iterations. Its sample mean then estimates the conditional mean within the visited mode, not the full posterior mean; tail probabilities and mixture weights can be severely wrong.
Run several chains from widely dispersed initial values, including starts near each plausible mode. Compare trace plots, rank or marginal density plots, mode occupancy and transitions, autocorrelation, and scale-reduction statistics. Where possible use reparameterisation or a sampler suited to multimodal targets and compare independent runs. Similar within-mode traces are not enough: chains should explore the same global posterior regions in compatible proportions. If they do not, withhold pooled posterior summaries.

5.2 Gibbs sampling

Problem 79

Original question (verbatim).
Let (X, Y) have a bivariate normal target distribution with zero means, unit variances, and correlation ρ. Derive the two full conditional distributions. Starting from a specified state $(x^{(0)}, y^{(0)})$, write one complete Gibbs sweep and explain the order in which the newly updated and previous values are used. Analyse qualitatively how the chain's mixing changes as $|\rho|$ approaches one.

Required theory (verbatim from the supplied lecture manual).
Gibbs sampling is especially natural when we can sample from full conditional distributions. Suppose the unknown parameter consists of several blocks. We hold all blocks except the first fixed and draw the first block from its conditional posterior. Then we update the second block conditional on the newly updated first block and the remaining current values, and continue through all blocks.
After one complete sweep, we have a new joint state. Repeating this process creates a Markov chain with the desired joint posterior as its stationary distribution under the usual conditions. The essential exam task is to derive the full conditional distributions correctly and state the update order. Remember that the chain draws are dependent: the next draw is based on the current state. This dependence is expected and must be considered when assessing effective sample size. **Board equations from the notes.**

$$\theta_1^{(t)} \sim p(\theta_1 \mid \theta_2^{(t-1)}, \ldots, \theta_d^{(t-1)}, y),$$
$$\theta_2^{(t)} \sim p(\theta_2 \mid \theta_1^{(t)}, \theta_3^{(t-1)}, \ldots, \theta_d^{(t-1)}, y),$$

Detailed solution and reasoning.
For a standardised bivariate normal target with correlation ρ,

$$X \mid Y = y \sim N(\rho y, 1-\rho^2), \qquad Y \mid X = x \sim N(\rho x, 1-\rho^2).$$

A systematic Gibbs sweep from $(x^{(0)}, y^{(0)})$ is

$$x^{(1)} \sim N(\rho y^{(0)}, 1-\rho^2), \qquad y^{(1)} \sim N(\rho x^{(1)}, 1-\rho^2).$$

The second update uses the newly drawn $x^{(1)}$, not $x^{(0)}$. The pair $(x^{(1)}, y^{(1)})$ is the next chain state. As $|\rho|$ approaches one, each conditional variance $1-\rho^2$ becomes small and the chain tends to move along a narrow, highly correlated ridge in small steps. Autocorrelation rises and mixing slows. At $\rho = 0$ the coordinates are independent and one complete sweep produces independent draws from both marginals.

Problem 80

Original question (verbatim).
Suppose $Y_i \mid \theta_i \sim N(\theta_i, s_i^2)$, $\theta_i \mid \mu, \tau^2 \sim N(\mu, \tau^2)$, and $\mu \sim N(m_0, v_0)$, with τ^2 known. Derive the full conditional distributions of each θ_i and of μ. Write a Gibbs algorithm for one iteration and explain why each update conditions on the most recently available values of the other blocks.

Required theory (verbatim from the supplied lecture manual).
Gibbs sampling is especially natural when we can sample from full conditional distributions. Suppose the unknown parameter consists of several blocks. We hold all blocks except the first fixed and draw the first block from its conditional posterior. Then we update the second block conditional on the newly updated first block and the remaining current values, and continue through all blocks.
After one complete sweep, we have a new joint state. Repeating this process creates a Markov chain with the desired joint posterior as its stationary distribution under the usual conditions. The essential exam task is to derive the full conditional distributions correctly and state the update order. Remember that the chain draws are dependent: the next draw is based on the current state. This dependence is expected and must be considered when assessing effective sample size. **Board equations from the notes.**

$$\theta_1^{(t)} \sim p(\theta_1 \mid \theta_2^{(t-1)}, \ldots, \theta_d^{(t-1)}, y),$$
$$\theta_2^{(t)} \sim p(\theta_2 \mid \theta_1^{(t)}, \theta_3^{(t-1)}, \ldots, \theta_d^{(t-1)}, y),$$

Detailed solution and reasoning.
For each observation,

$$\theta_i \mid y_i, \mu, \tau^2 \sim N(m_i, v_i), \quad v_i = \left(s_i^{-2} + \tau^{-2}\right)^{-1}, \quad m_i = v_i \left(\frac{y_i}{s_i^2} + \frac{\mu}{\tau^2}\right).$$

For the population mean,

$$\mu \mid \theta_{1:n} \sim N(m_\mu, v_\mu), \quad v_\mu = \left(v_0^{-1} + n\tau^{-2}\right)^{-1}, \quad m_\mu = v_\mu \left(\frac{m_0}{v_0} + \frac{\sum_i \theta_i}{\tau^2}\right).$$

One Gibbs iteration: (1) for $i = 1, \ldots, n$, draw θ_i from its full conditional, using the current μ and the observed y_i; (2) draw μ from its full conditional using the newly updated vector $\boldsymbol{\theta}$. If τ^2 were unknown, it would have another full conditional/prior update. Each block update conditions on the most recently available values of the other blocks, preserving the intended joint posterior as the stationary target.

Problem 81

Original question (verbatim).
A proposed Gibbs sampler uses the correct full conditional for θ_1 but accidentally conditions on the old value of θ_1 when updating θ_2, rather than the newly drawn value required by the specified scan. Analyse whether this changes the transition mechanism and what must be checked to establish the intended joint posterior as stationary. Explain why having individually recognisable conditional distributions is not, by itself, a complete proof that a proposed implementation is correct.

Required theory (verbatim from the supplied lecture manual).
Gibbs sampling is especially natural when we can sample from full conditional distributions. Suppose the unknown parameter consists of several blocks. We hold all blocks except the first fixed and draw the first block from its conditional posterior. Then we update the second block conditional on the newly updated first block and the remaining current values, and continue through all blocks.
After one complete sweep, we have a new joint state. Repeating this process creates a Markov chain with the desired joint posterior as its stationary distribution under the usual conditions. The essential exam task is to derive the full conditional distributions correctly and state the update order. Remember that the chain draws are dependent: the next draw is based on the current state. This dependence is expected and must be considered when assessing effective sample size. **Board equations from the notes.**

$$\theta_1^{(t)} \sim p(\theta_1 \mid \theta_2^{(t-1)}, \ldots, \theta_d^{(t-1)}, y),$$
$$\theta_2^{(t)} \sim p(\theta_2 \mid \theta_1^{(t)}, \theta_3^{(t-1)}, \ldots, \theta_d^{(t-1)}, y),$$

Detailed solution and reasoning.
A Gibbs sampler is a composition of transition kernels that each update one block from its full conditional. In a systematic scan, when a block is updated, it should condition on the latest values of all other blocks available in that sweep. Using an old value after a related block has already been updated defines a different transition mechanism from the stated scan.
The existence of familiar-looking one-dimensional conditionals is not enough to prove the implementation targets the joint posterior. Check whether each update is genuinely the full conditional of the intended joint density and whether the composition leaves that joint density invariant. For a two-block target, alternating exact conditionals in sequence has the desired invariant distribution; an ad hoc stale-value or parallel update can fail unless a valid transition-kernel argument establishes invariance. Test code on a target with known moments and compare simulated summaries with analytic results.

5.3 Metropolis-Hastings

Problem 82

Original question (verbatim).
A Metropolis-Hastings chain targets an unnormalised density $q(\theta)$. From current state $\theta = 1$, it proposes $\theta^* = 2$, with $q(1) = 0.10$, $q(2) = 0.06$, forward proposal density $g(2 \mid 1) = 0.4$, and reverse proposal density $g(1 \mid 2) = 0.2$. Calculate the acceptance probability. Repeat the calculation for a symmetric proposal and explain why omitting the proposal ratio in the asymmetric case produces the wrong stationary distribution.

Required theory (verbatim from the supplied lecture manual).
Metropolis–Hastings is useful when direct sampling from the target posterior is difficult but we can evaluate its density up to a constant. From the current state, propose a candidate using a proposal density. Then calculate an acceptance probability that compares the target density at the candidate and current state, while correcting for asymmetry in the proposal mechanism.
The correction is the reverse proposal probability divided by the forward proposal probability. If the proposal is symmetric, those terms cancel. We accept the candidate with the calculated probability; otherwise we remain at the current state. A common mistake is to omit the proposal ratio for an asymmetric proposal. Another mistake is to think that a candidate with lower posterior density must always be rejected. It may still be accepted, which allows the chain to explore the full target distribution. **Board equations from the notes.**

$$\boxed{\alpha(\theta, \theta^*) = \min\left\{1, \frac{q(\theta^*)g(\theta \mid \theta^*)}{q(\theta)g(\theta^* \mid \theta)}\right\}.}$$

$$\alpha = \min\left\{1, \frac{q(\theta^*)}{q(\theta)}\right\}.$$

Detailed solution and reasoning.
The Metropolis–Hastings acceptance probability is

$$\alpha(1, 2) = \min\left\{1, \frac{q(2)g(1 \mid 2)}{q(1)g(2 \mid 1)}\right\} = \min\left\{1, \frac{0.06(0.2)}{0.10(0.4)}\right\} = 0.03.$$

The proposal is accepted with probability 0.03; otherwise the chain stays at 1.
For a symmetric proposal, $g(1 \mid 2) = g(2 \mid 1)$, so the proposal terms cancel and

$$\alpha = \min(1, 0.06/0.10) = 0.6.$$

In the asymmetric case omitting the reverse/forward proposal ratio violates detailed balance in general and changes the stationary distribution. The Hastings correction compensates for unequal probabilities of proposing moves in opposite directions.

Problem 83

Original question (verbatim).
A random-walk Metropolis algorithm proposes a point with lower posterior density than the current point. A student insists that the proposal must be rejected. Use the acceptance rule to show why this is false. Explain the role of occasional downhill moves in exploring the target distribution,

and describe what happens to the chain state after rejection. Include the resulting effect on the sequence of dependent draws.

Required theory (verbatim from the supplied lecture manual).
Metropolis–Hastings is useful when direct sampling from the target posterior is difficult but we can evaluate its density up to a constant. From the current state, propose a candidate using a proposal density. Then calculate an acceptance probability that compares the target density at the candidate and current state, while correcting for asymmetry in the proposal mechanism.
The correction is the reverse proposal probability divided by the forward proposal probability. If the proposal is symmetric, those terms cancel. We accept the candidate with the calculated probability; otherwise we remain at the current state. A common mistake is to omit the proposal ratio for an asymmetric proposal. Another mistake is to think that a candidate with lower posterior density must always be rejected. It may still be accepted, which allows the chain to explore the full target distribution. **Board equations from the notes.**

$$\boxed{\alpha(\theta, \theta^*) = \min\left\{1, \frac{q(\theta^*)g(\theta \mid \theta^*)}{q(\theta)g(\theta^* \mid \theta)}\right\}.}$$

$$\alpha = \min\left\{1, \frac{q(\theta^*)}{q(\theta)}\right\}.$$

Detailed solution and reasoning.
For a proposed point θ^* with lower target density than current θ, a symmetric random-walk proposal accepts with probability

$$\alpha = \min\{1, q(\theta^*)/q(\theta)\},$$

which is positive whenever $q(\theta^*) > 0$. Therefore a lower-density proposal can still be accepted; it is not automatically rejected. Under asymmetric proposals the Hastings ratio also includes reverse and forward proposal densities.
Occasional downhill moves are essential: they allow the chain to leave a local high-density region and explore the full target while maintaining the correct stationary distribution. If rejected, the next chain state is the current state θ, so the stored sequence contains a repeated value. Such self-transitions are part of the chain and contribute to autocorrelation; they must not be removed when assessing sampling efficiency.

Problem 84

Original question (verbatim).
A Metropolis-Hastings algorithm has an acceptance rate of 0.02 with very large proposals. After reducing the proposal scale, its acceptance rate rises to 0.95, but the trace plot moves only slowly across the parameter space. Explain why neither extreme acceptance rate guarantees good mixing. Propose a tuning and diagnostic strategy that assesses movement, autocorrelation, and exploration of the target rather than optimising acceptance rate alone.

Required theory (verbatim from the supplied lecture manual).
Metropolis–Hastings is useful when direct sampling from the target posterior is difficult but we can evaluate its density up to a constant. From the current state, propose a candidate using a proposal density. Then calculate an acceptance probability that compares the target density at

the candidate and current state, while correcting for asymmetry in the proposal mechanism. The correction is the reverse proposal probability divided by the forward proposal probability. If the proposal is symmetric, those terms cancel. We accept the candidate with the calculated probability; otherwise we remain at the current state. A common mistake is to omit the proposal ratio for an asymmetric proposal. Another mistake is to think that a candidate with lower posterior density must always be rejected. It may still be accepted, which allows the chain to explore the full target distribution. **Board equations from the notes.**

$$\boxed{\alpha(\theta, \theta^*) = \min\left\{1, \frac{q(\theta^*)g(\theta \mid \theta^*)}{q(\theta)g(\theta^* \mid \theta)}\right\}.}$$

$$\alpha = \min\left\{1, \frac{q(\theta^*)}{q(\theta)}\right\}.$$

Detailed solution and reasoning.
An acceptance rate of 0.02 with very large random-walk steps suggests most proposals land in regions of very low target density. Reducing the step size to obtain 0.95 acceptance can make the steps so small that the chain explores the parameter space extremely slowly. Neither acceptance extreme guarantees good effective sample size.
Tune the proposal scale using pilot runs and assess trace plots, distance travelled, autocorrelation functions, effective sample sizes, and Monte Carlo errors for important posterior summaries. For multivariate targets, tune covariance structure or reparameterise when strong posterior correlation is present. The aim is efficient exploration of relevant posterior regions, not maximisation of acceptance rate as a stand-alone objective. Always retune only during warm-up/pilot phases, then run and diagnose the final sampler.

5.4 Convergence: what must be checked

Problem 85

Original question (verbatim).
Three chains are run from dispersed initial values. The trace plots show: chain 1 fluctuates around 0; chain 2 fluctuates around 4; chain 3 drifts slowly from 5 toward 2. The marginal density plots are visibly different. Diagnose the evidence and explain why pooling the draws would be misleading. Specify what additional graphical and numerical checks you would perform before deciding whether the chains have explored a common target distribution.

Required theory (verbatim from the supplied lecture manual).
A chain that looks busy is not necessarily converged. We want evidence that it has settled into the target distribution and is moving around that distribution effectively. Trace plots show the sequence over iterations. If there is a trend, long drift, or chains in different regions, that is a warning. Overlaid marginal density plots show whether separate chains are exploring similar distributions. Autocorrelation plots show how strongly nearby draws depend on each other.
Use several chains with dispersed starting points. If every chain begins at the same convenient point, we may fail to notice that the sampler has not explored another important region. Diagnostics should be applied to all important parameters and derived quantities, not only to one easy parameter. Convergence assessment is a collection of graphical and numerical checks rather than a single pass/fail test.

Detailed solution and reasoning.
The chains have not converged to a common stationary distribution: chain 1 remains near 0, chain 2 near 4, and chain 3 drifts from 5 toward 2. Their marginal density plots differ. Pooling these draws would combine different transient or mode-specific regions and create a misleading mixture that may not represent the target posterior.
Before pooling, inspect trace plots over time, rank plots, autocorrelation, between- and within-chain variation, and scale-reduction diagnostics. Run longer chains from dispersed starting values and verify that they overlap and explore the same regions. If the target is multimodal, determine whether chains transition between modes and whether observed mode frequencies are plausible. A numerical convergence statistic is supplemental; the visible disagreement already requires withholding posterior summaries.

Problem 86

Original question (verbatim).
For a multivariate posterior, the trace plot of each coefficient appears stationary, but a derived quantity $\beta_1 + \beta_2$ shows long-range drift and high autocorrelation. Explain why diagnostics must include derived quantities and not only the original parameters. Design a set of diagnostics for the quantity of scientific interest and state what patterns would make you withhold posterior summaries.

Required theory (verbatim from the supplied lecture manual).
A chain that looks busy is not necessarily converged. We want evidence that it has settled into the target distribution and is moving around that distribution effectively. Trace plots show the sequence over iterations. If there is a trend, long drift, or chains in different regions, that is a warning. Overlaid marginal density plots show whether separate chains are exploring similar distributions. Autocorrelation plots show how strongly nearby draws depend on each other.
Use several chains with dispersed starting points. If every chain begins at the same convenient point, we may fail to notice that the sampler has not explored another important region. Diagnostics should be applied to all important parameters and derived quantities, not only to one easy parameter. Convergence assessment is a collection of graphical and numerical checks rather than a single pass/fail test.

Detailed solution and reasoning.
A derived quantity such as $g(\beta) = \beta_1 + \beta_2$ can have a different autocorrelation and convergence behaviour from either coefficient separately. Strong negative posterior correlation may cause cancellation in the sum; strong positive correlation may magnify slow movement. A sampler can look stationary coordinatewise while drifting along a joint ridge that is visible in the derived quantity.
For the scientific target, calculate $g^{(m)} = \beta_1^{(m)} + \beta_2^{(m)}$ for every draw and inspect its trace, autocorrelation, chain-to-chain distributions, $\widehat{R}$, ESS, and MCSE. Also inspect joint scatter plots if relevant. Withhold a summary if chains disagree, drift persists, effective sample size is too small for the required precision, or tail probabilities are unstable. Diagnostics should cover all parameters and derived quantities used for inference.

Problem 87

Original question (verbatim).
A sampler appears to converge for the posterior mean but not for a 0.99 posterior quantile. Explain why tail functionals can be more difficult to estimate than central summaries. Describe how trace plots, autocorrelation plots, multiple chains, and Monte Carlo standard errors would be used to assess the tail probability. Explain why a single convergence statistic cannot certify all inferential quantities simultaneously.

Required theory (verbatim from the supplied lecture manual).
A chain that looks busy is not necessarily converged. We want evidence that it has settled into the target distribution and is moving around that distribution effectively. Trace plots show the sequence over iterations. If there is a trend, long drift, or chains in different regions, that is a warning. Overlaid marginal density plots show whether separate chains are exploring similar distributions. Autocorrelation plots show how strongly nearby draws depend on each other.
Use several chains with dispersed starting points. If every chain begins at the same convenient point, we may fail to notice that the sampler has not explored another important region. Diagnostics should be applied to all important parameters and derived quantities, not only to one easy parameter. Convergence assessment is a collection of graphical and numerical checks rather than a single pass/fail test.

Detailed solution and reasoning.
A 0.99 posterior quantile depends on a small fraction of extreme draws and is therefore more sensitive to poor tail exploration than a posterior mean. A chain may estimate central mass reasonably while failing to visit a heavy tail or a secondary mode, making extreme quantiles inaccurate.
Use multiple dispersed chains and inspect the trace and autocorrelation of the indicator $1\{\theta \leq q\}$ or of tail-relevant transformations. Compare chain-specific quantile estimates and use quantile-specific Monte Carlo standard errors. Check effective sample size in the tail and verify that all chains visit relevant extremes. A single $\widehat{R}$ or stable-looking trace cannot certify every posterior functional simultaneously; convergence and precision are quantity-specific.

5.5 Scale reduction and effective sample size

Problem 88

Original question (verbatim).
Four chains each retain 5,000 draws. Their lag autocorrelations are approximately $\rho_k = 0.7^k$. Approximate the integrated autocorrelation time and effective sample size for one chain, then for the combined retained draws under comparable mixing. Estimate the Monte Carlo standard error of a posterior mean when the posterior standard deviation is 2. Explain why 20,000 saved draws do not necessarily represent 20,000 independent observations.

Required theory (verbatim from the supplied lecture manual).
The scale-reduction diagnostic compares variation within chains with variation between chains. If chains have explored the same target distribution, those sources of variation should become similar. A value close to one is reassuring, but it is not a proof of convergence. We still need to inspect the plots and the scientific behaviour of the chain.

Effective sample size adjusts the raw number of draws for autocorrelation. If successive draws are highly correlated, ten thousand saved draws may contain much less than ten thousand independent-draws worth of information. The integrated autocorrelation time is one plus twice the sum of positive-lag autocorrelations, and effective sample size is approximately the raw sample size divided by that factor. The Monte Carlo standard error of a posterior mean is approximately the posterior standard deviation divided by the square root of effective sample size. Do not confuse Monte Carlo error with posterior uncertainty. **Board equations from the notes.**

$$B = \frac{n}{m-1}\sum_{j=1}^{m}(\bar{\theta}_j - \bar{\theta}_\cdot)^2,$$

$$W = \frac{1}{m}\sum_{j=1}^{m} s_j^2.$$

Detailed solution and reasoning.
With autocorrelation $\rho_k = 0.7^k$, the integrated autocorrelation time is

$$\tau_{int} = 1 + 2\sum_{k=1}^{\infty} 0.7^k = 1 + 2\frac{0.7}{1-0.7} = 5.6667.$$

For one chain with 5,000 retained draws,

$$ESS \approx 5000/5.6667 = 882.4.$$

For four similarly mixing chains and 20,000 total draws, the combined ESS is about $20,000/5.6667 = 3,529.4$ (assuming comparable mixing and valid convergence).
If posterior SD is 2, the Monte Carlo standard error of the mean is approximately

$$MCSE = 2/\sqrt{3529.4} = 0.0337$$

for the combined run (about 0.0673 for one chain). Twenty thousand saved draws are not twenty thousand independent observations because autocorrelation reduces effective information.

Problem 89

Original question (verbatim).
Two chains have similar within-chain variances but substantially different chain means. Explain how the between-chain and within-chain variation enter the scale-reduction diagnostic. A reported $\widehat{R} = 1.01$ is presented as proof of convergence. Critique this conclusion and identify the graphical checks and model-specific questions that should accompany the diagnostic.

Required theory (verbatim from the supplied lecture manual).
The scale-reduction diagnostic compares variation within chains with variation between chains. If chains have explored the same target distribution, those sources of variation should become similar. A value close to one is reassuring, but it is not a proof of convergence. We still need to inspect the plots and the scientific behaviour of the chain.
Effective sample size adjusts the raw number of draws for autocorrelation. If successive draws are highly correlated, ten thousand saved draws may contain much less than ten thousand independent-draws worth of information. The integrated autocorrelation time is one plus twice the sum of positive-lag autocorrelations, and effective sample size is approximately the raw sample size divided

by that factor. The Monte Carlo standard error of a posterior mean is approximately the posterior standard deviation divided by the square root of effective sample size. Do not confuse Monte Carlo error with posterior uncertainty. **Board equations from the notes.**

$$B = \frac{n}{m-1}\sum_{j=1}^{m}(\bar{\theta}_j - \bar{\theta}_\cdot)^2,$$

$$W = \frac{1}{m}\sum_{j=1}^{m} s_j^2.$$

Detailed solution and reasoning.
The scale-reduction diagnostic compares within-chain variation W with the variation between chain means B (scaled by chain length). If chains have similar means and are sampling the same stationary target, between-chain and within-chain variation should agree, and $\widehat{R}$ should be close to one. Similar within-chain variances do not compensate for different chain means: the between-chain component remains large.
A value $\widehat{R} = 1.01$ is encouraging but not proof. Check trace/rank plots, autocorrelation, ESS, MCSE, chain-specific summaries, tail and derived quantities, and whether chains explore all modes. Also ensure the model is identifiable and the sampler is correctly implemented. A statistic cannot replace substantive and graphical diagnostics.

Problem 90

Original question (verbatim).
A posterior mean is estimated as 0.42 with posterior standard deviation 0.15 and effective sample size 225. Calculate the approximate Monte Carlo standard error. The analyst reports the interval $0.42 \pm 1.96(0.15)$ as a Monte Carlo error interval. Explain the conceptual error, calculate the appropriate Monte Carlo precision scale, and distinguish uncertainty about the parameter from numerical error in estimating a posterior summary.

Required theory (verbatim from the supplied lecture manual).
The scale-reduction diagnostic compares variation within chains with variation between chains. If chains have explored the same target distribution, those sources of variation should become similar. A value close to one is reassuring, but it is not a proof of convergence. We still need to inspect the plots and the scientific behaviour of the chain.
Effective sample size adjusts the raw number of draws for autocorrelation. If successive draws are highly correlated, ten thousand saved draws may contain much less than ten thousand independent-draws worth of information. The integrated autocorrelation time is one plus twice the sum of positive-lag autocorrelations, and effective sample size is approximately the raw sample size divided by that factor. The Monte Carlo standard error of a posterior mean is approximately the posterior standard deviation divided by the square root of effective sample size. Do not confuse Monte Carlo error with posterior uncertainty. **Board equations from the notes.**

$$B = \frac{n}{m-1}\sum_{j=1}^{m}(\bar{\theta}_j - \bar{\theta}_\cdot)^2,$$

$$W = \frac{1}{m}\sum_{j=1}^{m} s_j^2.$$

Detailed solution and reasoning.
For posterior SD 0.15 and effective sample size 225,

$$MCSE(\bar{\theta}) \approx \frac{0.15}{\sqrt{225}} = 0.01.$$

An approximate 95% Monte Carlo error interval for the estimated posterior mean is therefore

$$0.42 \pm 1.96(0.01) = [0.4004, 0.4396].$$

The interval $0.42 \pm 1.96(0.15)$ instead describes posterior uncertainty about the parameter under a normal approximation; it is not the numerical error of the Monte Carlo estimate. Posterior SD is inferential uncertainty; MCSE is simulation error in estimating a posterior summary. They answer different questions.

5.6 Warm-up, thinning, and practical reporting

Problem 91

Original question (verbatim).
A researcher discards the first 1,000 iterations of a chain and claims that convergence is thereby guaranteed. Explain why warm-up removal is not a convergence test. Propose a defensible procedure for choosing warm-up length using trace plots, multiple chains, and numerical diagnostics. Explain how the decision should be revisited if different parameters or derived quantities show different convergence behaviour.

Required theory (verbatim from the supplied lecture manual).
Warm-up is the initial phase used to let the chain move away from its starting value and adapt if the algorithm requires adaptation. We do not include warm-up draws in posterior summaries. But deleting a fixed number of draws does not prove that the remaining chain has converged; diagnostics must support that decision.
Thinning means retaining every kth draw. It can reduce storage, but it does not create information and usually is not necessary merely to reduce autocorrelation. In a report, describe the sampler, number of chains, starting values, warm-up, retained iterations, convergence diagnostics, effective sample sizes, and Monte Carlo standard errors. The reader must be able to judge whether the reported summaries are based on reliable exploration of the posterior.

Detailed solution and reasoning.
Discarding the first 1,000 iterations is a warm-up decision, not a proof of convergence. The chain may still be trapped, drifting, or poorly mixing after that point. Warm-up removes dependence on initialisation only if the chain has actually reached its stationary regime; that must be assessed.
Run multiple chains from dispersed starting values. Examine trace and rank plots across the proposed warm-up boundary, autocorrelation, $\widehat{R}$, ESS, and MCSE for all key parameters and derived quantities. Increase warm-up or revise parameterisation/sampler if the chains do not stabilise. Different quantities can converge at different rates, so revisit the decision for tail functionals and scientifically important combinations rather than relying on one fixed cutoff.

Problem 92

Original question (verbatim).
A chain produces 100,000 highly autocorrelated draws. One analyst retains every 100th draw and claims that thinning has created 1,000 independent posterior draws. Evaluate the claim. Explain what thinning does and does not accomplish, and compare it with retaining all post-warm-up draws while calculating an effective sample size and Monte Carlo standard error.

Required theory (verbatim from the supplied lecture manual).
Warm-up is the initial phase used to let the chain move away from its starting value and adapt if the algorithm requires adaptation. We do not include warm-up draws in posterior summaries. But deleting a fixed number of draws does not prove that the remaining chain has converged; diagnostics must support that decision.
Thinning means retaining every kth draw. It can reduce storage, but it does not create information and usually is not necessary merely to reduce autocorrelation. In a report, describe the sampler, number of chains, starting values, warm-up, retained iterations, convergence diagnostics, effective sample sizes, and Monte Carlo standard errors. The reader must be able to judge whether the reported summaries are based on reliable exploration of the posterior.

Detailed solution and reasoning.
Thinning by retaining every 100th draw does not make the retained draws independent. It may reduce adjacent autocorrelation, but dependence can persist at lag 100 and thinning discards information. A chain with 100,000 highly autocorrelated draws may have an ESS much smaller than 100,000, but retaining 1,000 spaced draws does not guarantee ESS 1,000.
The usual approach is to retain all post-warm-up draws and calculate ESS and MCSE using the autocorrelation structure. Thin only for practical storage/communication constraints or where a specific computational reason justifies it, and report the effect on ESS. Independence is not created by a fixed thinning interval; it is an empirical property assessed through diagnostics.

Problem 93

Original question (verbatim).
Write a reproducibility report template for an MCMC analysis of a hierarchical model. It must include the sampler, target distribution, number of chains, starting values, warm-up, retained iterations, thinning decision, trace and autocorrelation checks, scale-reduction diagnostic, effective sample sizes, and Monte Carlo standard errors. Explain why each item helps a reader assess the reliability of the posterior summaries.

Required theory (verbatim from the supplied lecture manual).
Warm-up is the initial phase used to let the chain move away from its starting value and adapt if the algorithm requires adaptation. We do not include warm-up draws in posterior summaries. But deleting a fixed number of draws does not prove that the remaining chain has converged; diagnostics must support that decision.
Thinning means retaining every kth draw. It can reduce storage, but it does not create information and usually is not necessary merely to reduce autocorrelation. In a report, describe the sampler, number of chains, starting values, warm-up, retained iterations, convergence diagnostics, effective sample sizes, and Monte Carlo standard errors. The reader must be able to judge whether the reported summaries are based on reliable exploration of the posterior.

Detailed solution and reasoning.
A reproducibility report should include: (1) the exact posterior target and model parameterisation; (2) sampler name and transition/update equations; (3) number of chains and software/version/settings; (4) dispersed starting values and initialisation method; (5) warm-up length and rationale; (6) retained iterations per chain; (7) whether thinning was used and why; (8) trace/rank and autocorrelation checks; (9) scale-reduction values for all key parameters; (10) bulk and tail ESS; and (11) MCSE for means, probabilities, quantiles, and derived quantities that support the conclusions. Each item lets a reader assess whether the intended posterior was targeted, whether initialisation was forgotten, whether chains explored the same distribution, how much information the dependent draws contain, and whether Monte Carlo error is small relative to the reported precision. A report should show actual diagnostic values rather than merely state that convergence was checked.

Chapter 6

Bayesian linear models, ridge, LASSO, and spike-and-slab

6.1 The Gaussian linear regression model

Problem 94

Original question (verbatim).
Let $y \mid \beta, \sigma^2 \sim N_n(X\beta, \sigma^2 I_n)$. Derive the likelihood as a function of β, state the dimensions of every matrix, and obtain the posterior kernel under a flat prior on β when X has full column rank. Explain what changes when X is rank deficient and why the likelihood alone cannot identify coefficient combinations in the null space of X.

Required theory (verbatim from the supplied lecture manual).
In Bayesian linear regression, the response vector is modelled as a normal distribution centred at X beta. The design matrix contains the predictors, and beta contains the regression coefficients. The likelihood describes the data conditional on beta and the residual variance. Bayesian analysis adds a prior distribution to the coefficients and, when the variance is unknown, a prior to the variance as well.
The posterior is the combination of the regression likelihood and these priors. A flat prior on beta gives the familiar least-squares centre under standard rank conditions, while an informative prior regularises coefficients and can stabilise problems with many predictors. The important habit is to distinguish the design matrix, the coefficient vector, and residual variance, and keep dimensions consistent when completing the square. **Board equations from the notes.**

$$y \mid \beta, \sigma^2 \sim \mathcal{N}_n(X\beta, \sigma^2 I_n).$$

$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left\{ -\frac{(y - X\beta)^\top (y - X\beta)}{2\sigma^2} \right\}.$$

Detailed solution and reasoning.
Let $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, and $\beta \in \mathbb{R}^p$. Under $y \mid \beta, \sigma^2 \sim N_n(X\beta, \sigma^2 I_n)$,

$$p(y \mid \beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} (y - X\beta)^T (y - X\beta) \right].$$

The dimensions are $X : n \times p$, $\beta : p \times 1$, $X\beta : n \times 1$, and the residual quadratic form is scalar.

With a flat prior $\pi(\beta) \propto 1$ and full column rank $\operatorname{rank}(X) = p$, completing the square yields

$$\beta \mid y, \sigma^2 \sim N_p\left(\widehat{\beta}_{LS}, \sigma^2 (X^T X)^{-1}\right), \quad \widehat{\beta}_{LS} = (X^T X)^{-1} X^T y.$$

If X is rank deficient, $X^T X$ is singular and coefficient vectors differing by a vector in $\operatorname{Null}(X)$ produce identical fitted values. The likelihood cannot identify those directions; under a flat prior the posterior is generally improper along the null space. A proper prior can regularise those directions but does not create information that the data do not contain.

Problem 95

Original question (verbatim).
A regression has 30 observations and 40 predictors. The design matrix has rank 28. Explain why the ordinary least-squares coefficient vector is not uniquely identified. Show how a proper Gaussian prior on β can yield a proper posterior, and discuss which aspects of the result then come from the data and which are regularised by the prior. Do not confuse a numerically stable estimate with evidence that every coefficient is identified by the sample.

Required theory (verbatim from the supplied lecture manual).
In Bayesian linear regression, the response vector is modelled as a normal distribution centred at X beta. The design matrix contains the predictors, and beta contains the regression coefficients. The likelihood describes the data conditional on beta and the residual variance. Bayesian analysis adds a prior distribution to the coefficients and, when the variance is unknown, a prior to the variance as well.
The posterior is the combination of the regression likelihood and these priors. A flat prior on beta gives the familiar least-squares centre under standard rank conditions, while an informative prior regularises coefficients and can stabilise problems with many predictors. The important habit is to distinguish the design matrix, the coefficient vector, and residual variance, and keep dimensions consistent when completing the square. **Board equations from the notes.**

$$y \mid \beta, \sigma^2 \sim \mathcal{N}_n(X\beta, \sigma^2 I_n).$$

$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left\{-\frac{(y - X\beta)^\top (y - X\beta)}{2\sigma^2}\right\}.$$

Detailed solution and reasoning.
Here X is 30×40 but has rank 28. Since $\operatorname{rank}(X) < p$, the least-squares minimiser is not unique. If $\widehat{\beta}$ is one minimiser and $v \in \operatorname{Null}(X)$, then $X(\widehat{\beta} + v) = X\widehat{\beta}$, so all such vectors give the same fitted values and residual sum of squares.
Assign a proper Gaussian prior $\beta \sim N_p(b_0, V_0)$ with positive-definite V_0. For known σ^2, the posterior is proper with

$$V_n = (V_0^{-1} + X^T X/\sigma^2)^{-1}, \qquad b_n = V_n(V_0^{-1} b_0 + X^T y/\sigma^2).$$

The data inform coefficient combinations in the row space of X; the prior controls directions weakly or not at all identified by the likelihood. A stable numerical posterior is a consequence of prior regularisation and must not be described as proof that all 40 coefficients are data-identified.

Problem 96

Original question (verbatim).
Two design matrices X_1 and X_2 have the same dimensions but different predictor scaling. A colleague uses the same independent normal prior variance for every coefficient in both parameterisations. Explain why this may imply different prior beliefs about fitted effects. Construct a rescaling argument and state how the coefficient prior should transform to preserve the same prior distribution over the linear predictor.

Required theory (verbatim from the supplied lecture manual).
In Bayesian linear regression, the response vector is modelled as a normal distribution centred at X beta. The design matrix contains the predictors, and beta contains the regression coefficients. The likelihood describes the data conditional on beta and the residual variance. Bayesian analysis adds a prior distribution to the coefficients and, when the variance is unknown, a prior to the variance as well.
The posterior is the combination of the regression likelihood and these priors. A flat prior on beta gives the familiar least-squares centre under standard rank conditions, while an informative prior regularises coefficients and can stabilise problems with many predictors. The important habit is to distinguish the design matrix, the coefficient vector, and residual variance, and keep dimensions consistent when completing the square. **Board equations from the notes.**

$$y \mid \beta, \sigma^2 \sim \mathcal{N}_n(X\beta, \sigma^2 I_n).$$
$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left\{ -\frac{(y - X\beta)^\top (y - X\beta)}{2\sigma^2} \right\}.$$

Detailed solution and reasoning.
Suppose the second design uses rescaled predictors $X_2 = X_1 D$, where D is an invertible diagonal scaling matrix. To preserve the same linear predictor,

$$X_1\beta_1 = X_2\beta_2 = X_1 D\beta_2 \quad \Rightarrow \quad \beta_2 = D^{-1}\beta_1.$$

If $\beta_1 \sim N(b, V)$, the equivalent prior in the new coordinates is

$$\beta_2 \sim N(D^{-1}b, D^{-1}VD^{-T}).$$

If an independent common variance is assigned to each coefficient in both coordinate systems, this transformation rule is not followed, so the implied distribution of $X\beta$ changes. For example, doubling a predictor halves the coefficient required to produce the same effect; the same coefficient prior variance therefore implies a different prior spread for its fitted contribution. Priors should be transformed with the parameterisation, or predictor standardisation and prior scaling should be made explicit.

6.2 Normal-Inverse-Gamma prior and posterior

Problem 97

Original question (verbatim).
Assume $\beta \mid \sigma^2 \sim N_p(b_0, \sigma^2 V_0)$ and $\sigma^2 \sim \text{InvGamma}(a_0, d_0)$. Starting from the Gaussian regression likelihood, derive the posterior quantities V_n, b_n, a_n, and d_n by completing the matrix square. Show every matrix dimension and identify the residual and prior quadratic forms. Explain why simply quoting the updated hyperparameters is insufficient evidence of a correct derivation.

Required theory (verbatim from the supplied lecture manual).
The Normal–Inverse-Gamma prior couples the coefficient vector and variance. Conditional on sigma-squared, beta is multivariate normal. The variance itself is inverse-Gamma. This prior is conjugate for the Gaussian linear regression likelihood, so the posterior has the same family.
To derive the update, combine the residual quadratic form from the likelihood with the prior quadratic form for beta. Collect the coefficient terms to identify the posterior precision matrix and posterior mean. Then complete the square. The remaining terms update the inverse-Gamma shape and rate. I want students to show the matrix algebra, because simply quoting the posterior parameter names does not demonstrate why the update is correct. **Board equations from the notes.**

$$\beta \mid \sigma^2 \sim \mathcal{N}_p(b_0, \sigma^2 V_0), \qquad \sigma^2 \sim \text{InvGamma}(a_0, d_0),$$
$$V_n = (V_0^{-1} + X^\top X)^{-1}, \qquad b_n = V_n(V_0^{-1} b_0 + X^\top y),$$

Detailed solution and reasoning.
The likelihood and prior kernels are

$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\{-\tfrac{1}{2\sigma^2}(y - X\beta)^T(y - X\beta)\},$$

$$\beta \mid \sigma^2 \sim N_p(b_0, \sigma^2 V_0), \quad \sigma^2 \sim IG(a_0, d_0).$$

The updated coefficient covariance matrix and mean are

$$V_n = (V_0^{-1} + X^T X)^{-1}, \qquad b_n = V_n(V_0^{-1} b_0 + X^T y).$$

Complete the square:

$$(y - X\beta)^T(y - X\beta) + (\beta - b_0)^T V_0^{-1}(\beta - b_0) = (\beta - b_n)^T V_n^{-1}(\beta - b_n) + y^T y + b_0^T V_0^{-1} b_0 - b_n^T V_n^{-1} b_n.$$

Collecting powers of σ^2 gives

$$a_n = a_0 + n/2, \qquad d_n = d_0 + \tfrac{1}{2}(y^T y + b_0^T V_0^{-1} b_0 - b_n^T V_n^{-1} b_n).$$

Therefore $\beta \mid \sigma^2, y \sim N_p(b_n, \sigma^2 V_n)$ and $\sigma^2 \mid y \sim IG(a_n, d_n)$. Dimensions: V_0, V_n are $p \times p$; $X^T X$ is $p \times p$; b_0, b_n are $p \times 1$; each quadratic form in d_n is scalar. Showing the square-completion identity demonstrates how every updated term arises.

Problem 98

Original question (verbatim).
In a regression problem, the prior mean b_0 is far from the least-squares estimate. Derive the posterior inverse-Gamma scale and identify the term measuring disagreement between the prior centre and the data-supported coefficient vector. Explain how this term affects posterior uncertainty about σ^2, and analyse what happens when the prior covariance becomes very large or very small.

Required theory (verbatim from the supplied lecture manual).
The Normal–Inverse-Gamma prior couples the coefficient vector and variance. Conditional on sigma-squared, beta is multivariate normal. The variance itself is inverse-Gamma. This prior is conjugate for the Gaussian linear regression likelihood, so the posterior has the same family.
To derive the update, combine the residual quadratic form from the likelihood with the prior quadratic form for beta. Collect the coefficient terms to identify the posterior precision matrix and posterior mean. Then complete the square. The remaining terms update the inverse-Gamma shape and rate. I want students to show the matrix algebra, because simply quoting the posterior parameter names does not demonstrate why the update is correct. **Board equations from the notes.**

$$\beta \mid \sigma^2 \sim \mathcal{N}_p(b_0, \sigma^2 V_0), \qquad \sigma^2 \sim \text{InvGamma}(a_0, d_0),$$
$$V_n = (V_0^{-1} + X^\top X)^{-1}, \qquad b_n = V_n(V_0^{-1} b_0 + X^\top y),$$

Detailed solution and reasoning.
The updated inverse-Gamma scale can be written as

$$d_n = d_0 + \frac{1}{2}\left[(y - Xb_n)^T(y - Xb_n) + (b_n - b_0)^T V_0^{-1}(b_n - b_0)\right].$$

The second quadratic term measures the discrepancy between the posterior coefficient centre and the prior centre, weighted by prior precision. If the data-supported fit lies far from b_0 relative to the prior covariance, this term increases the posterior scale for σ^2.
As V_0 becomes very large (weak coefficient prior), $V_0^{-1} \to 0$, b_n approaches the least-squares estimate (when identifiable), and d_n approaches $d_0 + SSE/2$. As V_0 becomes very small (strong prior), b_n is pulled toward b_0 and the scale approaches $d_0 + \frac{1}{2}(y - Xb_0)^T(y - Xb_0)$ when the prior is tightly concentrated. A badly centred restrictive prior can therefore inflate posterior uncertainty about the residual variance because the fixed prior fit leaves substantial residual error.

Problem 99

Original question (verbatim).
Two analysts use the same data but one works with centred predictors and the other with uncentred predictors. Derive the transformation of the coefficient vector and explain how a Normal-Inverse-Gamma prior must be transformed to represent the same prior over the regression function. Discuss how failing to transform the prior can produce different posterior conclusions even though the likelihood models describe the same fitted values.

Required theory (verbatim from the supplied lecture manual).
The Normal–Inverse-Gamma prior couples the coefficient vector and variance. Conditional on sigma-squared, beta is multivariate normal. The variance itself is inverse-Gamma. This prior is

conjugate for the Gaussian linear regression likelihood, so the posterior has the same family.
To derive the update, combine the residual quadratic form from the likelihood with the prior quadratic form for beta. Collect the coefficient terms to identify the posterior precision matrix and posterior mean. Then complete the square. The remaining terms update the inverse-Gamma shape and rate. I want students to show the matrix algebra, because simply quoting the posterior parameter names does not demonstrate why the update is correct. **Board equations from the notes.**

$$\beta \mid \sigma^2 \sim \mathcal{N}_p(b_0, \sigma^2 V_0), \qquad \sigma^2 \sim \text{InvGamma}(a_0, d_0),$$
$$V_n = (V_0^{-1} + X^\top X)^{-1}, \qquad b_n = V_n(V_0^{-1} b_0 + X^\top y),$$

Detailed solution and reasoning.
Write the uncentred regression as $y = \alpha + X\beta + \epsilon$. If predictors are centred at vector $\bar{x}$, then

$$y = \alpha_c + (X - \mathbf{1}\bar{x}^T)\beta + \epsilon = (\alpha_c - \bar{x}^T\beta)\mathbf{1} + X\beta + \epsilon.$$

Thus the uncentred intercept is $\alpha = \alpha_c - \bar{x}^T\beta$, while slopes are unchanged (for centring only, not rescaling). This is a linear transformation of the full coefficient vector, say $\beta_{unc} = T\beta_{cen}$.
To represent the same prior on fitted regression functions, transform the prior jointly: if $\beta_{cen} \sim N(b, V)$, then $\beta_{unc} \sim N(Tb, TVT^T)$. The intercept and slopes may become correlated even if the centred-coordinate prior was independent. Giving unrelated independent priors in both coordinate systems changes the prior on fitted values and can change the posterior, despite the likelihood's equivalent fitted-value space. Transform the prior, including its covariance and any variance hierarchy, to preserve the same model.

6.3 Posterior prediction in linear regression

Problem 100

Original question (verbatim).
For a new predictor vector x_*, derive the posterior predictive distribution of a future response under the Normal-Inverse-Gamma regression model. Integrate over both β and σ^2, and state the resulting degrees of freedom, location, and scale. Compare the predictive uncertainty for an individual response with uncertainty about the conditional mean $x_*^\top\beta$. Explain why the individual prediction distribution contains an additional observation-noise contribution.

Required theory (verbatim from the supplied lecture manual).
Prediction requires one more step beyond estimating the regression coefficients. For a new predictor vector x-star, the conditional mean response is x-star transpose beta. But beta is uncertain, and sigma-squared may also be uncertain. Therefore the predictive distribution averages the sampling distribution of the new response over the posterior distribution of beta and sigma-squared.
There are two sources of uncertainty: uncertainty about the fitted regression mean, and new observation noise around that mean. This is why a prediction interval for an individual future response is wider than an interval for the conditional mean. Under the Normal–Inverse-Gamma model, integrating out beta and sigma-squared gives a Student-t predictive distribution. When writing its scale, distinguish the parameterisation of the Student-t distribution from its variance. **Board equations from the notes.**

$$y_* \mid \beta, \sigma^2 \sim \mathcal{N}(x_*^\top \beta, \sigma^2).$$
$$y_* \mid \sigma^2, y \sim \mathcal{N}\left(x_*^\top b_n, \sigma^2(1 + x_*^\top V_n x_*)\right).$$

Detailed solution and reasoning.
Under the Normal–Inverse-Gamma regression posterior,

$$\beta \mid \sigma^2, y \sim N_p(b_n, \sigma^2 V_n), \qquad \sigma^2 \mid y \sim IG(a_n, d_n).$$

For a new design vector $x_* \in \mathbb{R}^p$, the conditional response is $Y_* \mid \beta, \sigma^2 \sim N(x_*^T\beta, \sigma^2)$. Integrating over β first yields

$$Y_* \mid \sigma^2, y \sim N\left(x_*^T b_n, \sigma^2(1 + x_*^T V_n x_*)\right).$$

Integrating over σ^2 gives

$$Y_* \mid y \sim t_{2a_n}\left(x_*^T b_n,\ s_*\right), \qquad s_*^2 = \frac{d_n}{a_n}(1 + x_*^T V_n x_*).$$

The degrees of freedom are $2a_n$, location is $x_*^T b_n$, and s_* is the Student-t scale.
For the conditional mean $x_*^T\beta$, the factor 1 from observation noise is absent; its scale multiplier is $x_*^T V_n x_*$. An individual response includes both coefficient uncertainty and new observation noise, so its predictive interval is wider.

Problem 101

Original question (verbatim).
A new observation is to be predicted at a predictor vector x_* with very high leverage, so that $x_*^\top V_n x_*$ is much larger than for points near the centre of the training design. Use the posterior predictive variance to explain why the prediction interval widens. Discuss why reporting only the fitted mean or posterior mean coefficient vector hides this uncertainty.

Required theory (verbatim from the supplied lecture manual).
Prediction requires one more step beyond estimating the regression coefficients. For a new predictor vector x-star, the conditional mean response is x-star transpose beta. But beta is uncertain, and sigma-squared may also be uncertain. Therefore the predictive distribution averages the sampling distribution of the new response over the posterior distribution of beta and sigma-squared.
There are two sources of uncertainty: uncertainty about the fitted regression mean, and new observation noise around that mean. This is why a prediction interval for an individual future response is wider than an interval for the conditional mean. Under the Normal–Inverse-Gamma model, integrating out beta and sigma-squared gives a Student-t predictive distribution. When writing its scale, distinguish the parameterisation of the Student-t distribution from its variance.
Board equations from the notes.

$$y_* \mid \beta, \sigma^2 \sim \mathcal{N}(x_*^\top \beta, \sigma^2).$$
$$y_* \mid \sigma^2, y \sim \mathcal{N}\left(x_*^\top b_n, \sigma^2(1 + x_*^\top V_n x_*)\right).$$

Detailed solution and reasoning.

For $a_n > 1$, the Student-t predictive variance is

$$\mathrm{Var}(Y_* \mid y) = \frac{d_n}{a_n - 1}(1 + x_*^T V_n x_*).$$

The design-dependent term $x_*^T V_n x_*$ measures uncertainty in the linear combination of coefficients used at x_*. High-leverage predictor vectors lie far from the information-rich region of the training design, so this quadratic form can be much larger and the predictive interval widens.
Reporting only $x_*^T b_n$ or the posterior mean coefficient vector hides how uncertain the extrapolated linear combination is. Two points can have similar fitted means but very different predictive uncertainty. Always report a predictive interval and explain whether the new point is inside or far outside the observed predictor range.

Problem 102

Original question (verbatim).
A fitted regression is used to predict far outside the range of observed predictor values. The posterior predictive interval is computed correctly under the assumed linear model but is still narrow. Explain why correct propagation of parameter uncertainty does not guarantee reliable extrapolation. Identify the modelling assumption being extended beyond the data and propose posterior predictive or prior predictive checks that could reveal implausible extrapolation behaviour.

Required theory (verbatim from the supplied lecture manual).
Prediction requires one more step beyond estimating the regression coefficients. For a new predictor vector x-star, the conditional mean response is x-star transpose beta. But beta is uncertain, and sigma-squared may also be uncertain. Therefore the predictive distribution averages the sampling distribution of the new response over the posterior distribution of beta and sigma-squared.
There are two sources of uncertainty: uncertainty about the fitted regression mean, and new observation noise around that mean. This is why a prediction interval for an individual future response is wider than an interval for the conditional mean. Under the Normal–Inverse-Gamma model, integrating out beta and sigma-squared gives a Student-t predictive distribution. When writing its scale, distinguish the parameterisation of the Student-t distribution from its variance.
Board equations from the notes.

$$y_* \mid \beta, \sigma^2 \sim \mathcal{N}(x_*^\top \beta, \sigma^2).$$
$$y_* \mid \sigma^2, y \sim \mathcal{N}\left(x_*^\top b_n, \sigma^2(1 + x_*^\top V_n x_*)\right).$$

Detailed solution and reasoning.
A posterior predictive interval can correctly propagate uncertainty about β and σ^2 and still be misleading far outside the training design. The calculation is conditional on the assumed linear mean function, error distribution, and prior. Extrapolation extends that functional form into a region where the data provide little or no direct evidence; the model may be wrong there even if parameter uncertainty under the model is small.
Useful checks include prior predictive plots of regression curves over and beyond the observed range, posterior predictive checks for residual shape and heteroscedasticity, leave-one-range-out or validation near the boundary, and sensitivity to plausible nonlinear terms or alternative mean functions. Compare predictions under several scientifically defensible models. Correct integration answers 'what follows if this model is true'; it does not guarantee that the model remains true outside observed support.

6.4 Ridge regression as a Gaussian-prior MAP estimate

Problem 103

Original question (verbatim).
Assume a Gaussian regression likelihood and independent priors $\beta_j \sim N(0, \tau^2)$. Derive the negative log posterior and show that its minimiser is the ridge estimator. Express the ridge penalty in terms of σ^2 and τ^2. Explain how changing the prior variance changes shrinkage and why the intercept is often treated separately. Then discuss what is lost when only the MAP estimate is reported instead of the full posterior.

Required theory (verbatim from the supplied lecture manual).
Ridge regression can be interpreted as Bayesian regression with a zero-centred normal prior on the coefficients. The normal prior contributes a quadratic penalty on the coefficient magnitudes. When we combine it with the Gaussian likelihood and maximise the posterior, we obtain the ridge estimator.
The amount of shrinkage is controlled by the prior variance. A smaller prior variance means stronger prior precision and stronger pull toward zero. In the conventional objective, the ridge penalty parameter is proportional to the likelihood variance divided by the prior variance. The intercept is often treated separately rather than penalised. The Bayesian interpretation is valuable because it explains the penalty as prior information and, in a full Bayesian analysis, retains uncertainty rather than reporting only the penalised point estimate. **Board equations from the notes.**

$$\beta \sim \mathcal{N}_p(0, \tau^2 I).$$

$$\exp\left\{-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta) - \frac{1}{2\tau^2}\beta^\top \beta\right\}.$$

Detailed solution and reasoning.
With Gaussian likelihood and independent priors $\beta_j \sim N(0, \tau^2)$,

$$-\log p(\beta \mid y) = \frac{1}{2\sigma^2}\|y - X\beta\|^2 + \frac{1}{2\tau^2}\|\beta\|^2 + \text{constant}.$$

Multiplying by $2\sigma^2$ does not change the minimiser, so the MAP estimate minimises

$$\|y - X\beta\|^2 + \lambda\|\beta\|^2, \qquad \lambda = \sigma^2/\tau^2.$$

(If the prior is scaled as $N(0, \sigma^2\tau^2 I)$, the penalty is $1/\tau^2$.) A smaller prior variance means a larger penalty and stronger shrinkage; an infinite prior variance approaches unpenalised least squares when identifiable.
The intercept is often left unpenalised because it represents the baseline level rather than a slope to shrink toward zero, and because centring predictors changes its interpretation. Reporting only the MAP discards posterior uncertainty, coefficient correlations, credible intervals, and uncertainty in predictions. Ridge is continuous shrinkage and generally does not make coefficients exactly zero.

Problem 104

Original question (verbatim).
Two predictors are highly correlated and both have large ordinary least-squares coefficients with opposite signs. Explain how a ridge prior changes the optimisation problem and why it can stabilise the coefficient estimates. Analyse the limiting behaviour as the prior variance tends to zero and infinity. Explain why ridge usually shrinks coefficients continuously rather than setting them exactly to zero.

Required theory (verbatim from the supplied lecture manual).
Ridge regression can be interpreted as Bayesian regression with a zero-centred normal prior on the coefficients. The normal prior contributes a quadratic penalty on the coefficient magnitudes. When we combine it with the Gaussian likelihood and maximise the posterior, we obtain the ridge estimator.
The amount of shrinkage is controlled by the prior variance. A smaller prior variance means stronger prior precision and stronger pull toward zero. In the conventional objective, the ridge penalty parameter is proportional to the likelihood variance divided by the prior variance. The intercept is often treated separately rather than penalised. The Bayesian interpretation is valuable because it explains the penalty as prior information and, in a full Bayesian analysis, retains uncertainty rather than reporting only the penalised point estimate. **Board equations from the notes.**

$$\beta \sim \mathcal{N}_p(0, \tau^2 I).$$

$$\exp\left\{-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta) - \frac{1}{2\tau^2}\beta^\top \beta\right\}.$$

Detailed solution and reasoning.
When predictors are highly correlated, the likelihood may have a long shallow ridge: large positive and negative coefficients can compensate while producing similar fitted values. A Gaussian prior adds curvature through $\lambda\|\beta\|^2$, making the posterior objective better conditioned and reducing sensitivity to small data perturbations. The ridge solution is

$$\widehat{\beta}_R = (X^T X + \lambda I)^{-1} X^T y$$

(for a penalised least-squares scaling with unpenalised intercept handled separately).
As the prior variance tends to zero, $\lambda \to \infty$ and penalised coefficients tend to zero. As the prior variance tends to infinity, $\lambda \to 0$ and the solution approaches ordinary least squares (if uniquely identified). The Gaussian density is continuous and has no point mass at zero; its posterior mode may be small but is not exactly zero except in special cases. Ridge stabilises estimation by continuous shrinkage rather than variable deletion.

Problem 105

Original question (verbatim).
A researcher standardises predictors, fits ridge regression, and then converts coefficients back to their original units. Another researcher fits ridge without standardising and uses the same numerical penalty. Explain why these procedures generally impose different prior information. Derive the effect of rescaling a predictor on the coefficient prior and discuss how prior scales should be chosen when predictors have different units.

Required theory (verbatim from the supplied lecture manual).
Ridge regression can be interpreted as Bayesian regression with a zero-centred normal prior on the coefficients. The normal prior contributes a quadratic penalty on the coefficient magnitudes. When we combine it with the Gaussian likelihood and maximise the posterior, we obtain the ridge estimator.
The amount of shrinkage is controlled by the prior variance. A smaller prior variance means stronger prior precision and stronger pull toward zero. In the conventional objective, the ridge penalty parameter is proportional to the likelihood variance divided by the prior variance. The intercept is often treated separately rather than penalised. The Bayesian interpretation is valuable because it explains the penalty as prior information and, in a full Bayesian analysis, retains uncertainty rather than reporting only the penalised point estimate. **Board equations from the notes.**

$$\beta \sim \mathcal{N}_p(0, \tau^2 I).$$

$$\exp\left\{-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta) - \frac{1}{2\tau^2}\beta^\top \beta\right\}.$$

Detailed solution and reasoning.
Suppose predictor x_j is replaced by $x_j^* = cx_j$. To preserve the same fitted contribution, its coefficient must be $\beta_j^* = \beta_j/c$. If the original coefficient prior is $N(0, \tau^2)$, then the equivalent prior after rescaling is

$$\beta_j^* \sim N(0, \tau^2/c^2).$$

Using the same numerical prior variance or ridge penalty after changing predictor scale therefore changes the implied prior distribution over fitted effects. Predictors measured in metres, years, and dollars should not automatically receive identical coefficient scales without considering units. Standardising predictors makes a common prior scale more interpretable, but coefficients must be transformed back consistently. A fair analysis either scales predictors before fitting and specifies priors on standardised effects, or transforms the prior covariance with the predictor scaling matrix. The numerical penalty is meaningful only relative to the parameterisation.

6.5 LASSO as a Laplace-prior MAP estimate

Problem 106

Original question (verbatim).
Assume independent Laplace priors $p(\beta_j) = (\lambda/2)\exp(-\lambda|\beta_j|)$ in a Gaussian regression model. Derive the negative log posterior and show that its MAP estimate minimises a least-squares criterion plus an L_1 penalty. In the one-dimensional orthogonal case, derive the soft-thresholding rule and identify the threshold below which the MAP estimate is exactly zero.

Required theory (verbatim from the supplied lecture manual).
The LASSO penalty is the sum of the absolute values of the coefficients. In Bayesian terms, this corresponds to independent Laplace, or double-exponential, priors centred at zero. The posterior mode minimises the negative log likelihood plus the absolute-value penalty.
The sharp corner of the absolute-value penalty at zero is what allows the MAP estimate to become exactly zero for some coefficients. In a simple one-dimensional orthogonal case, this gives a soft-thresholding rule: coefficients smaller in magnitude than the threshold are set to zero,

while larger coefficients are reduced by the threshold amount. In a multidimensional model with correlated predictors, the solution is not generally obtained by applying this rule independently to each coefficient. Also, for a continuous Laplace prior, the posterior probability that a coefficient is exactly zero remains zero; exact posterior point mass at zero requires a discrete spike component.
Board equations from the notes.

$$p(\beta_j) = \frac{\lambda}{2} e^{-\lambda|\beta_j|}.$$

$$\frac{1}{2\sigma^2}\|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

Detailed solution and reasoning.
With Gaussian likelihood, the negative log posterior (up to a constant) is

$$\frac{1}{2\sigma^2}\|y - X\beta\|^2 + \lambda \sum_{j} |\beta_j|,$$

so the MAP estimator minimises a least-squares loss plus an L_1 penalty. In the one-dimensional orthogonal standardised problem, take the objective

$$Q(\beta) = \frac{1}{2}(z - \beta)^2 + \lambda|\beta|.$$

For $\beta > 0$, setting derivative $\beta - z + \lambda = 0$ gives $\beta = z - \lambda$; for $\beta < 0$, it gives $\beta = z + \lambda$. At zero, the subgradient condition is $|z| \leq \lambda$. Combining cases,

$$\widehat{\beta}_L = \text{sign}(z)(|z| - \lambda)_+.$$

Thus the threshold is λ: if $|z| \leq \lambda$, the MAP coefficient is exactly zero. Under another scaling of the likelihood, the threshold is the corresponding penalty multiplied by the effective sampling variance; the objective convention must be stated.

Problem 107

Original question (verbatim).
A student applies the scalar soft-thresholding formula separately to two highly correlated predictors and claims to have solved the multivariable LASSO problem. Explain why the coordinate-wise rule does not generally solve the correlated multivariable problem in one independent step. Discuss how correlation can make variable selection unstable and why the optimisation must account for the joint design matrix.

Required theory (verbatim from the supplied lecture manual).
The LASSO penalty is the sum of the absolute values of the coefficients. In Bayesian terms, this corresponds to independent Laplace, or double-exponential, priors centred at zero. The posterior mode minimises the negative log likelihood plus the absolute-value penalty.
The sharp corner of the absolute-value penalty at zero is what allows the MAP estimate to become exactly zero for some coefficients. In a simple one-dimensional orthogonal case, this gives a soft-thresholding rule: coefficients smaller in magnitude than the threshold are set to zero, while larger coefficients are reduced by the threshold amount. In a multidimensional model with correlated predictors, the solution is not generally obtained by applying this rule independently to

each coefficient. Also, for a continuous Laplace prior, the posterior probability that a coefficient is exactly zero remains zero; exact posterior point mass at zero requires a discrete spike component.
Board equations from the notes.

$$p(\beta_j) = \frac{\lambda}{2} e^{-\lambda|\beta_j|}.$$

$$\frac{1}{2\sigma^2}\|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

Detailed solution and reasoning.
The scalar soft-thresholding rule assumes an orthogonal coordinate: the least-squares loss separates into a sum of independent terms. With correlated predictors,

$$\|y - X\beta\|^2 = y^T y - 2\beta^T X^T y + \beta^T X^T X \beta$$

contains cross-products $\beta_j \beta_k x_j^T x_k$. The optimisation no longer separates into independent scalar problems. Applying the one-dimensional rule once to each marginal score ignores these cross-terms and generally does not minimise the multivariable LASSO objective.
Correlated predictors can substitute for one another, so small data changes may cause LASSO to select one variable rather than another. Solve the joint convex optimisation (for example by coordinate descent with iterative updates or another valid algorithm), check convergence, and examine stability under resampling or alternative predictor sets. The scalar formula is exact only in the stated orthogonal case.

Problem 108

Original question (verbatim).
A continuous Laplace prior is said to assign positive posterior probability to the event $\beta_j = 0$ because the LASSO MAP estimate is sometimes exactly zero. Determine whether this is correct. Contrast the density at zero with a point mass at zero, and explain why an exact posterior inclusion probability requires a spike component rather than an ordinary continuous Laplace prior.

Required theory (verbatim from the supplied lecture manual).
The LASSO penalty is the sum of the absolute values of the coefficients. In Bayesian terms, this corresponds to independent Laplace, or double-exponential, priors centred at zero. The posterior mode minimises the negative log likelihood plus the absolute-value penalty.
The sharp corner of the absolute-value penalty at zero is what allows the MAP estimate to become exactly zero for some coefficients. In a simple one-dimensional orthogonal case, this gives a soft-thresholding rule: coefficients smaller in magnitude than the threshold are set to zero, while larger coefficients are reduced by the threshold amount. In a multidimensional model with correlated predictors, the solution is not generally obtained by applying this rule independently to each coefficient. Also, for a continuous Laplace prior, the posterior probability that a coefficient is exactly zero remains zero; exact posterior point mass at zero requires a discrete spike component.
Board equations from the notes.

$$p(\beta_j) = \frac{\lambda}{2} e^{-\lambda|\beta_j|}.$$

$$\frac{1}{2\sigma^2}\|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

Detailed solution and reasoning.
The claim is false. A continuous Laplace prior has density

$$p(\beta_j) = \frac{\lambda}{2} e^{-\lambda|\beta_j|},$$

which is finite and positive at zero, but for any continuous posterior

$$\Pr(\beta_j = 0 \mid y) = 0.$$

A point has zero probability under a continuous distribution, regardless of how high its density is or where the MAP estimate lies. LASSO may have a MAP estimate exactly at zero because the absolute-value penalty creates a kink; this does not imply posterior point mass at zero.
An exact inclusion probability requires a spike-and-slab prior with a discrete atom, such as $\Pr(\beta_j = 0) = 1 - q$ and a continuous slab when included. Then the posterior can assign positive probability to exclusion and report $\Pr(\gamma_j = 1 \mid y)$.

6.6 Spike-and-slab priors and Bayesian variable selection

Problem 109

Original question (verbatim).
For one coefficient, let $\gamma \sim \text{Bernoulli}(q)$, with $\beta = 0$ when $\gamma = 0$ and $\beta \sim N(0, \tau^2)$ when $\gamma = 1$. Derive the posterior inclusion odds in terms of the prior inclusion odds and the Bayes factor comparing inclusion with exclusion. Explain how the posterior inclusion probability differs from a continuous shrinkage estimate of the coefficient.

Required theory (verbatim from the supplied lecture manual).
Spike-and-slab priors introduce an explicit indicator for whether each predictor is included. The spike is a point mass at zero, while the slab is a continuous distribution for nonzero coefficients. The model indicator therefore has a direct posterior inclusion probability.
We can compare inclusion and exclusion through prior odds and a Bayes factor. The posterior inclusion odds equal prior inclusion odds multiplied by the Bayes factor for including the predictor. Unlike a continuous shrinkage prior, spike-and-slab gives genuine posterior probability to a coefficient being exactly zero. When many predictors are available, there can be an enormous number of possible models, so computation may require stochastic search or other approximations. Model averaging can preserve uncertainty about which variables belong in the model. **Board equations from the notes.**

$$\gamma_j \sim \text{Bernoulli}(q_j), \qquad \beta_j \mid \gamma_j = 0 = 0, \qquad \beta_j \mid \gamma_j = 1 \sim \mathcal{N}(0, \tau_j^2).$$
$$\Pr(\gamma_j = 1 \mid y).$$

Detailed solution and reasoning.
Let $\gamma \sim \text{Bernoulli}(q)$, with $\beta = 0$ under exclusion and $\beta \sim N(0, \tau^2)$ under inclusion. The marginal

likelihoods are

$$p(y \mid \gamma = 0) = p(y \mid \beta = 0), \qquad p(y \mid \gamma = 1) = \int p(y \mid \beta)\phi(\beta; 0, \tau^2)d\beta.$$

The Bayes factor for inclusion is

$$BF_{10} = \frac{p(y \mid \gamma = 1)}{p(y \mid \gamma = 0)}.$$

Prior and posterior inclusion odds satisfy

$$\frac{\Pr(\gamma = 1 \mid y)}{\Pr(\gamma = 0 \mid y)} = \frac{q}{1-q} BF_{10},$$

so $\Pr(\gamma = 1 \mid y) = qBF_{10}/(1 - q + qBF_{10})$.
This probability quantifies posterior uncertainty about whether the coefficient is included. A continuous shrinkage estimate, by contrast, reports a coefficient value under a model that does not assign an exact point mass to exclusion. Inclusion uncertainty and coefficient magnitude are distinct inferential targets.

Problem 110

Original question (verbatim).
Two predictors are strongly correlated and each has a high marginal association with the response. Under a spike-and-slab prior, explain why the posterior inclusion probabilities may be sensitive to whether the predictors are considered jointly or separately. Formulate the model indicators for the two predictors, identify the four possible models, and discuss why model averaging can be more informative than reporting only the single most probable model.

Required theory (verbatim from the supplied lecture manual).
Spike-and-slab priors introduce an explicit indicator for whether each predictor is included. The spike is a point mass at zero, while the slab is a continuous distribution for nonzero coefficients. The model indicator therefore has a direct posterior inclusion probability.
We can compare inclusion and exclusion through prior odds and a Bayes factor. The posterior inclusion odds equal prior inclusion odds multiplied by the Bayes factor for including the predictor. Unlike a continuous shrinkage prior, spike-and-slab gives genuine posterior probability to a coefficient being exactly zero. When many predictors are available, there can be an enormous number of possible models, so computation may require stochastic search or other approximations. Model averaging can preserve uncertainty about which variables belong in the model. **Board equations from the notes.**

$$\gamma_j \sim \text{Bernoulli}(q_j), \qquad \beta_j \mid \gamma_j = 0 = 0, \qquad \beta_j \mid \gamma_j = 1 \sim \mathcal{N}(0, \tau_j^2).$$
$$\Pr(\gamma_j = 1 \mid y).$$

Detailed solution and reasoning.
For two predictors, define indicators $(\gamma_1, \gamma_2) \in \{(0,0), (1,0), (0,1), (1,1)\}$. These correspond to the null model, predictor 1 only, predictor 2 only, and both predictors. With independent Bernoulli(q_j) prior indicators, prior model probabilities factorise, but posterior probabilities depend on the joint likelihood and slab priors.
If predictors are strongly correlated, models $(1,0)$ and $(0,1)$ may explain the response similarly. Conditional inclusion probabilities can be sensitive to whether the other variable is included;

evidence may be shared or divided between them. Model averaging predicts using

$$p(\tilde{y} \mid y) = \sum_{\gamma} p(\tilde{y} \mid y, \gamma) \Pr(\gamma \mid y),$$

which propagates uncertainty across all four models. Reporting only the single most probable model hides uncertainty among correlated alternatives and can overstate certainty about which predictor is responsible.

Problem 111

Original question (verbatim).
A model has 20 candidate predictors and independent prior inclusion probability $q = 0.1$ for each. Calculate the prior expected number of included predictors and the prior probability of the null model. Explain how the prior inclusion probability affects multiplicity and model uncertainty. Discuss the computational challenge of exploring 2^{20} possible models and how posterior inclusion probabilities summarise uncertainty across them.

Required theory (verbatim from the supplied lecture manual).
Spike-and-slab priors introduce an explicit indicator for whether each predictor is included. The spike is a point mass at zero, while the slab is a continuous distribution for nonzero coefficients. The model indicator therefore has a direct posterior inclusion probability.
We can compare inclusion and exclusion through prior odds and a Bayes factor. The posterior inclusion odds equal prior inclusion odds multiplied by the Bayes factor for including the predictor. Unlike a continuous shrinkage prior, spike-and-slab gives genuine posterior probability to a coefficient being exactly zero. When many predictors are available, there can be an enormous number of possible models, so computation may require stochastic search or other approximations. Model averaging can preserve uncertainty about which variables belong in the model. **Board equations from the notes.**

$$\gamma_j \sim \text{Bernoulli}(q_j), \qquad \beta_j \mid \gamma_j = 0 = 0, \qquad \beta_j \mid \gamma_j = 1 \sim \mathcal{N}(0, \tau_j^2).$$
$$\Pr(\gamma_j = 1 \mid y).$$

Detailed solution and reasoning.
With 20 independent inclusion indicators each having probability $q = 0.1$, the prior expected number included is

$$E(K) = 20(0.1) = 2.$$

The prior probability that all predictors are excluded is

$$\Pr(K = 0) = (1 - q)^{20} = 0.9^{20} = 0.121577.$$

Thus the prior expects a sparse model on average but still assigns substantial probability to models with some included predictors. Smaller q imposes stronger multiplicity control by making complex models less probable a priori.
There are $2^{20} = 1,048,576$ possible models, so exhaustive model evaluation can be computationally demanding. Stochastic search or MCMC over model indicators may be required. Posterior inclusion probabilities $\Pr(\gamma_j = 1 \mid y)$ summarise how often each predictor appears across posterior model uncertainty; they are not the same as coefficients from one selected model.

6.7 Ridge, LASSO, and spike-and-slab compared

Problem 112

Original question (verbatim).
A prediction team wants stable predictions from highly correlated predictors but does not need a sparse model. A scientific team wants an interpretable sparse point estimate. A third team wants posterior probabilities that each predictor belongs in the model. For each objective, choose among ridge, LASSO, and spike-and-slab, justify the prior structure, and state what uncertainty the selected approach preserves or discards. Explain why the three procedures answer related but non-identical questions.

Required theory (verbatim from the supplied lecture manual).
Let us compare the three approaches in terms of what their priors are doing. Ridge uses a normal prior and usually shrinks all coefficients continuously toward zero. LASSO uses a Laplace prior; its posterior mode can be sparse because of the sharp corner at zero, but the full continuous posterior does not assign positive probability to exact zero. Spike-and-slab has an explicit point mass at zero and therefore supports posterior inclusion and exclusion probabilities.
These approaches answer related but different questions. Ridge emphasises stable estimation. LASSO combines shrinkage and sparse point estimation. Spike-and-slab directly models uncertainty about variable inclusion. The right choice depends on whether we want stabilisation, a sparse point estimate, or a probability distribution over candidate models.

Detailed solution and reasoning.
For stable prediction with correlated predictors and no need for sparsity, ridge is appropriate: the zero-centred Gaussian prior shrinks coefficients continuously and typically keeps correlated variables together. It retains a full posterior if fitted Bayesianly, though a ridge MAP-only report discards uncertainty.
For an interpretable sparse point estimate, LASSO corresponds to independent Laplace priors and an L_1 MAP penalty; it can set estimates exactly to zero but ordinary continuous Laplace posteriors do not assign positive posterior probability to exact exclusion. For posterior probabilities that predictors belong in the model, spike-and-slab uses discrete inclusion indicators and a continuous slab, preserving model uncertainty and supporting model averaging.
The objectives differ: prediction stability, sparse point estimation, and uncertainty about model inclusion are not the same question. The chosen method should match the inferential goal and the uncertainty to be reported.

Problem 113

Original question (verbatim).
Consider one orthogonalised predictor with standardised score z. Compare the MAP estimates under a zero-centred Gaussian prior and a Laplace prior. Derive the continuous shrinkage rule for ridge and the thresholding rule for LASSO. Then explain how a spike-and-slab prior adds a discrete inclusion decision that neither continuous prior provides. Identify situations in which the three estimates may coincide and situations in which they differ sharply.

Required theory (verbatim from the supplied lecture manual).
Let us compare the three approaches in terms of what their priors are doing. Ridge uses a normal prior and usually shrinks all coefficients continuously toward zero. LASSO uses a Laplace prior; its

posterior mode can be sparse because of the sharp corner at zero, but the full continuous posterior does not assign positive probability to exact zero. Spike-and-slab has an explicit point mass at zero and therefore supports posterior inclusion and exclusion probabilities.
These approaches answer related but different questions. Ridge emphasises stable estimation. LASSO combines shrinkage and sparse point estimation. Spike-and-slab directly models uncertainty about variable inclusion. The right choice depends on whether we want stabilisation, a sparse point estimate, or a probability distribution over candidate models.

Detailed solution and reasoning.
In the orthogonal scalar problem, let the sufficient score be z with likelihood $z \mid \beta \sim N(\beta, s^2)$. Under a Gaussian prior $\beta \sim N(0, \tau^2)$, the posterior mean/MAP is

$$\widehat{\beta}_R = \frac{\tau^2}{\tau^2 + s^2} z,$$

a continuous shrinkage rule. Under a Laplace prior with penalty parameter λ, the MAP solves $\frac{1}{2}(z - \beta)^2 + \lambda s^2 |\beta|$ (under this variance scaling), giving

$$\widehat{\beta}_L = \operatorname{sign}(z)(|z| - \lambda s^2)_+.$$

It is exactly zero below the threshold.
Spike-and-slab adds an indicator γ: the posterior can put positive mass at $\beta = 0$ and separately assign probability to inclusion. Ridge and LASSO are continuous priors; neither provides a posterior inclusion probability. Estimates can be close when $|z|$ is large and penalties are weak; they differ sharply near zero, where LASSO thresholds but ridge only shrinks, and in the spike-and-slab model where model inclusion can switch discretely.

Problem 114

Original question (verbatim).
A report compares ridge, LASSO, and spike-and-slab solely by the number of zero coefficients. Critique this comparison. Propose a fair evaluation that distinguishes predictive stability, sparse point estimation, posterior inclusion uncertainty, and model averaging. State what should be reported for each method and explain why a common numerical penalty value does not make the prior assumptions comparable.

Required theory (verbatim from the supplied lecture manual).
Let us compare the three approaches in terms of what their priors are doing. Ridge uses a normal prior and usually shrinks all coefficients continuously toward zero. LASSO uses a Laplace prior; its posterior mode can be sparse because of the sharp corner at zero, but the full continuous posterior does not assign positive probability to exact zero. Spike-and-slab has an explicit point mass at zero and therefore supports posterior inclusion and exclusion probabilities.
These approaches answer related but different questions. Ridge emphasises stable estimation. LASSO combines shrinkage and sparse point estimation. Spike-and-slab directly models uncertainty about variable inclusion. The right choice depends on whether we want stabilisation, a sparse point estimate, or a probability distribution over candidate models.

Detailed solution and reasoning.
Counting exact zeros is not a fair comparison. Ridge generally produces no exact zeros, LASSO does so through a continuous-prior MAP kink, and spike-and-slab produces posterior exclusion prob-

abilities because it contains a point mass. A zero count therefore measures different mathematical features in the three procedures.
A fair comparison should use held-out predictive performance and calibration for ridge; sparsity, selection stability, and predictive performance for LASSO; and posterior inclusion probabilities, model uncertainty, and model-averaged prediction for spike-and-slab. Report coefficient scales, prior/penalty definitions, uncertainty intervals where available, and sensitivity to predictor scaling/correlation. Equal numerical penalty values are not comparable because the Gaussian variance, Laplace rate, and slab/inclusion priors induce different prior distributions and parameter scales.

Chapter 7

Hierarchical Bayes, empirical Bayes, and profile likelihood

7.1 Why hierarchical models are needed

Problem 115

Original question (verbatim).
Ten hospitals each have a small sample and their observed success proportions vary widely. Specify a three-level hierarchical model with patient-level outcomes, hospital-specific probabilities, and population-level hyperparameters. Write the joint posterior kernel and explain how information flows from patient observations to hospitals and then across hospitals. Compare the model with no pooling and complete pooling, identifying the inferential cost of each extreme.

Required theory (verbatim from the supplied lecture manual).
Suppose several groups each have their own parameter, but we believe the groups are related. Estimating each group independently can be unstable when group sample sizes are small. Forcing every group to have exactly the same parameter can ignore real differences. A hierarchical model provides a middle path: each group has its own parameter, and those parameters come from a shared population distribution.
This adds another level of uncertainty. We have the observation model at the data level, the group-parameter distribution at the next level, and hyperparameters that govern the population distribution. Information can flow between groups through the shared hyperparameters. This is the basic reason hierarchical models can stabilise estimates while still allowing genuine group variation. **Board equations from the notes.**

$$y_j \mid \theta_j \sim p(y_j \mid \theta_j), \qquad \theta_j \mid \eta \sim p(\theta_j \mid \eta), \qquad \eta \sim p(\eta),$$

$$p(\theta_1, \ldots, \theta_J, \eta \mid y_1, \ldots, y_J) \propto p(\eta) \prod_{j=1}^{J} p(\theta_j \mid \eta) p(y_j \mid \theta_j).$$

Detailed solution and reasoning.
Let Y_{ij} indicate success for patient i in hospital j, $j = 1, \ldots, 10$. A three-level model is

$$Y_{ij} \mid p_j \sim \text{Bernoulli}(p_j), \qquad p_j \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta), \qquad (\alpha, \beta) \sim \pi(\alpha, \beta),$$

with a proper hyperprior supported on positive values. Let $s_j = \sum_i Y_{ij}$ and n_j be the hospital

sample size. The joint posterior kernel is

$$\pi(\alpha,\beta)\prod_{j=1}^{10}\frac{p_j^{s_j+\alpha-1}(1-p_j)^{n_j-s_j+\beta-1}}{B(\alpha,\beta)}.$$

Patient outcomes update each p_j; the hospital-level posterior distributions inform the shared hyperparameters; the hyperparameter posterior then feeds back to each hospital through partial pooling. No pooling estimates each p_j independently and can be unstable for small n_j. Complete pooling forces one common probability and can erase real hospital differences. Hierarchical partial pooling shares information while permitting heterogeneity, but relies on the assumption that hospitals are plausibly exchangeable conditional on the hierarchy.

Problem 116

Original question (verbatim).
A national programme compares treatment effects across districts. Some districts have 5 observations and others have 500. Explain why estimating every district independently can be unstable and why forcing a single common effect can be unrealistic. Formulate a partial-pooling model and predict which districts will be shrunk more strongly toward the population distribution. State which assumptions justify borrowing information across districts.

Required theory (verbatim from the supplied lecture manual).
Suppose several groups each have their own parameter, but we believe the groups are related. Estimating each group independently can be unstable when group sample sizes are small. Forcing every group to have exactly the same parameter can ignore real differences. A hierarchical model provides a middle path: each group has its own parameter, and those parameters come from a shared population distribution.
This adds another level of uncertainty. We have the observation model at the data level, the group-parameter distribution at the next level, and hyperparameters that govern the population distribution. Information can flow between groups through the shared hyperparameters. This is the basic reason hierarchical models can stabilise estimates while still allowing genuine group variation. **Board equations from the notes.**

$$y_j \mid \theta_j \sim p(y_j \mid \theta_j), \qquad \theta_j \mid \eta \sim p(\theta_j \mid \eta), \qquad \eta \sim p(\eta),$$

$$p(\theta_1,\ldots,\theta_J,\eta \mid y_1,\ldots,y_J) \propto p(\eta)\prod_{j=1}^{J} p(\theta_j \mid \eta)p(y_j \mid \theta_j).$$

Detailed solution and reasoning.
Independent district estimates use only each district's data. With five observations, the estimate can be extremely noisy; with 500 it is much more stable. Complete pooling forces a single treatment effect and can conceal genuine heterogeneity. A partial-pooling model is

$$\widehat{\theta}_j \mid \theta_j \sim N(\theta_j, s_j^2), \qquad \theta_j \mid \mu,\tau^2 \sim N(\mu,\tau^2),$$

with hyperpriors on (μ,τ^2).
Conditional posterior mean is $w_j\widehat{\theta}_j + (1-w_j)\mu$, with

$$w_j = \frac{\tau^2}{\tau^2 + s_j^2}.$$

A district with fewer observations or larger standard error has larger s_j^2, smaller w_j, and therefore stronger shrinkage toward the population mean. Borrowing information is justified only if the groups are sufficiently similar/exchangeable under the model and relevant differences (such as covariates or policy regimes) are represented.

Problem 117

Original question (verbatim).
A grouped dataset is analysed with a hierarchical model, but one group appears to follow a very different data-generating process from all others. Explain why exchangeability is a modelling assumption rather than a guaranteed property of grouped data. Propose diagnostics or model changes that could accommodate systematic group differences without abandoning the idea of shared population structure.

Required theory (verbatim from the supplied lecture manual).
Suppose several groups each have their own parameter, but we believe the groups are related. Estimating each group independently can be unstable when group sample sizes are small. Forcing every group to have exactly the same parameter can ignore real differences. A hierarchical model provides a middle path: each group has its own parameter, and those parameters come from a shared population distribution.
This adds another level of uncertainty. We have the observation model at the data level, the group-parameter distribution at the next level, and hyperparameters that govern the population distribution. Information can flow between groups through the shared hyperparameters. This is the basic reason hierarchical models can stabilise estimates while still allowing genuine group variation. **Board equations from the notes.**

$$y_j \mid \theta_j \sim p(y_j \mid \theta_j), \qquad \theta_j \mid \eta \sim p(\theta_j \mid \eta), \qquad \eta \sim p(\eta),$$

$$p(\theta_1, \ldots, \theta_J, \eta \mid y_1, \ldots, y_J) \propto p(\eta) \prod_{j=1}^{J} p(\theta_j \mid \eta) p(y_j \mid \theta_j).$$

Detailed solution and reasoning.
Exchangeability means the joint model is invariant to permutations of group labels. It is a modelling assumption, not a guarantee that real groups are interchangeable. A group may differ because it belongs to another population, follows a different intervention protocol, has different case mix, or has a different data-generating process.
Inspect group-level residuals and posterior predictive distributions, compare observed and replicated between-group variation, and check whether one group has persistently unusual outcomes. Possible revisions include group-level covariates, known subgroup-specific hyperparameters, mixture or robust heavy-tailed random effects, a separate model for the outlying group, or a structured spatial/temporal hierarchy. These retain shared population structure while relaxing the assumption that every group arises identically from one simple distribution. Strong shrinkage should not be imposed blindly on a non-exchangeable group.

7.2 Exchangeability and partial pooling

Problem 118

Original question (verbatim).
For group j, suppose $\bar{y}_j \mid \theta_j \sim N(\theta_j, s_j^2)$ and $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$. Derive the posterior mean and variance of θ_j conditional on (μ, τ^2). Calculate the shrinkage weight for two groups with the same observed mean but sampling variances 1 and 9. Explain why the less precise group is pulled more strongly toward μ.

Required theory (verbatim from the supplied lecture manual).
Exchangeability at the group level means that if we relabel the groups, the joint distribution of their parameters does not change. It is not the same as saying the group parameters are equal. A shared population distribution provides a common structure without forcing every parameter to the same value.
No pooling estimates each group separately. Complete pooling assumes a common parameter. Partial pooling allows differences but shrinks noisy group estimates toward the population distribution. The degree of shrinkage depends on the balance between each group's sampling variance and the between-group variance. A group with imprecise data is pulled more strongly toward the population mean than a group with precise data.

Detailed solution and reasoning.
The likelihood for $\bar{y}_j$ is $N(\theta_j, s_j^2)$ and the conditional prior is $N(\mu, \tau^2)$. Multiplying the normal kernels gives posterior precision $s_j^{-2} + \tau^{-2}$, hence

$$\theta_j \mid \bar{y}_j, \mu, \tau^2 \sim N(m_j, v_j), \quad v_j = \left(s_j^{-2} + \tau^{-2}\right)^{-1} = \frac{s_j^2 \tau^2}{s_j^2 + \tau^2},$$

$$m_j = w_j \bar{y}_j + (1 - w_j)\mu, \qquad w_j = \frac{\tau^2}{\tau^2 + s_j^2}.$$

For sampling variances 1 and 9, the weights are respectively

$$w_1 = \frac{\tau^2}{\tau^2 + 1}, \qquad w_2 = \frac{\tau^2}{\tau^2 + 9}.$$

For every $\tau^2 > 0$, $w_2 < w_1$. Thus the group with variance 9 places less weight on its noisy observed mean and more weight on the population mean. A numerical weight needs a specified τ^2; the relative comparison is determined without it.

Problem 119

Original question (verbatim).
A set of group labels is randomly permuted, and the hierarchical prior assigns the same joint probability before and after relabelling. Explain why this is exchangeability and why it does not imply that all group parameters are equal. Compare no pooling, complete pooling, and partial pooling in terms of the assumptions each makes and the information each uses.

Required theory (verbatim from the supplied lecture manual).
Exchangeability at the group level means that if we relabel the groups, the joint distribution of their parameters does not change. It is not the same as saying the group parameters are equal. A shared population distribution provides a common structure without forcing every parameter to the same value.
No pooling estimates each group separately. Complete pooling assumes a common parameter. Partial pooling allows differences but shrinks noisy group estimates toward the population distribution. The degree of shrinkage depends on the balance between each group's sampling variance and the between-group variance. A group with imprecise data is pulled more strongly toward the population mean than a group with precise data.

Detailed solution and reasoning.
If a prior has the same joint probability after any permutation of group labels, it is exchangeable. This property says the labels do not matter to the joint distribution; it does not assert $\theta_1 = \cdots = \theta_J$. In a normal hierarchy, group effects are conditionally iid draws from a population distribution and can differ while their joint prior remains permutation-invariant.
No pooling gives separate group parameters and no borrowing across groups. Complete pooling sets all group parameters equal, eliminating between-group variation. Partial pooling gives group parameters a shared population distribution, so each group's estimate is pulled toward the common distribution according to its precision and the estimated heterogeneity. The three are different assumptions about the structure of group differences and the information shared.

Problem 120

Original question (verbatim).
Suppose the between-group variance τ^2 is estimated to be very small, but one group has an extreme estimate with a small standard error. Determine how the Normal-Normal model treats this group. Discuss whether strong shrinkage is automatically appropriate, how a non-exchangeable outlier could affect the population fit, and what model checking would reveal.

Required theory (verbatim from the supplied lecture manual).
Exchangeability at the group level means that if we relabel the groups, the joint distribution of their parameters does not change. It is not the same as saying the group parameters are equal. A shared population distribution provides a common structure without forcing every parameter to the same value.
No pooling estimates each group separately. Complete pooling assumes a common parameter. Partial pooling allows differences but shrinks noisy group estimates toward the population distribution. The degree of shrinkage depends on the balance between each group's sampling variance and the between-group variance. A group with imprecise data is pulled more strongly toward the population mean than a group with precise data.

Detailed solution and reasoning.
In the Normal–Normal model the shrinkage weight is $w_j = \tau^2/(\tau^2 + s_j^2)$. A group with small standard error (small s_j^2) receives a relatively large weight on its own estimate, even when the estimated between-group variance is small. Thus an extreme but highly precise group may resist shrinkage more than an imprecise one. If τ^2 is extremely close to zero relative to s_j^2, shrinkage can still be substantial, but the small within-group variance makes the group's evidence influential.
Strong shrinkage is not automatically appropriate. If the group is genuinely non-exchangeable, forcing it into the common distribution can pull the estimated population mean/variance and

misrepresent the group. Check its residual and posterior predictive tail probability, leave-one-group-out influence, group-specific covariates, and sensitivity to robust or mixture random-effects distributions. A systematic outlier suggests revisiting the hierarchy rather than simply accepting a tiny τ^2 estimate.

7.3 The Normal-Normal hierarchical model

Problem 121

Original question (verbatim).
Three studies report $\bar{y}_j$ with known sampling variances s_j^2. Conditional on (μ, τ^2), assume $\bar{y}_j \mid \theta_j \sim N(\theta_j, s_j^2)$ and $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$. Derive the conditional posterior distribution of each θ_j. Then derive its precision-weighted posterior mean and explain how the weights change as s_j^2 and τ^2 vary.

Required theory (verbatim from the supplied lecture manual).
The Normal–Normal model makes partial pooling algebraically transparent. We observe a group estimate y-bar-j with sampling variance s-j-squared. The true group effect theta-j is itself normal with population mean mu and between-group variance tau-squared. If we condition on mu and tau-squared, the posterior for theta-j is normal.
Its precision is the sum of sampling precision and population-prior precision. Its mean is a precision-weighted average of the group's observed estimate and the population mean. If the group's sampling variance is large, its own observation receives less weight and shrinkage is stronger. If the between-group variance is large, the groups are allowed to differ substantially and the shrinkage is weak. This gives us a direct algebraic expression for partial pooling. **Board equations from the notes.**

$$\bar{y}_j \mid \theta_j \sim \mathcal{N}(\theta_j, s_j^2), \qquad \theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2).$$
$$\theta_j \mid \bar{y}_j, \mu, \tau^2 \sim \mathcal{N}(m_j, v_j),$$

Detailed solution and reasoning.
For each study, the conditional posterior is obtained by multiplying $N(\bar{y}_j; \theta_j, s_j^2)$ and $N(\theta_j; \mu, \tau^2)$:

$$\theta_j \mid \bar{y}_j, \mu, \tau^2 \sim N\left(\frac{\bar{y}_j/s_j^2 + \mu/\tau^2}{1/s_j^2 + 1/\tau^2}, \frac{1}{1/s_j^2 + 1/\tau^2}\right).$$

Equivalently, the posterior mean is $w_j \bar{y}_j + (1 - w_j)\mu$ with $w_j = \tau^2/(\tau^2 + s_j^2)$.
Increasing sampling variance s_j^2 reduces the data precision and lowers w_j, producing stronger shrinkage toward μ. Increasing between-study variance τ^2 weakens the population prior precision, raises w_j, and moves the posterior mean toward the study's own estimate. The posterior variance is the reciprocal of the sum of precisions; both data precision and population precision contribute.

Problem 122

Original question (verbatim).
For a Normal-Normal hierarchy, analyse the limits $\tau^2 \to 0$ and $\tau^2 \to \infty$. Show how the posterior estimates approach complete pooling and no pooling, respectively. Explain why these are limiting cases of one model rather than unrelated estimators, and discuss what the observed group variability tells us about the plausibility of each limit.

Required theory (verbatim from the supplied lecture manual).
The Normal–Normal model makes partial pooling algebraically transparent. We observe a group estimate y-bar-j with sampling variance s-j-squared. The true group effect theta-j is itself normal with population mean mu and between-group variance tau-squared. If we condition on mu and tau-squared, the posterior for theta-j is normal.
Its precision is the sum of sampling precision and population-prior precision. Its mean is a precision-weighted average of the group's observed estimate and the population mean. If the group's sampling variance is large, its own observation receives less weight and shrinkage is stronger. If the between-group variance is large, the groups are allowed to differ substantially and the shrinkage is weak. This gives us a direct algebraic expression for partial pooling. **Board equations from the notes.**

$$\bar{y}_j \mid \theta_j \sim \mathcal{N}(\theta_j, s_j^2), \qquad \theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2).$$
$$\theta_j \mid \bar{y}_j, \mu, \tau^2 \sim \mathcal{N}(m_j, v_j),$$

Detailed solution and reasoning.
The conditional posterior mean is $m_j = w_j \bar{y}_j + (1 - w_j)\mu$ with $w_j = \tau^2/(\tau^2 + s_j^2)$. If $\tau^2 \to 0$, then $w_j \to 0$ and $m_j \to \mu$: all groups share the population mean, the complete-pooling limit. If $\tau^2 \to \infty$, then $w_j \to 1$ and $m_j \to \bar{y}_j$: each group is estimated independently, the no-pooling limit. The posterior variance $s_j^2\tau^2/(s_j^2 + \tau^2)$ tends to 0 in the complete-pooling conditional limit and to s_j^2 in the no-pooling limit.
These are limiting cases of one hierarchical model, controlled by between-group variance. Observed between-group variability relative to sampling error informs whether τ^2 is near zero or large. If group estimates vary no more than expected from their sampling errors, complete pooling may be plausible; substantial residual heterogeneity supports partial pooling with positive τ^2.

Problem 123

Original question (verbatim).
Two groups have the same observed estimate, but Group A has standard error 0.1 and Group B has standard error 2.0. Under a common Normal population distribution, derive the two posterior means and variances conditional on the same (μ, τ^2). Explain why equal observed estimates need not lead to equal posterior uncertainty and why the data precision, not just the point estimate, determines shrinkage.

Required theory (verbatim from the supplied lecture manual).
The Normal–Normal model makes partial pooling algebraically transparent. We observe a group estimate y-bar-j with sampling variance s-j-squared. The true group effect theta-j is itself normal with population mean mu and between-group variance tau-squared. If we condition on mu and tau-squared, the posterior for theta-j is normal.

Its precision is the sum of sampling precision and population-prior precision. Its mean is a precision-weighted average of the group's observed estimate and the population mean. If the group's sampling variance is large, its own observation receives less weight and shrinkage is stronger. If the between-group variance is large, the groups are allowed to differ substantially and the shrinkage is weak. This gives us a direct algebraic expression for partial pooling. **Board equations from the notes.**

$$\bar{y}_j \mid \theta_j \sim \mathcal{N}(\theta_j, s_j^2), \qquad \theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2).$$
$$\theta_j \mid \bar{y}_j, \mu, \tau^2 \sim \mathcal{N}(m_j, v_j),$$

Detailed solution and reasoning.
The two sampling variances are $s_A^2 = 0.1^2 = 0.01$ and $s_B^2 = 2^2 = 4$. Conditional posterior variances and means are

$$v_A = \frac{0.01\tau^2}{0.01 + \tau^2}, \quad m_A = \mu + \frac{\tau^2}{\tau^2 + 0.01}(\bar{y} - \mu),$$
$$v_B = \frac{4\tau^2}{4 + \tau^2}, \quad m_B = \mu + \frac{\tau^2}{\tau^2 + 4}(\bar{y} - \mu).$$

Since $0.01 < 4$, Group A has a larger data weight and a smaller posterior variance, while Group B is pulled more strongly toward μ and remains more uncertain. Equal observed point estimates do not imply equal posterior distributions: the sampling precision determines how much each estimate should influence its group effect.

7.4 Marginal likelihood for hyperparameters

Problem 124

Original question (verbatim).
Starting from $\bar{y}_j \mid \theta_j \sim N(\theta_j, s_j^2)$ and $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$, integrate out θ_j and derive the marginal distribution of $\bar{y}_j \mid \mu, \tau^2$. Write the marginal likelihood across all groups. Explain why this reduces the dimension of the problem and how it can be used to learn the population mean and between-group variance.

Required theory (verbatim from the supplied lecture manual).
The hyperparameters describe the population distribution, so they must be learned from the collection of groups. One way to do that is to integrate the group-specific parameters out of the joint model. The resulting marginal likelihood describes the observed group data conditional on the hyperparameters.
In the Normal–Normal case, integrating theta-j gives a normal distribution for each observed group mean with variance equal to the sampling variance plus the between-group variance. This calculation is useful because it reduces the dimension of the problem and shows how the group summaries inform the population-level parameters. Do not confuse marginalising a parameter by integration with profiling it out by maximisation. Those are different operations. **Board equations from the notes.**

$$p(y \mid \eta) = \prod_{j=1}^{J} \int p(y_j \mid \theta_j) p(\theta_j \mid \eta) \, \mathrm{d}\theta_j.$$

$$\bar{y}_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, s_j^2 + \tau^2),$$

Detailed solution and reasoning.
The model is $\bar{y}_j \mid \theta_j \sim N(\theta_j, s_j^2)$ and $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$. The sum of two independent normal variables gives

$$\bar{y}_j = \theta_j + \epsilon_j, \quad \theta_j \sim N(\mu, \tau^2), \quad \epsilon_j \sim N(0, s_j^2),$$

so after integrating θ_j,

$$\bar{y}_j \mid \mu, \tau^2 \sim N(\mu, s_j^2 + \tau^2).$$

For conditionally independent groups, the marginal likelihood is

$$L_m(\mu, \tau^2) = \prod_{j=1}^{J} \frac{1}{\sqrt{2\pi(s_j^2 + \tau^2)}} \exp\left[-\frac{(\bar{y}_j - \mu)^2}{2(s_j^2 + \tau^2)}\right].$$

Integrating out the J group parameters reduces the optimisation/integration dimension to the shared hyperparameters (μ, τ^2). The marginal likelihood contains evidence about the population mean and between-group variance while accounting for known sampling variances.

Problem 125

Original question (verbatim).
A researcher maximises the marginal likelihood for (μ, τ^2) after integrating out all group effects. Another researcher conditions on the group effects and maximises the joint likelihood. Explain why the two optimisations target different likelihood functions. Derive the marginal likelihood in the Normal-Normal model and discuss how integrating out group parameters changes the information available about the hyperparameters.

Required theory (verbatim from the supplied lecture manual).
The hyperparameters describe the population distribution, so they must be learned from the collection of groups. One way to do that is to integrate the group-specific parameters out of the joint model. The resulting marginal likelihood describes the observed group data conditional on the hyperparameters.
In the Normal–Normal case, integrating theta-j gives a normal distribution for each observed group mean with variance equal to the sampling variance plus the between-group variance. This calculation is useful because it reduces the dimension of the problem and shows how the group summaries inform the population-level parameters. Do not confuse marginalising a parameter by integration with profiling it out by maximisation. Those are different operations. **Board equations from the notes.**

$$p(y \mid \eta) = \prod_{j=1}^{J} \int p(y_j \mid \theta_j) p(\theta_j \mid \eta) \, \mathrm{d}\theta_j.$$

$$\bar{y}_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, s_j^2 + \tau^2),$$

Detailed solution and reasoning.
The conditional joint likelihood is $L(y \mid \theta_1, \ldots, \theta_J, \mu, \tau^2)$; maximising it over group effects treats them as nuisance parameters chosen to fit the data best. The marginal likelihood instead averages

over their conditional prior distribution:

$$L_m(\mu, \tau^2) = \int L(y \mid \boldsymbol{\theta}) \prod_j p(\theta_j \mid \mu, \tau^2) d\boldsymbol{\theta}.$$

For the Normal–Normal model this integral is

$$L_m(\mu, \tau^2) = \prod_j N(\bar{y}_j; \mu, s_j^2 + \tau^2).$$

Maximisation and integration are different operations. A profile or conditional likelihood uses the best-fitting group effects; the marginal likelihood accounts for the full distribution of plausible group effects and includes the dispersion penalty implied by that distribution. The resulting functions of (μ, τ^2) need not have the same shape or maximiser. The marginal likelihood is appropriate for Bayesian hyperparameter updating; conditional likelihood optimisation targets a different likelihood objective.

Problem 126

Original question (verbatim).
A colleague says that integrating out a group parameter and maximising over it are equivalent because both remove a nuisance quantity. Use a simple normal hierarchical model to compare marginalisation and profiling. Identify what distribution is averaged over in the former and what optimisation is performed in the latter. Give a concrete circumstance in which the resulting functions of the hyperparameters differ.

Required theory (verbatim from the supplied lecture manual).
The hyperparameters describe the population distribution, so they must be learned from the collection of groups. One way to do that is to integrate the group-specific parameters out of the joint model. The resulting marginal likelihood describes the observed group data conditional on the hyperparameters.
In the Normal–Normal case, integrating theta-j gives a normal distribution for each observed group mean with variance equal to the sampling variance plus the between-group variance. This calculation is useful because it reduces the dimension of the problem and shows how the group summaries inform the population-level parameters. Do not confuse marginalising a parameter by integration with profiling it out by maximisation. Those are different operations. **Board equations from the notes.**

$$p(y \mid \eta) = \prod_{j=1}^{J} \int p(y_j \mid \theta_j) p(\theta_j \mid \eta) \, \mathrm{d}\theta_j.$$

$$\bar{y}_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, s_j^2 + \tau^2),$$

Detailed solution and reasoning.
Consider $Y \mid \theta, \eta \sim N(\theta, 1)$ and $\theta \mid \eta \sim N(\mu, \tau^2)$, where $\eta = (\mu, \tau^2)$ are hyperparameters. Marginalisation computes

$$p(y \mid \eta) = \int p(y \mid \theta, \eta) p(\theta \mid \eta) d\theta,$$

which averages the likelihood over all possible θ values weighted by their prior distribution under

η. Profiling instead computes

$$L_p(\eta) = \max_{\theta} p(y \mid \theta, \eta),$$

which keeps only the best-fitting value.
For example, in the normal-normal summary model marginalisation produces $\bar{y} \mid \mu, \tau^2 \sim N(\mu, s^2 + \tau^2)$, while profiling over θ in $\bar{y} \mid \theta \sim N(\theta, s^2)$ and $\theta \sim N(\mu, \tau^2)$ selects the conditional likelihood maximum at $\theta = \bar{y}$ (with the prior treatment depending on whether it is included in the joint objective). An integral averages a range of values and changes with the prior spread; a maximum evaluates one best-fitting value. They are not equivalent nuisance-removal operations.

7.5 Empirical Bayes

Problem 127

Original question (verbatim).
Suppose observed group estimates follow $\bar{y}_j \mid \mu, \tau^2 \sim N(\mu, s_j^2 + \tau^2)$, for $j = 1, \ldots, J$. Derive a method-of-moments estimator for τ^2 in the balanced case with common known sampling variance. Use the estimator in the conditional posterior of a group effect and explain how the estimated between-group variance controls shrinkage. Discuss what happens when the moment estimate is negative and why a boundary estimate of zero corresponds to complete pooling.

Required theory (verbatim from the supplied lecture manual).
Empirical Bayes estimates the hyperparameters from the data and then plugs those estimates into the group-level prior. One common method maximises the marginal likelihood after integrating out the group effects. Another uses moments of the observed group estimates. This is convenient because the data themselves help determine the amount of pooling.
The limitation is that the hyperparameters are treated as if their estimates were known exactly. That ignores uncertainty about the population distribution, which may be substantial when there are only a few groups. Consequently, empirical-Bayes intervals for group effects can be too narrow. Full hierarchical Bayes addresses this by assigning a prior to the hyperparameters and averaging over their posterior uncertainty. **Board equations from the notes.**

$$\hat{\eta}_{EB} = \arg\max_{\eta} p(y \mid \eta), \qquad p(y \mid \eta) = \int p(y \mid \theta, \eta) p(\theta \mid \eta)\, \mathrm{d}\theta.$$

Detailed solution and reasoning.
In the balanced case $s_j^2 = s^2$, the marginal model is

$$\bar{y}_j \sim N(\mu, s^2 + \tau^2).$$

The method-of-moments equation matches the variance of group estimates:

$$S_{\bar{y}}^2 = \frac{1}{J-1} \sum_j (\bar{y}_j - \bar{y}_\cdot)^2 \approx s^2 + \tau^2,$$

so

$$\widehat{\tau^2}_{MOM} = S_{\bar{y}}^2 - s^2,$$

usually truncated at zero to respect $\tau^2 \geq 0$. Plug this estimate into

$$\theta_j \mid \bar{y}_j, \mu, \hat{\tau}^2 \sim N\left(w_j \bar{y}_j + (1 - w_j)\mu, \frac{s^2 \hat{\tau}^2}{s^2 + \hat{\tau}^2} \right), \quad w_j = \frac{\hat{\tau}^2}{s^2 + \hat{\tau}^2}.$$

A negative moment estimate means observed dispersion is no larger than expected sampling variance; setting the constrained variance estimate to zero gives $w_j = 0$, the complete-pooling boundary. This is an estimate, not proof that groups are exactly identical.

Problem 128

Original question (verbatim).
An empirical-Bayes analysis first integrates out group effects, maximises the marginal likelihood for hyperparameters, and then plugs the estimates into each group posterior. Write the steps mathematically for the Normal-Normal model. Explain the computational advantage and identify the uncertainty that is ignored when hyperparameters are treated as fixed at their estimates.

Required theory (verbatim from the supplied lecture manual).
Empirical Bayes estimates the hyperparameters from the data and then plugs those estimates into the group-level prior. One common method maximises the marginal likelihood after integrating out the group effects. Another uses moments of the observed group estimates. This is convenient because the data themselves help determine the amount of pooling.
The limitation is that the hyperparameters are treated as if their estimates were known exactly. That ignores uncertainty about the population distribution, which may be substantial when there are only a few groups. Consequently, empirical-Bayes intervals for group effects can be too narrow. Full hierarchical Bayes addresses this by assigning a prior to the hyperparameters and averaging over their posterior uncertainty. **Board equations from the notes.**

$$\hat{\eta}_{EB} = \arg\max_{\eta} p(y \mid \eta), \qquad p(y \mid \eta) = \int p(y \mid \theta, \eta) p(\theta \mid \eta)\, \mathrm{d}\theta.$$

Detailed solution and reasoning.
For the Normal–Normal model, first integrate the group effects:

$$\bar{y}_j \mid \eta = (\mu, \tau^2) \sim N(\mu, s_j^2 + \tau^2), \quad L_m(\eta) = \prod_j N(\bar{y}_j; \mu, s_j^2 + \tau^2).$$

Second, estimate hyperparameters by

$$\widehat{\eta}_{EB} = \arg\max_{\eta} L_m(\eta).$$

Third, plug them into each conditional group posterior:

$$p_{EB}(\theta_j \mid y) = p(\theta_j \mid \bar{y}_j, \widehat{\mu}_{EB}, \widehat{\tau}^2_{EB}).$$

This avoids sampling/integrating a full hyperparameter posterior and can be computationally efficient. However, it treats $\widehat{\eta}_{EB}$ as fixed. Uncertainty about μ and especially τ^2 is not propagated into the group posterior, so intervals may be too narrow, particularly with few groups or a variance estimate near the boundary.

Problem 129

Original question (verbatim).
Only four groups are available, and the estimated between-group variance is close to zero. An empirical-Bayes report gives very narrow intervals for each group effect. Explain why the small number of groups makes hyperparameter uncertainty particularly important. Propose a comparison with full hierarchical Bayes and discuss how plugging in a point estimate can understate posterior uncertainty.

Required theory (verbatim from the supplied lecture manual).
Empirical Bayes estimates the hyperparameters from the data and then plugs those estimates into the group-level prior. One common method maximises the marginal likelihood after integrating out the group effects. Another uses moments of the observed group estimates. This is convenient because the data themselves help determine the amount of pooling.
The limitation is that the hyperparameters are treated as if their estimates were known exactly. That ignores uncertainty about the population distribution, which may be substantial when there are only a few groups. Consequently, empirical-Bayes intervals for group effects can be too narrow. Full hierarchical Bayes addresses this by assigning a prior to the hyperparameters and averaging over their posterior uncertainty. **Board equations from the notes.**

$$\hat{\eta}_{EB} = \arg\max_{\eta} p(y \mid \eta), \qquad p(y \mid \eta) = \int p(y \mid \theta, \eta) p(\theta \mid \eta)\, \mathrm{d}\theta.$$

Detailed solution and reasoning.
With only four groups, the marginal likelihood for (μ, τ^2) is based on very limited between-group information. It may be relatively flat or have substantial posterior mass near zero and away from zero, even when the empirical-Bayes maximiser is near zero. A plug-in analysis replaces this distribution with a single point, discarding uncertainty about the amount of heterogeneity.
A full hierarchical analysis assigns a proper hyperprior and computes

$$p(\theta_j \mid y) = \int p(\theta_j \mid y, \mu, \tau^2) p(\mu, \tau^2 \mid y) d\mu d\tau^2.$$

Compare posterior means, intervals, tail probabilities, and predictions under full Bayes and EB. Full Bayes will often produce wider intervals because it averages over uncertain τ^2; the exact difference depends on hyperpriors and data. With few groups, an EB interval based on a boundary estimate can materially understate uncertainty and should be presented as a plug-in approximation.

7.6 Profile likelihood: the syllabus connection

Problem 130

Original question (verbatim).
For the Normal-Normal group-summary model, suppose $\bar{y}_j \mid \mu, \tau^2 \sim N(\mu, s_j^2 + \tau^2)$, with known $s_j^2 = s^2$. For each fixed $\tau^2 \geq 0$, maximise the likelihood over μ, substitute the maximiser back into the likelihood, and derive the profile likelihood for τ^2. In the balanced case, identify the maximum-likelihood estimate of τ^2 and explain why the divisor is J, not $J - 1$, in the likelihood maximisation result.

Required theory (verbatim from the supplied lecture manual).
Profile likelihood is a likelihood-based method for handling nuisance parameters. Suppose psi is the parameter we care about and lambda is a nuisance parameter. For every fixed value of psi, we maximise the likelihood over lambda. The resulting function of psi is called the profile likelihood. This is different from Bayesian marginalisation. Bayesian marginalisation integrates over lambda using a probability distribution. Profiling selects the best-fitting lambda for each psi, while integration averages over possible lambda values. We will work through the Normal group-summary example: for fixed tau-squared, maximise over mu, substitute the estimate back into the likelihood, and then maximise the profile function over the nonnegative range of tau-squared. The divisor J in the resulting maximum-likelihood expression comes from likelihood maximisation; the familiar J-minus-one divisor belongs to the unbiased sample variance. **Board equations from the notes.**

$$L_p(\psi) = \max_{\lambda} L(\psi, \lambda).$$

$$p(y \mid \psi) = \int p(y \mid \lambda, \psi) p(\lambda \mid \psi)\, \mathrm{d}\lambda,$$

Detailed solution and reasoning.
Write $v = s^2 + \tau^2$ and $S = \sum_{j=1}^{J} (\bar{y}_j - \bar{y}_\cdot)^2$. The likelihood is

$$L(\mu, v) \propto v^{-J/2} \exp\left[-\frac{1}{2v} \sum_j (\bar{y}_j - \mu)^2\right].$$

For fixed v, differentiating the log likelihood with respect to μ gives $\widehat{\mu}(v) = \bar{y}_\cdot$. Substitution yields the profile likelihood

$$L_p(\tau^2) \propto (s^2 + \tau^2)^{-J/2} \exp\left[-\frac{S}{2(s^2 + \tau^2)}\right], \quad \tau^2 \geq 0.$$

The profile log-likelihood derivative with respect to v is

$$-\frac{J}{2v} + \frac{S}{2v^2} = 0 \quad \Rightarrow \quad \widehat{v} = S/J.$$

Thus

$$\widehat{\tau^2}_{ML} = \max\{0, S/J - s^2\}.$$

The divisor is J because it comes from maximising the Gaussian likelihood with an estimated mean. The unbiased sample variance uses $J - 1$; it is a different estimator with a different objective.

Problem 131

Original question (verbatim).
Let ψ be a parameter of interest and λ a nuisance parameter. Define the profile likelihood and compare it with a Bayesian marginal likelihood. Use a normal model with unknown mean and variance to show explicitly which operation is performed on the nuisance parameter in each method. Explain why profiling does not average over uncertainty in λ, whereas marginalisation does.

Required theory (verbatim from the supplied lecture manual).
Profile likelihood is a likelihood-based method for handling nuisance parameters. Suppose psi is

the parameter we care about and lambda is a nuisance parameter. For every fixed value of psi, we maximise the likelihood over lambda. The resulting function of psi is called the profile likelihood. This is different from Bayesian marginalisation. Bayesian marginalisation integrates over lambda using a probability distribution. Profiling selects the best-fitting lambda for each psi, while integration averages over possible lambda values. We will work through the Normal group-summary example: for fixed tau-squared, maximise over mu, substitute the estimate back into the likelihood, and then maximise the profile function over the nonnegative range of tau-squared. The divisor J in the resulting maximum-likelihood expression comes from likelihood maximisation; the familiar J-minus-one divisor belongs to the unbiased sample variance. **Board equations from the notes.**

$$L_p(\psi) = \max_{\lambda} L(\psi, \lambda).$$

$$p(y \mid \psi) = \int p(y \mid \lambda, \psi) p(\lambda \mid \psi)\, \mathrm{d}\lambda,$$

Detailed solution and reasoning.
The profile likelihood for parameter of interest ψ and nuisance parameter λ is

$$L_p(\psi) = \max_{\lambda} L(\psi, \lambda).$$

For $Y_i \sim N(\mu, \sigma^2)$, treating μ as interest and σ^2 as nuisance, the maximiser for fixed μ is

$$\widehat{\sigma}^2(\mu) = \frac{1}{n} \sum_i (y_i - \mu)^2,$$

which is substituted into the likelihood to get $L_p(\mu)$.
A Bayesian marginal likelihood instead integrates against a conditional prior:

$$p(y \mid \mu) = \int p(y \mid \mu, \sigma^2) \pi(\sigma^2 \mid \mu) d\sigma^2.$$

Profiling selects one best-fitting nuisance value for every μ; marginalisation averages over the distribution of nuisance values. The latter requires a prior and retains uncertainty through integration, while the former is likelihood-based and does not average over nuisance-parameter uncertainty. They generally produce different functions of μ.

Problem 132

Original question (verbatim).
A report presents a profile likelihood curve for a variance component and interprets the height at each value as the posterior probability of that variance. Explain why this interpretation is incorrect. Describe how a profile likelihood can be used to study the likelihood support for a parameter of interest, what additional ingredients a Bayesian posterior requires, and why the two curves need not have the same shape.

Required theory (verbatim from the supplied lecture manual).
Profile likelihood is a likelihood-based method for handling nuisance parameters. Suppose psi is the parameter we care about and lambda is a nuisance parameter. For every fixed value of psi, we maximise the likelihood over lambda. The resulting function of psi is called the profile likelihood. This is different from Bayesian marginalisation. Bayesian marginalisation integrates over lambda

using a probability distribution. Profiling selects the best-fitting lambda for each psi, while integration averages over possible lambda values. We will work through the Normal group-summary example: for fixed tau-squared, maximise over mu, substitute the estimate back into the likelihood, and then maximise the profile function over the nonnegative range of tau-squared. The divisor J in the resulting maximum-likelihood expression comes from likelihood maximisation; the familiar J-minus-one divisor belongs to the unbiased sample variance. **Board equations from the notes.**

$$L_p(\psi) = \max_{\lambda} L(\psi, \lambda).$$

$$p(y \mid \psi) = \int p(y \mid \lambda, \psi) p(\lambda \mid \psi) \, \mathrm{d}\lambda,$$

Detailed solution and reasoning.
A profile likelihood $L_p(\tau^2)$ is a maximised likelihood value, not a probability density for τ^2. Its vertical height is not normalised to integrate to one and contains no prior information. A profile curve can show how well the data support different values of a variance component after optimising nuisance parameters; likelihood ratios or confidence procedures can be built from it.
A Bayesian posterior requires a prior density and Bayes' rule:

$$p(\tau^2 \mid y) \propto p(y \mid \tau^2) \pi(\tau^2),$$

where the likelihood may be a marginal likelihood after integrating other parameters. The posterior also requires normalisation. Prior weighting, integration versus profiling, and parameter boundaries can all change curve shape. Therefore the profile height cannot be read as posterior probability.

7.7 Full hierarchical Bayes versus empirical Bayes

Problem 133

Original question (verbatim).
Write the posterior for a group parameter θ_j under full hierarchical Bayes by integrating its conditional posterior over the posterior distribution of hyperparameters. Write the empirical-Bayes approximation obtained by plugging in $\widehat{\eta}_{EB}$. Explain which uncertainty is propagated in each approach and construct a setting with few groups in which the two resulting intervals differ substantially.

Required theory (verbatim from the supplied lecture manual).
The central difference is how uncertainty about the hyperparameters is handled. In full hierarchical Bayes, we assign a hyperprior and calculate the joint posterior. To obtain the posterior for a particular group parameter, we integrate its conditional posterior over the posterior distribution of the hyperparameters. In empirical Bayes, we estimate the hyperparameters and plug in one value. Both approaches use the observed data. The difference is not whether data are used, but whether hyperparameter uncertainty is propagated. Full Bayes is more complete but can be computationally more demanding. Empirical Bayes can be efficient and useful, but it may understate uncertainty, especially when the number of groups is small. **Board equations from the notes.**

$$p(\theta_j \mid y) = \int p(\theta_j \mid \eta, y) p(\eta \mid y) \, \mathrm{d}\eta.$$

$$p_{EB}(\theta_j \mid y) = p(\theta_j \mid \hat{\eta}_{EB}, y).$$

Detailed solution and reasoning.
Let $\eta = (\mu, \tau^2)$ denote hyperparameters. Full hierarchical Bayes gives

$$p(\theta_j \mid y) = \int p(\theta_j \mid y, \eta) p(\eta \mid y) d\eta.$$

This averages the conditional group posterior over the posterior uncertainty in the hyperparameters. Empirical Bayes substitutes one estimate:

$$p_{EB}(\theta_j \mid y) = p(\theta_j \mid y, \widehat{\eta}_{EB}).$$

The full posterior propagates both within-group and hyperparameter uncertainty; EB propagates the former conditional on a fixed estimated hierarchy. With few groups, $p(\tau^2 \mid y)$ may be broad or include values near zero and appreciably larger values. Integrating over this distribution can yield wider or differently shaped intervals than a plug-in posterior. The difference is especially important when the EB estimate lies on the boundary $\widehat{\tau}^2 = 0$.

Problem 134

Original question (verbatim).
A full Bayesian analysis assigns a hyperprior to (μ, τ^2), while an empirical-Bayes analysis estimates these quantities from the marginal likelihood. Compare the information each method uses and the computational burden of each. Explain why full Bayes is not automatically immune to poor prior or likelihood choices, and why empirical Bayes is not necessarily invalid merely because it plugs in an estimate.

Required theory (verbatim from the supplied lecture manual).
The central difference is how uncertainty about the hyperparameters is handled. In full hierarchical Bayes, we assign a hyperprior and calculate the joint posterior. To obtain the posterior for a particular group parameter, we integrate its conditional posterior over the posterior distribution of the hyperparameters. In empirical Bayes, we estimate the hyperparameters and plug in one value. Both approaches use the observed data. The difference is not whether data are used, but whether hyperparameter uncertainty is propagated. Full Bayes is more complete but can be computationally more demanding. Empirical Bayes can be efficient and useful, but it may understate uncertainty, especially when the number of groups is small. **Board equations from the notes.**

$$p(\theta_j \mid y) = \int p(\theta_j \mid \eta, y) p(\eta \mid y) \, \mathrm{d}\eta.$$

$$p_{EB}(\theta_j \mid y) = p(\theta_j \mid \hat{\eta}_{EB}, y).$$

Detailed solution and reasoning.
Full Bayes assigns a hyperprior $\pi(\mu, \tau^2)$ and updates

$$p(\boldsymbol{\theta}, \mu, \tau^2 \mid y) \propto p(y \mid \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mu, \tau^2) \pi(\mu, \tau^2).$$

It uses the data at all levels and propagates hyperparameter uncertainty, but may require more computation and careful MCMC diagnostics. Empirical Bayes maximises the integrated group-effect likelihood and plugs in estimates; it is often faster and can be reasonable when many groups make the hyperparameters well estimated or when the goal is a practical approximation.

Full Bayes is not automatically protected from a poor likelihood or unjustified hyperprior; it can give precise answers under a misspecified model. EB is not inherently invalid, but its plug-in uncertainty omission should be acknowledged and assessed, especially with few groups. The distinction is how hyperparameter uncertainty is treated, not a simple valid/invalid dichotomy.

Problem 135

Original question (verbatim).
A collaborator argues that empirical Bayes, full hierarchical Bayes, and profile likelihood are three names for the same operation of 'removing nuisance parameters'. Construct a mathematical comparison showing (i) integration over hyperparameters, (ii) plug-in estimation of hyperparameters, and (iii) maximisation over nuisance parameters. Explain the different inferential targets and identify which uncertainty is retained or discarded in each procedure.

Required theory (verbatim from the supplied lecture manual).
The central difference is how uncertainty about the hyperparameters is handled. In full hierarchical Bayes, we assign a hyperprior and calculate the joint posterior. To obtain the posterior for a particular group parameter, we integrate its conditional posterior over the posterior distribution of the hyperparameters. In empirical Bayes, we estimate the hyperparameters and plug in one value. Both approaches use the observed data. The difference is not whether data are used, but whether hyperparameter uncertainty is propagated. Full Bayes is more complete but can be computationally more demanding. Empirical Bayes can be efficient and useful, but it may understate uncertainty, especially when the number of groups is small. **Board equations from the notes.**

$$p(\theta_j \mid y) = \int p(\theta_j \mid \eta, y) p(\eta \mid y)\, \mathrm{d}\eta.$$

$$p_{EB}(\theta_j \mid y) = p(\theta_j \mid \hat{\eta}_{EB}, y).$$

Detailed solution and reasoning.
The three operations are mathematically distinct. **(i) Full hierarchical Bayes integrates:**

$$p(\theta_j \mid y) = \int p(\theta_j \mid \eta, y) p(\eta \mid y) d\eta.$$

This averages over the posterior distribution of hyperparameters and propagates their uncertainty.
(ii) Empirical Bayes plugs in:

$$p_{EB}(\theta_j \mid y) = p(\theta_j \mid \widehat{\eta}_{EB}, y), \qquad \widehat{\eta}_{EB} = \arg\max_{\eta} p(y \mid \eta),$$

where group effects may first be integrated out to obtain the marginal likelihood. It retains conditional group uncertainty but discards hyperparameter uncertainty beyond the point estimate.
(iii) Profile likelihood maximises over nuisance parameters:

$$L_p(\psi) = \max_{\lambda} L(\psi, \lambda).$$

It keeps the best-fitting nuisance value for each ψ and is a likelihood function, not a posterior distribution. It does not average over a probability distribution.
Thus integration, plug-in estimation, and maximisation have different targets and retain different uncertainty. Calling all three 'removing nuisance parameters' hides the essential distinction: full Bayes propagates uncertainty, EB approximates it by a fixed estimate, and profiling optimises it away.

Final derivation checklist

Before submitting an exam solution, verify that you have (i) stated the sampling model and unknown parameter, (ii) written the likelihood with the correct conditioning, (iii) specified the prior and its parameterisation, (iv) shown the algebra leading to the posterior or inferential target, (v) integrated over posterior uncertainty for prediction or marginal likelihood when required, (vi) distinguished conditional from marginal independence, (vii) stated the uncertainty represented by each interval or diagnostic, and (viii) interpreted the result in context. For MCMC and hierarchical models, separately justify computation, convergence, model adequacy, and the treatment of hyperparameter uncertainty.